Combinatorial and topological games

Baire spaces and applications in classical mathematics

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Contents

Introduction 2

Part I. Topological games and Baire spaces 6
1. Baire category 7
2. Infinite games 9
3. The Banach–Mazur game 12
4. The strong Choquet game 16
5. The converse of Choquet's theorem 22
6. Summary: from complete metrizability to Baire category 28
7. Notes and historical remarks 28

Part II. Applications of Banach–Mazur and Choquet games 30
8. Applications to Diophantine approximation 32
9. Applications to homogeneous dynamics 37
10. Applications to computational complexity 41
11. Applications to topological groups 44
12. Applications to infinite-dimensional Ramsey theory 47
13. Applications to Banach space theory 51
14. Applications to ultrafilters 55

Part III. Applications of other topological games 62
15. Applications to set-theoretic topology 63
16. Applications to descriptive set theory 65

Part IV. Applications of non-topological games 70
17. Applications to the probabilistic method 70
18. Applications to fixed point theory 73

Part V. Ehrenfeucht–Fraïssé games 76
19. Elementary equivalence and the game 77
20. The Ehrenfeucht–Fraïssé theorem 79
21. Linear orders, parity and connectivity 83
22. Back-and-forth systems and ultrapowers 88
23. Extensions to other logics 89
24. Notes and historical remarks 91

Part VI. The Scottish Book, von Neumann, Ulam, Texas, Toronto, and the triumph of games, logic, topology and combinatorics 92
25. The marble tables of the Scottish Café 93
26. Problem 43 94
27. Von Neumann's invitation and the bridge to America 95
28. War, and the survival of the book 95
29. Ulam after Lwów 96
30. The 1957 Los Alamos mimeograph 96
31. Problem 153: a live goose 96
32. The road to Denton: Dan Mauldin and North Texas 97
33. The New Scottish Book 97

34. Set-theoretic topology and Ramsey spaces at Toronto 97
35. The triumph of games: combinatorics, logic and the real-valued setting 98
36. Epilogue: A Scottish Book of our own 99
Further reading 99
References 99

Introduction

In 1935 Stanisław Mazur entered a problem in the Scottish Book (see Figure 1). The book was a notebook kept at the Scottish Café, in the beautiful city of Lwów (see Figure 2), and the mathematicians of Lwów did a good part of their thinking at the café's tables. The problem concerns a game in which two players take turns choosing intervals of the real line, each inside the previous one. Mazur asked when the first player can force the intersection of the intervals to meet a given set, and when the second player can keep the intersection disjoint from it. Banach found the answer but never published a proof. The first published proof is Oxtoby's, in a paper of 1957 whose title calls the game by the name it has kept: the Banach–Mazur game.

A game is, at first sight, a curious way of asking a question about a set. Its interest lies in the answer. In Part I we show that the second player can force the intersection of the moves to lie inside a set exactly when the set is large in the sense of Baire category (see Theorem 3.5). Category and the game are thus one notion under two modes of presentation: the one given by countable unions of nowhere dense sets, the other by what a player can enforce. The second presentation proved the more fertile. Over the last sixty years the rules have been changed in many ways: the sizes of the moves have been prescribed, the players have been limited in how much they may compute, and they have been allowed to play vectors, points or covers instead of intervals. Each change has found its problem in some unexpected corner of mathematics. We first develop the theory of the game and then tell twelve stories in which a game, often one of these variants, settles a problem.

The theory occupies Part I: the Banach–Mazur game, its strengthening, the strong Choquet game, and what the two games reveal about Baire spaces and complete metrizability. The part is self-contained and can be read as a short course on its own.

In Part II we collect applications of the Banach–Mazur game itself and of its close relatives, the games of Choquet type, in which the two players take turns shrinking a set:

Number theory: Schmidt's games and badly approximable numbers (Schmidt, 1966).

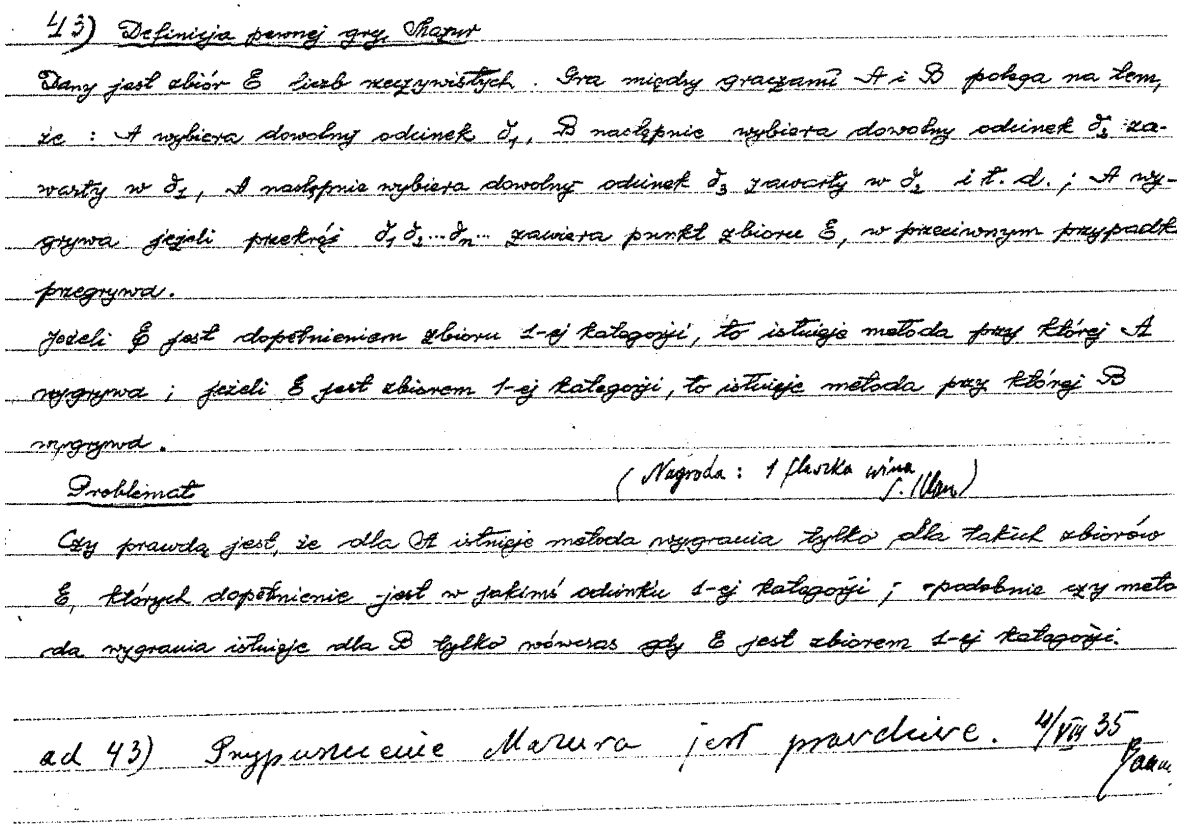
Dynamical systems and Lie theory: Bounded orbits of flows on spaces of lattices, and Margulis's conjecture, proved by Kleinbock and Margulis (1996).

Theoretical computer science: Baire category inside complexity classes (Mehlhorn, 1973; Lutz, 1990).

Topological groups: When separate continuity implies joint continuity (Ellis, 1957; Namioka, 1974; Kenderov, Kortezov and Moors, 2001).

Infinite-dimensional Ramsey theory: The Ellentuck topology (Ellentuck, 1974) and the Kastanas game (Kastanas, 1983).

Functional analysis: Asplund spaces (Namioka–Phelps and Stegall, 1975; the game of Kenderov and Moors, 1999) and Gowers's dichotomy for Banach spaces (Gowers, 1996).


43) Definicja pewnej gry. Mazur

Dany jest zbiór E liczb rzeczywistych. Gra między graczami A i B polega na tem, że: A wybiera dowolny odcinek δ_1, B następnie wybiera dowolny odcinek δ_2 zawarty w δ_1, A następnie wybiera dowolny odcinek δ_3 zawarty w δ_2 i t. d.; A wygrywa jeżeli przekrój $\delta_1 \delta_2 \ldots \delta_n \ldots$ zawiera punkt zbioru E, w przeciwnym przypadku przegrywa.

Jeżeli E jest dopełnieniem zbioru 1-ej kategorji, to istnieje metoda przy której A wygrywa i jeżeli E jest zbiorem 1-ej kategorji, to istnieje metoda przy której B wygrywa.

Problemat (Nagroda: 1 flaszka wina S. Ulam)

Czy prawdą jest, że dla A istnieje metoda wygrania tylko dla takich zbiorów E, których dopełnienie jest w jakimś odcinku 1-ej kategorji; podobnie czy metoda wygrania istnieje dla B tylko wówczas gdy E jest zbiorem 1-ej kategorji.

ad 43) Przypuszczenie Mazura jest prawdziwe. 4/VIII 35 Banach

FIGURE 1. Mazur's Problem 43 in the Scottish Book, with the prize he offered and his signature: "Nagroda: 1 flaszka wina" (a bottle of wine). Below, from the opposite page: "ad 43) Przypuszczenie Mazura jest prawdziwe. 4/VIII 35" ("Mazur's conjecture is true," 4 August 1935), signed by Banach. Courtesy of the Banach family, via the Stefan Banach Portal, Adam Mickiewicz University, Poznań (`kielich.amu.edu.pl/Stefan_Banach`).

Set theory: Nonprincipal ultrafilters on the set ω of natural numbers, the Baire property and the axiom of choice (Sierpiński, 1938; Jalali-Naini, 1976; Talagrand, 1980), and Galvin's games for selective ultrafilters and P-points.

The last of these stories brings the Banach–Mazur game face to face with the axiom of choice: a nonprincipal ultrafilter on ω defines a game in which neither player has a winning strategy, and consequently, if every game of this kind is determined, there is no nonprincipal ultrafilter on ω (see Theorem 14.1).

In Part III we turn to other famous topological games, in which the players do not simply shrink a set:

Set-theoretic topology: The Menger game and the refutation of Menger's conjecture (Fremlin and Miller, 1988).

Descriptive set theory: Borel determinacy (Martin, 1975) and the Wadge hierarchy (Wadge, 1972).

Two beautiful applications of games that are not topological at all make up Part IV:

Probabilistic combinatorics: The Erdős–Selfridge theorem and derandomization (Erdős and Selfridge, 1973).

Fixed point theory: Hex and the Brouwer fixed point theorem (Gale, 1979).

Logic has a game of its own, and Part V is devoted to it. Two structures are *elementarily equivalent* when they satisfy the same first-order sentences, and a check of this sentence by sentence would never end. The Ehrenfeucht–Fraïssé game (Fraïssé, 1954; Ehrenfeucht, 1961) replaces the check by a contest: two players choose elements of the two structures in turn, each round answering one quantifier, and the second player can survive n rounds exactly when the structures satisfy the same sentences with at most n nested quantifiers. We prove this theorem and use it to show that neither the parity of a finite linear order nor the connectivity of a finite graph can be expressed by a first-order sentence; we then relate the game to back-and-forth systems and to ultrapowers, where the ultrafilters of Part II return, and we close with the games that characterize other logics. The game was found in Warsaw, the other great center of Polish mathematics between the wars, and the notes that begin with a game from Lwów thus end with one from Warsaw. Like Part IV, the part draws on Part I only for the language of games of Section 2.

The epilogue, Part VI, is the story of the Scottish Book: how it survived the war, how its problems traveled to Los Alamos, Denton and Toronto, and how the games it started came back to settle a problem of Banach himself.

Once Part I has been read, the order is the reader's to choose: most sections of Parts II–IV stand on their own, and at the opening of Part II we indicate which of its sections lean on which. These sections share roughly the same plan (the problem, its history, the game, the main result, and how the game solves the problem), so that a reader who knows one of them knows how to read the others. The part on logic, Part V, is built differently, as a chapter of a textbook: definitions, theorems and proofs come first, the applications follow, and the history is gathered at the end.

Since every game in Part II is the original one with a few rules changed, we record the original here, in the form in which we will keep modifying it. Mazur's question was about a set E of real numbers at which the first player aims; we write $S = \mathbb{R} \setminus E$ for the set at which the second player aims, and give the payoff to her (see the conventions below). In Part I we prove the theorem in the box below for an arbitrary topological space in place of the real line (see Theorem 3.5).

The Banach–Mazur game. Fix a set $S \subseteq \mathbb{R}$. Player I chooses a closed interval U_0 of positive length; Player II chooses a closed interval $V_0 \subseteq U_0$ of positive length; Player I chooses $U_1 \subseteq V_0$; and so on, forever. Player II wins if every point of $\bigcap_{k=0}^{\infty} V_k$ belongs to S. Theorem (Banach; Oxtoby, 1957): Player II has a winning strategy if and only if the complement $\mathbb{R} \setminus S$ is meager (see Definition 1.1), that is, if and only if the set S is "large" in the sense of Baire category.

Behind all the variations of the rules, two features of games recur, and between them they account for much of what follows. The first is that a winning strategy is built one move at a time: a global question, whether a set is large, becomes a local one, what to play next. The second is that strategies can be *combined*: a player can pursue several goals at once by taking turns between them, one round for this goal, the next round for that one. This second feature explains why the sets for which Player II has a winning strategy survive countable intersections. Comeager sets, whose complements are countable unions of nowhere dense sets,

and sets of full measure survive them too, but largeness in the sense of dimension does not: two sets of full Hausdorff dimension can be disjoint. The badly approximable numbers, the real numbers x for which there is $c > 0$ with $|x - p/q| \geq c/q^2$ for all integers p and $q \geq 1$, form one such set, and their complement another. For each value of the shrinking factor that Schmidt's game prescribes to the second player, the winning sets of the game have both virtues: they have full dimension, and they are closed under countable intersections. We show both facts in Section 8, where we also tell the history of the problem.

Conventions. We follow the usage of Part I throughout: Player I moves first; in games of Banach–Mazur type, Player II answers inside Player I's move; and the payoff set belongs to Player II, whom we call "she". In most of our games she is the one trying to produce an object with a given property, and Player I, whom we call "he", is the one trying to stop her; in the Maker–Breaker games of Section 17 the roles are reversed, and Maker, who builds, is Player I. Where an author used other names for the players (β and α, Black and White, ONE and TWO, Maker and Breaker, Spoiler and Duplicator), we say which of them is our Player I. Among the exercises at the end of the sections, the more demanding ones are marked $(*)$.



(a) (b)

FIGURE 2. The Scottish Café. (a) Plac Akademicki in Lwów, 1925; the café occupied the ground floor of the turreted corner building on the right, today Prospekt Shevchenka 27. Photograph by Marek Münz; National Library of Poland (Polona), public domain. (b) The interior of the café in the 1930s; photographer unknown, public domain.

Part

I

Topological games and Baire spaces

A set of real numbers may be large in more than one sense: in measure, in dimension, or in category. Of these, the third is the one most naturally described by a game. A set of the first category is a countable union of pieces that are thin everywhere, and a player who answers each open set with a smaller one can step clear of one such piece at every move; by dodging the pieces one after another, the player keeps the intersection of the moves outside the whole set. That only sets of the first category can be dodged in this way is the deeper half of the theorem of Banach and Oxtoby, which we prove in Section 3. In this part we turn this observation into a theory of *topological games*: infinite games in which two players alternately choose open sets (in one of them Player I also chooses points), and in which a topological property of the underlying space is characterized by which player has a winning strategy.

Roadmap

Two theorems carry this part, and each of them turns a property of a space into a fact about a game. The first is that a non-empty space is a Baire space exactly when the first player has no winning strategy in the Banach–Mazur game (see Theorem 3.3): that the space is Baire and that Player I is powerless are one fact, stated once in the language of topology and once in the language of play. The second is that a non-empty metrizable space is completely metrizable exactly when the second player has a winning strategy in the strong Choquet game (see Theorem 4.4).

In Section 1 we recall category and Baire spaces, and in Section 2 we say what a game is. The first theorem is proved in Section 3. In Section 4 the moves of Player I acquire points; there we prove the easy half of the second theorem and derive from it the Baire category theorem (see Corollary 4.5), and we extend that half to Čech-complete spaces (see Subsection 4.1). The hard half, which we prove in Section 5, is that a winning strategy for Player II produces a complete metric that induces the topology of the space. The proof uses three classical tools that are worth knowing for their own sake: Alexandrov's theorem on countable intersections of open sets, A. H. Stone's theorem on locally finite refinements, and Kőnig's lemma. We close with a single chain

of implications that gathers all these results (see Section 6), and with notes on their history (see Section 7). A reader who knows Baire category may begin with Section 2. The later parts draw on this one unequally: the applications in Part II rest on the Banach–Mazur and strong Choquet games of Sections 3 and 4, whereas Parts III–V need little more than the language of Section 2. If the part has a moral, it is that a strategy can stand in for a construction: the complete metric of Section 5 is read off from what Player II knows how to do.

Prerequisites. We assume a first course in metric spaces and the basic vocabulary of point-set topology: open and closed sets, closure and interior, dense sets, continuity, and Cauchy sequences and completeness. We also use the following notions:

(1) A topological space X is *metrizable* if there is a metric on X whose open sets are exactly the open sets of X; such a metric is called *compatible*. The space is *completely metrizable* if it has a compatible metric that is complete. Completeness is a property of the metric, whereas complete metrizability is a property of the topology: the set of irrational numbers is not complete in the usual metric of $\mathbb{R}$, yet it carries a complete compatible metric (see Section 5).
(2) A G_δ *set* is a countable intersection of open sets.
(3) Every metric space (X, d) embeds isometrically as a dense subspace of a complete metric space, its *completion*, which is unique up to an isometry that fixes X; for instance, the completion of $\mathbb{Q}$ is $\mathbb{R}$.
(4) *Zorn's lemma*: if every chain in a non-empty partially ordered set has an upper bound, then the set has a maximal element. We use it to find maximal families of pairwise disjoint open sets.

1. Baire category

Baire category measures the size of a set by topology alone: a set is small when it is a countable union of sets that are thin everywhere. In this section we recall the notions of category, and we show that a space is a Baire space exactly when none of its non-empty open subsets is small in this sense, that is, of the first category (see Theorem 1.3 below).

Definition 1.1. Let X be a topological space, and let $E \subseteq X$ be given. We denote the closure of E by $\overline{E}$, and its interior by E°.

(1) The set E is *nowhere dense* in X if $(\overline{E})^\circ = \emptyset$, that is, if its closure has empty interior.
(2) The set E is of the *first category* in X (or *meager*) if it is the union of a countable family of nowhere dense sets. A set that is not of the first category is of the *second category* (or *non-meager*).
(3) The set E is *comeager* (or *residual*) in X if its complement $X \setminus E$ is of the first category.
(4) The space X is called a *Baire space* if the intersection of every countable family of dense open subsets of X is dense in X.
(5) The set E has the *Baire property* if there is an open set O such that the symmetric difference $E \triangle O$ is of the first category; that is, if E is an open set up to an error of the first category.

Remark 1.2. The notions in Definition 1.1 classify sets by their topological "size"; they form a qualitative analogue of measure theory.

(1) A nowhere dense set is thin everywhere. Equivalently (see Exercise 1 at the end of this section), a set E is nowhere dense exactly when every non-empty open set contains a non-empty open set disjoint from E. Every finite subset of $\mathbb{R}$ is nowhere dense, and so is the Cantor ternary set (see Figure 3).
(2) A set of the first category is topologically negligible, and a countable union of such sets is again of the first category, just as a countable union of Lebesgue-null sets is null. The rationals $\mathbb{Q} = \bigcup_{q\in\mathbb{Q}}\{q\}$ form a set of the first category in $\mathbb{R}$. The analogy has limits: the line $\mathbb{R}$ is the union of a set of the first category and a set of Lebesgue measure zero [Oxt80, Chapter 1], so a set that is small in one sense may be large in the other.
(3) A comeager set contains "most" points of the space. A property is said to hold of the *generic* point of a topological space X if the set of points that have it is comeager. In $\mathbb{R}$, the set of irrational numbers is comeager, so the generic real number is irrational.
(4) In Theorem 1.3 below we show that a space is a Baire space exactly when none of its non-empty open subsets is negligible. This *local* condition is stronger than asking that the space itself be of the second category in itself. Consider the disjoint union $X = (\mathbb{Q}\times\{0\})\cup(\mathbb{R}\times\{1\}) \subseteq \mathbb{R}^2$ (see Figure 4). Its open subset $\mathbb{R}\times\{1\}$, a copy of $\mathbb{R}$, is of the second category in itself by the Baire category theorem (see Corollary 4.5). Since a subset of an open set $U \subseteq X$ that is nowhere dense in X is also nowhere dense in U, the set $\mathbb{R}\times\{1\}$ is of the second category in X, and therefore so is X; yet the space X is not a Baire space, since its open subset $\mathbb{Q}\times\{0\}$ is of the first category.
(5) Every open set O has the Baire property, because $O \mathbin{\triangle} O = \emptyset$. Every closed set F has it as well: the set $F \mathbin{\triangle} F^\circ = F \setminus F^\circ$ is closed and has empty interior, hence it is nowhere dense. The sets with the Baire property form a σ-algebra (see [Oxt80]), and therefore every Borel set has the Baire property.



FIGURE 3. The first stages of the Cantor ternary set $C = \bigcap_{n=0}^{\infty} C_n$: starting from $C_0 = [0,1]$, the set C_{n+1} is obtained from C_n by removing the open middle third of each of its intervals. The set C is closed and uncountable, yet every open interval contains a smaller open interval that misses C; it is therefore nowhere dense.



FIGURE 4. The space $X = (\mathbb{Q}\times\{0\})\cup(\mathbb{R}\times\{1\})$ of Remark 1.2(4). Each line is an open subset of X. The upper line makes X of the second category in itself; the lower line, a countable union of points, is a non-empty open set of the first category, which keeps X from being a Baire space.

Theorem 1.3. *A topological space X is a Baire space if and only if no non-empty open subset of X is of the first category.*

Proof. The idea is to pass to complements: a closed set is nowhere dense exactly when its complement is a dense open set.

Suppose first that X is a Baire space, and let $G \subseteq X$ be open and non-empty. If $G = \bigcup_{n=1}^{\infty} E_n$ with each E_n nowhere dense, then each set $V_n = X \setminus \overline{E_n}$ is open and dense in X. Since X is a Baire space, the intersection $\bigcap_{n=1}^{\infty} V_n$ is dense in X, and it therefore meets the non-empty open set G. However, we have

$$G \cap \bigcap_{n=1}^{\infty} V_n = G \setminus \bigcup_{n=1}^{\infty} \overline{E_n} = \emptyset,$$

a contradiction, which shows that G is of the second category.

Conversely, suppose that no non-empty open subset of X is of the first category. Let $(V_n)_{n\geq 1}$ be a sequence of dense open subsets of X, and let $O \subseteq X$ be open and non-empty; we show that O meets $\bigcap_{n=1}^{\infty} V_n$. For each n, the set $E_n = X \setminus V_n$ is closed, and it has empty interior because V_n is dense; thus each E_n is nowhere dense. If O missed $\bigcap_{n=1}^{\infty} V_n$, it would be contained in $\bigcup_{n=1}^{\infty} E_n$, and would therefore be a non-empty open set of the first category, contrary to our hypothesis. Hence $O \cap \bigcap_{n=1}^{\infty} V_n \neq \emptyset$. Since O was arbitrary, the intersection $\bigcap_{n=1}^{\infty} V_n$ is dense, which proves that X is a Baire space. ☺

Exercises.

(1) Let X be a topological space, and let $E \subseteq X$ be given. Show that E is nowhere dense in the sense of Definition 1.1(1) if and only if every non-empty open subset of X contains a non-empty open set disjoint from E (see Remark 1.2(1)), and that if E is nowhere dense, then so is its closure.

(2) Show that the union of two nowhere dense subsets of a topological space is nowhere dense. Show by an example in $\mathbb{R}$ that the union of countably many nowhere dense sets need not be nowhere dense.

(3) Let X be a topological space. Show that X is a Baire space if and only if every comeager subset of X is dense in X. Deduce that in a non-empty Baire space no comeager set is of the first category.

(4) Let X be a Baire space, and let $U \subseteq X$ be a non-empty open set. Show that U, with the subspace topology, is a Baire space. (Hint: show that a subset of U that is nowhere dense in U is nowhere dense in X, and use Theorem 1.3.)

(5) $(*)$ Let X be a Baire space, and let $D \subseteq X$ be a dense G_δ set. Show that D, with the subspace topology, is a Baire space. (Hint: a dense open subset of D has the form $W \cap D$ with W open and dense in X.)

2. Infinite games

In this section we fix the formal framework for all the games of these notes, from the Banach–Mazur game to the Ehrenfeucht–Fraïssé game: trees of positions, plays, payoff sets, and strategies.

A definition of a game must settle three matters. The first is the state of play: what is known when a move is made, namely the finite sequence of moves so far, and which moves are allowed at that point. The second is the outcome: an infinite play is won by Player II exactly when it belongs to a given set. The third is a method of play: a rule that assigns a

move to every position at which a player is to move. Trees of positions settle the first matter, payoff sets of plays the second, and strategies the third.

Definition 2.1. Let A be a non-empty set. As usual, we write $\omega = \{0, 1, 2, \dots\}$ for the set of natural numbers. We denote by $A^{<\omega}$ the collection of all finite sequences $s = (s_0, \dots, s_{n-1})$ of elements of A, with $n \in \omega$, and by A^ω the set of all infinite sequences $(a_n)_{n\in\omega}$ of elements of A. The length n of s is denoted by $|s|$; the empty sequence $\langle\rangle$ has length 0. If $s, t \in A^{<\omega}$, $|t| \le |s|$, and $t_i = s_i$ for $0 \le i < |t|$, we write $t \subseteq s$ and say that s *extends* t; when sequences are regarded as functions defined on initial segments of ω, the relation is literally inclusion. For $s \in A^{<\omega}$ and $a \in A$, we write $s^\frown a = (s_0, \dots, s_{|s|-1}, a)$ for the extension of s by a.

A non-empty subset $T \subseteq A^{<\omega}$ is called a *tree* if $t \in T$ whenever $s \in T$ and $t \subseteq s$. For each $s \in T$, we write

$$\mathrm{Succ}(s) = \{a \in A : s^\frown a \in T\}.$$

We assume throughout that $\mathrm{Succ}(s) \neq \emptyset$ for every $s \in T$. An infinite sequence $x = (a_n)_{n=0}^\infty \in A^\omega$ is a *play* in T if $(a_0, \dots, a_{n-1}) \in T$ for all $n \ge 1$. The set of all such plays is denoted by $[T]$.

Definition 2.2. An *infinite game* is a pair (T, W), where $T \subseteq A^{<\omega}$ is a tree on a non-empty set A and $W \subseteq [T]$ is a set of plays, the *payoff set*. Two players, I and II, alternately choose elements of A: Player I chooses a_{2n}, and Player II chooses a_{2n+1}, for $n \in \omega$, subject to the rule that $(a_0, \dots, a_k) \in T$ for every $k \in \omega$. Player II *wins* the resulting play $x = (a_n)_{n\in\omega} \in [T]$ if $x \in W$; otherwise, Player I wins. The elements of T are called *positions*.

A *strategy* for Player I is a function σ, defined on the positions of even length, such that $\sigma(s) \in \mathrm{Succ}(s)$ for every such position s. A play or position is *consistent* with σ if $a_{2n} = \sigma(a_0, \dots, a_{2n-1})$ for every $n \in \omega$ for which a_{2n} is defined. We say that σ is a *winning strategy* for Player I if every play consistent with σ lies in $[T] \setminus W$.

A strategy τ for Player II and the notion of a winning strategy for Player II are defined analogously.

Remarks 2.3.

(1) In these notes, the payoff set W is the set of plays won by Player II. Many texts, among them Kechris's book [Kec95], use the opposite convention, in which the payoff set is Player I's. Ours suits the games below, in which Player II is the player who tries to build something: a point common to all the moves, or an intersection that lies inside a given set. As in the Introduction, we call Player I "he" and Player II "she".

(2) The two players need not choose from the same kind of object: if Player I chooses from a set A_{I} and Player II from a set A_{II}, we take $A = A_{\mathrm{I}} \cup A_{\mathrm{II}}$ and let the tree T record which moves are legal.

(3) At most one player has a winning strategy: if σ and τ were winning strategies for Players I and II, respectively, the unique play consistent with both (see Exercise 2 at the end of this section) would lie both outside and inside W.

(4) A game is called *determined* if one of the two players has a winning strategy. Not every game is determined. There are Baire spaces X in which neither player has a winning strategy in the Banach–Mazur game $\mathrm{BM}(X)$ of Section 3 (see the remark after Theorem 3.3), and the axiom of choice produces undetermined games in which the players choose finite binary strings (see Section 14). On the other side stands Martin's theorem (see Section 16): every game in which the players choose natural numbers, and whose payoff set is a Borel subset of the space ω^ω (the product of countably many copies of the discrete space ω), is determined.

(5) A strategy sees the whole position, not only the last move: the move that a strategy σ for Player I prescribes at a position $(a_0, \dots, a_{2n-1})$, with $n \in \omega$, may depend on everything played so far (see Figure 5). This memory can be indispensable (see Exercise 5 at the end of this section).

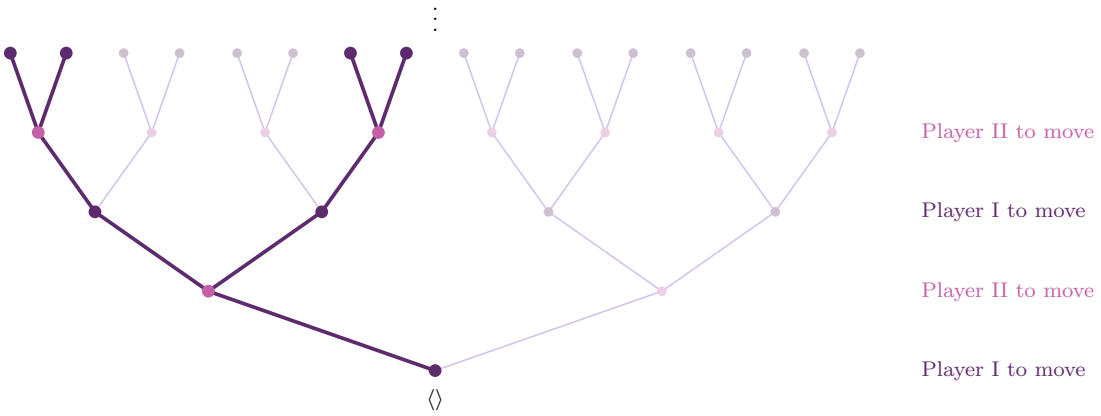


FIGURE 5. A game tree, drawn with two moves available at each position; the root, at the bottom, is the empty position $\langle\rangle$. The levels alternate between Player I (plum) and Player II (pink). A strategy σ for Player I picks one move at each position at which he is to move. The positions consistent with σ (dark) form a subtree that branches only where Player II is to move, and its infinite branches are the plays consistent with σ. The strategy is winning if all of these plays lie outside the payoff set W of Player II.

Exercises.

(1) Let $A = \{0, 1\}$ and $T = A^{<\omega}$, and write each play as $(a_n)_{n\in\omega}$. For each of the following payoff sets $W \subseteq [T]$, decide which player has a winning strategy in (T, W) and describe one: (a) the plays containing at least one 1; (b) the plays with $a_n = 0$ for every $n \in \omega$; (c) the plays with $a_1 = a_0$; (d) the plays in which Player I plays 1 infinitely often.

(2) Let (T, W) be an infinite game, and let σ and τ be strategies for Player I and Player II, respectively. Show that there is exactly one play consistent with both σ and τ, a fact used in Remarks 2.3(3).

(3) Let T be a tree, and let $W \subseteq W' \subseteq [T]$ be given. Show that a winning strategy for Player II in (T, W) is winning in (T, W'), and that a winning strategy for Player I in (T, W') is winning in (T, W).

(4) Let (T, W) be an infinite game, and suppose that there is a natural number N such that whether a play lies in W depends only on its first N moves. Show that the game is determined. (Hint: decide who wins from each position of length N, and work backwards to $\langle\rangle$.)

(5) $(*)$ Let $A = \{0, 1\}$ and $T = A^{<\omega}$, and let W be the set of plays $(a_n)_{n\in\omega}$ with $a_{2n+1} = a_0$ for every $n \in \omega$. Show that Player II has a winning strategy, but that no strategy for her whose move depends only on the last move of Player I is winning.

3. The Banach–Mazur game

In the Banach–Mazur game, two players take turns shrinking an open set, and the second player wants the sets to keep a point in common. In this section we define the game and show that a non-empty space is a Baire space exactly when the first player has no winning strategy (see Theorem 3.3 below). We then let the players aim at a given set, and prove the theorem of Banach and Oxtoby: the second player can force the intersection of the moves to lie inside the set exactly when the complement of the set is of the first category (see Theorem 3.5 below).

Definition 3.1. Let X be a non-empty topological space, and let $\mathcal{O}^+(X)$ be the collection of all non-empty open subsets of X. The *Banach–Mazur game* on X, denoted by $\mathrm{BM}(X)$, is the game (T, W) where:

(1) The set of moves is $A = \mathcal{O}^+(X)$.
(2) The tree T consists of all finite sequences $(U_0, V_0, U_1, V_1, \dots)$ of sets in $\mathcal{O}^+(X)$ with $U_0 \supseteq V_0 \supseteq U_1 \supseteq V_1 \supseteq \cdots$.
(3) The payoff set is $W = \{(U_0, V_0, U_1, \dots) \in [T] : \bigcap_{n=0}^{\infty} U_n \neq \emptyset\}$.

Thus Player I plays the sets U_n, Player II plays the sets V_n, and Player II wins if the sets played have a point in common.

Example 3.2. (1) In $\mathrm{BM}(\mathbb{Q})$, Player I has a winning strategy. Fix an enumeration $q_0, q_1, \dots$ of $\mathbb{Q}$, and let Player I play $U_0 = \mathbb{Q} \setminus \{q_0\}$ and $U_n = V_{n-1} \setminus \{q_n\}$ for $n \geq 1$. These sets are open because the points of $\mathbb{Q}$ are closed, and non-empty because $\mathbb{Q}$ has no isolated points; since every q_n is missing from U_n, the intersection $\bigcap_n U_n$ is empty (see Figure 6).

(2) In $\mathrm{BM}(\mathbb{R})$, Player II has a winning strategy: in answer to U_n, she plays a bounded open interval V_n with $\overline{V_n} \subseteq U_n$. The compact sets $\overline{V_n}$ are nested and non-empty, hence they have a common point, and that point lies in every U_n because $\overline{V_n} \subseteq U_n$.

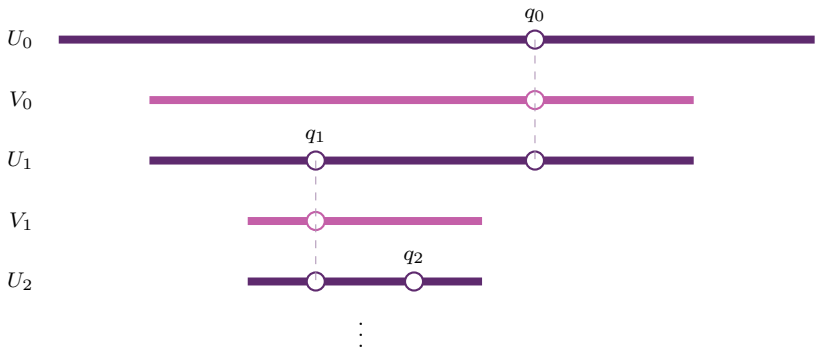


Figure 6. Player I wins $\mathrm{BM}(\mathbb{Q})$ (see Example 3.2(1)). The moves of Player I are drawn in plum and those of Player II in pink, as intervals of rationals. At each step $n \in \omega$, Player I removes the rational q_n, where $q_0, q_1, \dots$ is the enumeration of $\mathbb{Q}$ fixed in the example; each removal leaves a hole, which persists in every later set. Every rational is removed at some step, so the sets have no point in common.

Theorem 3.3. *Let X be a non-empty topological space.*

(1) *Player I has a winning strategy in* $\mathrm{BM}(X)$ *if and only if X contains a non-empty open set of the first category.*
(2) *The space X is a Baire space if and only if Player I has no winning strategy in* $\mathrm{BM}(X)$.

Proof. By Theorem 1.3, part (2) is an immediate consequence of part (1), which we now prove. The idea is that a nowhere dense set can be avoided in a single move. If a non-empty open set is a countable union of nowhere dense sets, Player I opens with it and avoids one of the pieces at each later move; conversely, the plays of a winning strategy for Player I, laid side by side, show that his first move is of the first category.

Suppose first that $G \subseteq X$ is open, non-empty, and of the first category. Then there are nowhere dense sets $E_0, E_1, \ldots$ with $G = \bigcup_{n=0}^{\infty} E_n$. Player I plays $U_0 = G$. If Player II responds with $V_0 \subseteq U_0$, the set $V_0 \setminus \overline{E_0}$ is open, and it is non-empty because $\overline{E_0}$ has empty interior. Player I may therefore choose a non-empty open set $U_1 \subseteq V_0 \setminus \overline{E_0}$. In general, for $n \geq 1$, given V_{n-1}, Player I chooses an open set

$$U_n \subseteq V_{n-1} \setminus \overline{E_{n-1}}, \qquad U_n \neq \emptyset.$$

This defines a strategy σ for Player I. Every play consistent with σ satisfies $U_n \cap E_{n-1} = \emptyset$ for all $n \geq 1$, whence

$$\bigcap_{n=0}^{\infty} U_n \subseteq G \setminus \bigcup_{n=0}^{\infty} E_n = \emptyset.$$

Thus the strategy σ is winning for Player I.

Conversely, suppose that Player I has a winning strategy σ, and put $U_0 = \sigma(\langle\rangle)$. We show that the non-empty open set U_0 is of the first category. For a position p of odd length $2n+1$ consistent with σ, we shall write U^p for the last entry of p, the set U_n played by Player I.

We build, for each $n \geq 0$, a family $\mathcal{P}_n$ of positions of length $2n+1$ consistent with σ, such that:

(a) the sets U^p, $p \in \mathcal{P}_n$, are pairwise disjoint;
(b) their union $D_n = \bigcup_{p \in \mathcal{P}_n} U^p$ is dense in U_0;
(c) every $p \in \mathcal{P}_{n+1}$ extends some $q \in \mathcal{P}_n$.

Let $\mathcal{P}_0 = \{(U_0)\}$. Suppose that $\mathcal{P}_n$ has been built, and let $q \in \mathcal{P}_n$ be given. Each non-empty open $V \subseteq U^q$ is a legal move for Player II at q, and leads to the position $q^\frown V^\frown \sigma(q^\frown V)$ of length $2n+3$. By Zorn's lemma (the union of a chain of such families keeps the property), we may choose a family $\mathcal{P}(q)$ of such positions, maximal with respect to the property that their last entries are pairwise disjoint. The union of these last entries is dense in U^q: otherwise some non-empty open $O \subseteq U^q$ would miss all of them, and the position $q^\frown O^\frown \sigma(q^\frown O)$, whose last entry lies in O, could be added to $\mathcal{P}(q)$, contradicting maximality. Let $\mathcal{P}_{n+1}$ be the union $\bigcup_{q \in \mathcal{P}_n} \mathcal{P}(q)$. Property (c) holds by construction. Property (a) holds because last entries of positions in the same $\mathcal{P}(q)$ are disjoint, and those in $\mathcal{P}(q)$ and $\mathcal{P}(q')$ with $q \neq q'$ lie in the disjoint sets U^q and $U^{q'}$. For (b), we argue by induction on n; the case $n = 0$ is trivial, since $D_0 = U_0$. By the inductive hypothesis, every non-empty open set $O \subseteq U_0$ meets D_n, hence meets U^q for some $q \in \mathcal{P}_n$. The union of the last entries of the positions in $\mathcal{P}(q)$ is dense in U^q, and it therefore meets the non-empty open set $O \cap U^q$; as this union is contained in D_{n+1}, the set O meets D_{n+1}. The first two levels are shown in Figure 7.

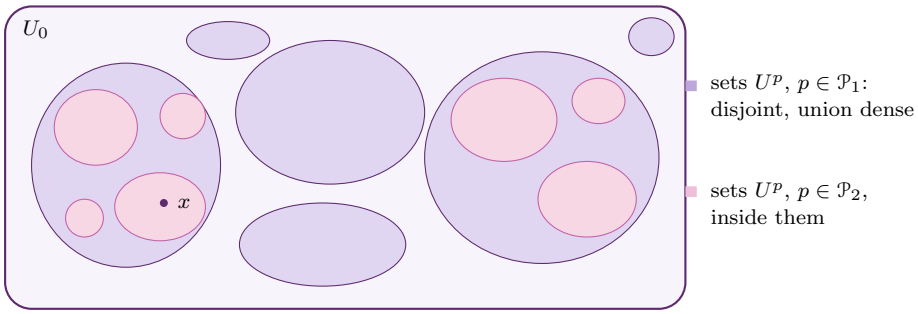


FIGURE 7. The proof of Theorem 3.3: the first two levels of the construction, for a winning strategy σ of Player I with opening move U_0. Inside U_0, the sets U^p, the last entries of the positions p of the first family $\mathcal{P}_1$ and thus moves prescribed by σ, form a maximal family of pairwise disjoint open sets, whose union D_1 is dense in U_0; inside each set U^q with $q \in \mathcal{P}_1$, the last entries of the positions of $\mathcal{P}_2$ that extend q form such a family again. A point x lying in every D_n, $n \in \omega$, picks out exactly one set at each level, and these sets are the moves of Player I in a single play consistent with σ.

Each D_n is open and dense in U_0; hence the set $U_0 \setminus D_n$ is nowhere dense in X. Suppose, toward a contradiction, that U_0 is of the second category. Then the first-category set $\bigcup_n (U_0 \setminus D_n)$ does not contain U_0, and there is a point $x \in \bigcap_n D_n$. By (a), for each n there is exactly one $p_n \in \mathcal{P}_n$ with $x \in U^{p_n}$. Property (c) provides a position $q \in \mathcal{P}_n$ extended by p_{n+1}; since the moves along a play are nested, we have $x \in U^{p_{n+1}} \subseteq U^q$, and property (a) forces $q = p_n$. Thus the positions $p_0 \subseteq p_1 \subseteq \cdots$ fit together into a single play consistent with σ, and the point x lies in every set U^{p_n} that Player I plays in it. This contradicts the assumption that σ is winning. Hence the opening move U_0 is a non-empty open set of the first category, and the proof is complete.

Remark 3.4. When a non-empty topological space X is a Baire space, part (2) of Theorem 3.3 rules out a winning strategy for Player I in $\mathrm{BM}(X)$, but Player II need not have one either: there are Baire spaces in which neither player has a winning strategy (see [Tel87]). This happens already for subspaces of the real line: if $B \subseteq \mathbb{R}$ meets every uncountable closed subset of $\mathbb{R}$, and so does its complement (Bernstein constructed such sets in 1908), then the subspace B is a Baire space on which neither player of $\mathrm{BM}(B)$ has a winning strategy.

In Mazur's original game, and in most applications in Part II, the players aim at a given set. Let X be a non-empty topological space, and let $S \subseteq X$ be given. The game $\mathrm{BM}(X, S)$ has the same moves as $\mathrm{BM}(X)$, but Player II wins the play $(U_0, V_0, U_1, \dots)$ if

$$\bigcap_{n=0}^{\infty} U_n \subseteq S,$$

that is, if no point outside S survives. Since the moves are nested, the moves of the two players have the same intersection, $\bigcap_n U_n = \bigcap_n V_n$. Unlike in $\mathrm{BM}(X)$, Player II also wins when the intersection is empty; Player I wins only if he produces a point outside S.

Theorem 3.5 (Banach; Oxtoby). *Let X be a non-empty topological space, and let $S \subseteq X$ be given.*

(1) *Player II has a winning strategy in* $\mathrm{BM}(X, S)$ *if and only if $X \setminus S$ is of the first category in X.*

(2) *If X is a complete metric space, then Player I has a winning strategy in* $\mathrm{BM}(X, S)$ *if and only if there is a non-empty open set U such that $U \cap S$ is of the first category in X.*

Proof. Both parts rest on the two ideas of the proof of Theorem 3.3: a nowhere dense set can be avoided in a single move, and a winning strategy, unfolded into maximal families of positions, exhibits a set of the first category. (1) Suppose first that $X \setminus S = \bigcup_n E_n$ with each E_n nowhere dense. In answer to U_n, Player II plays a non-empty open set $V_n \subseteq U_n \setminus \overline{E_n}$. A point of $\bigcap_n V_n$ lies outside every E_n, hence in S, so this strategy is winning.

Conversely, let τ be a winning strategy for Player II. We repeat the construction in the proof of Theorem 3.3, now with the moves of Player II. For a position p of non-zero even length consistent with τ, we write V^p for its last entry. As in that proof, Zorn's lemma provides families $\mathcal{P}_n$, for $n \in \omega$, of positions of length $2n + 2$ consistent with τ, such that the sets V^p, $p \in \mathcal{P}_n$, are pairwise disjoint, their union D_n is dense in X, and every $p \in \mathcal{P}_{n+1}$ extends some $q \in \mathcal{P}_n$. For $\mathcal{P}_0$ we take a maximal family of positions $(U, \tau(U))$ whose last entries are pairwise disjoint; the union of these last entries is dense, because for every non-empty open set O the answer $\tau(O)$ lies in O. At a position $q \in \mathcal{P}_n$, every non-empty open $U \subseteq V^q$ is a legal move for Player I, and the answer of τ lies in U, so the maximality argument goes through unchanged. Now let $x \in \bigcap_n D_n$ be given. Exactly as before, the positions $p_n \in \mathcal{P}_n$ with $x \in V^{p_n}$ fit together into a single play consistent with τ, and every set played in it contains x. Since τ is winning, the point x lies in S. Thus the intersection $\bigcap_n D_n$ is contained in S, whence $X \setminus S \subseteq \bigcup_n (X \setminus D_n)$. Each $X \setminus D_n$ is closed with empty interior, hence nowhere dense, and therefore the complement $X \setminus S$ is of the first category.

(2) Let d be the metric of X, which is complete by hypothesis. Suppose first that U is a non-empty open set and that $U \cap S = \bigcup_n E_n$ with each E_n nowhere dense. Player I plays $U_0 = U$ and, in answer to V_{n-1} for each $n \geq 1$, a non-empty open set U_n with $\overline{U_n} \subseteq V_{n-1} \setminus \overline{E_{n-1}}$ and with diameter $\operatorname{diam} U_n = \sup\{d(x, y) : x, y \in U_n\}$ less than 2^{-n}. The closed sets $\overline{U_n}$ are nested and their diameters tend to 0, so by completeness they have a point x in common. The point x lies in every U_n, because $x \in \overline{U_{n+1}} \subseteq V_n \subseteq U_n$; in particular, it lies in $U = U_0$. It lies in no E_n, because $\overline{U_{n+1}}$ misses $\overline{E_n}$. Hence Player I has produced a point of $U \setminus S$, namely x, and he has won.

Conversely, let σ be a winning strategy for Player I, and put $U_0 = \sigma(\langle\rangle)$. We show that $U_0 \cap S$ is of the first category. Once more we follow the construction in the proof of Theorem 3.3, with one change: at level n, the families $\mathcal{P}(q)$ use only moves V of Player II with $\operatorname{diam} V < 2^{-n}$. Every non-empty open set contains such a set V (a small open ball), and the maximality argument shows, as before, that the unions D_n are dense in U_0. Every point $x \in \bigcap_n D_n$ again picks out a play consistent with σ whose intersection contains x. The diameters of the moves tend to 0, so this intersection is exactly $\{x\}$. Since σ is winning, the intersection contains a point outside S, and that point can only be x. Thus the intersection $\bigcap_n D_n$ lies in $U_0 \setminus S$, and the set $U_0 \cap S$ is contained in $\bigcup_n (U_0 \setminus D_n)$, which is of the first category because each $U_0 \setminus D_n$ is nowhere dense. Hence Player I's first move U_0 is the required non-empty open set. ☺

Remark 3.6. By part (1) of Theorem 3.5, Player II has a winning strategy in $\mathrm{BM}(\mathbb{R}, S)$ when S is the set of irrational numbers: she removes the rationals one at a time, just as Player I did in Example 3.2(1). Part (1) needs no hypothesis on the space. In part (2), the "only if" direction uses the metric but not its completeness, which is needed only in the "if" direction, where Player I must produce a point. Without completeness the "if" direction fails: in $X = \mathbb{Q}$, with the usual metric, and with $S = \emptyset$, the set $U \cap S$ is empty for every non-empty open set U, yet Player II has a winning strategy by part (1), because $\mathbb{Q}$ is of the first category in itself.

Exercises.

(1) Let C be the Cantor ternary set (see Figure 3). For each of the sets S below, decide which player has a winning strategy in $\mathrm{BM}(\mathbb{R}, S)$, and describe one explicitly: (a) the irrational numbers; (b) the rational numbers; (c) the set C; (d) the set $\mathbb{R} \setminus C$. Check your answers against Theorem 3.5.

(2) Let X be a non-empty compact Hausdorff space. Show that Player II has a winning strategy in $\mathrm{BM}(X)$. (Hint: such a space is regular, so every point of a non-empty open set U has an open neighborhood whose closure lies in U.)

(3) Let X be a non-empty topological space, and let $S_0, S_1, \dots \subseteq X$ be given. Suppose that Player II has a winning strategy τ_k in $\mathrm{BM}(X, S_k)$ for every $k \in \omega$. Show directly that she has a winning strategy in $\mathrm{BM}(X, \bigcap_k S_k)$. (Hint: split the set ω of rounds into pairwise disjoint infinite sets R_k, for $k \in \omega$, and let her follow τ_k in the rounds that belong to R_k.) Then give a second proof using Theorem 3.5.

(4) Let X be a Baire space, and let $U \subseteq X$ be a non-empty open set. Use Theorem 3.3 to show that U is a Baire space, by turning a winning strategy for Player I in $\mathrm{BM}(U)$ into one in $\mathrm{BM}(X)$.

(5) $(*)$ Let X be a non-empty complete metric space, and let $S \subseteq X$ have the Baire property. Show that $\mathrm{BM}(X, S)$ is determined. (Hint: if $U \cap S$ is of the first category for no non-empty open set U, show that every open set O with $S \triangle O$ of the first category is dense.)

4. The strong Choquet game

The Banach–Mazur game detects category, but not completeness: a metrizable space for which Player II has a winning strategy in the Banach–Mazur game need not be completely metrizable (see Remark 4.6 below). In this section we let Player I name a point with each of his moves, and we show that the resulting game characterizes complete metrizability among metrizable spaces (see Theorem 4.4 below). Here we prove the easy half of the theorem and derive from it the Baire category theorem (see Corollary 4.5 below); the hard half is proved in Section 5. At the end of the section we leave metric spaces and turn to Čech-completeness, a form of completeness that makes sense in every Tychonoff space: in Subsection 4.1 we show that Player II has a winning strategy in the new game on every non-empty Čech-complete space, and that the converse fails.

Definition 4.1. Let X be a non-empty topological space, and recall that $\mathcal{O}^+(X)$ is the collection of all non-empty open subsets of X. The *strong Choquet game* on X, denoted by $\mathrm{Ch}(X)$, is played as follows. Player I chooses a pair (x_0, U_0) with $U_0 \in \mathcal{O}^+(X)$ and $x_0 \in U_0$. Player II responds with $V_0 \in \mathcal{O}^+(X)$ such that $x_0 \in V_0 \subseteq U_0$. At step $n \geq 1$, Player I chooses (x_n, U_n) such that $x_n \in U_n \subseteq V_{n-1}$, and Player II responds with $V_n \in \mathcal{O}^+(X)$ such that $x_n \in V_n \subseteq U_n$. Player II wins the play if $\bigcap_{n=0}^{\infty} U_n \neq \emptyset$.

In the format of Definition 2.2, Player I chooses from $X \times \mathcal{O}^+(X)$ and Player II from $\mathcal{O}^+(X)$. If the points x_n are omitted, what remains is exactly the Banach–Mazur game $\mathrm{BM}(X)$, which in this context is also called the *Choquet game*. Choquet called the two players β (the first) and α (the second). Following him, we call a non-empty topological space X *α-favorable* if Player II has a winning strategy in $\mathrm{BM}(X)$, and *strongly α-favorable* if she has a winning strategy in $\mathrm{Ch}(X)$; Kechris [Kec95] calls these spaces *Choquet spaces* and *strong Choquet spaces*, respectively.

Lemma 4.2. *Every strongly α-favorable space is α-favorable.*

Proof. The idea is that Player II can supply the missing points herself. Let X be a strongly α-favorable space, and let τ be a winning strategy for Player II in $\mathrm{Ch}(X)$. In $\mathrm{BM}(X)$, when Player I plays U_n, Player II picks a point $x_n \in U_n$ and answers with the set that τ assigns to the position $(x_0, U_0), V_0, \ldots, V_{n-1}, (x_n, U_n)$ of $\mathrm{Ch}(X)$. Since $x_n \in U_n \subseteq V_{n-1}$, the moves (x_n, U_n) are legal in $\mathrm{Ch}(X)$. Every play of $\mathrm{BM}(X)$ that follows this rule therefore arises, by omitting the points, from a play of $\mathrm{Ch}(X)$ consistent with τ, and the sets U_n have a common point because τ is winning. Hence this rule is a winning strategy for Player II in $\mathrm{BM}(X)$. ☺

Example 4.3. Player I wins $\mathrm{Ch}(\mathbb{Q})$: he follows the strategy of Example 3.2(1), adding any point of each set he plays.

Theorem 4.4 (Choquet). *A non-empty metrizable space is completely metrizable if and only if it is strongly α-favorable.*

Proof. Let X be a metrizable space, and suppose first that X is completely metrizable, with a compatible complete metric d. We define a strategy τ for Player II whose answers are balls with radii tending to 0; the points of Player I then form a Cauchy sequence, and its limit is a common point of the play. When Player I plays (x_n, U_n), Player II chooses r_n with $0 < r_n < 2^{-n}$ such that the closed ball $B[x_n, r_n] = \{y \in X : d(x_n, y) \le r_n\}$ is contained in U_n, and plays the open ball $V_n = B(x_n, r_n) = \{y \in X : d(x_n, y) < r_n\}$.

Let $(x_0, U_0), V_0, (x_1, U_1), V_1, \ldots$ be a play consistent with τ. For $m > n$ we have $x_m \in U_m \subseteq V_n = B(x_n, r_n)$, whence $d(x_m, x_n) < 2^{-n}$. Thus the points x_n form a Cauchy sequence, which converges to some $x \in X$ because d is complete. For each n, the points x_m with $m \ge n$ all lie in the closed ball $B[x_n, r_n]$; so does their limit, and that ball is contained in U_n. Consequently, the limit x lies in every U_n, and the strategy τ is winning for Player II (see Figure 8).

The converse, that a strongly α-favorable metrizable space is completely metrizable, is proved in Section 5. ☺

Corollary 4.5 (Baire category theorem). *Every completely metrizable space is a Baire space.*

Proof. Let X be completely metrizable. The empty space is trivially a Baire space, so we may assume that X is non-empty. By Theorem 4.4 and Lemma 4.2, Player II has a winning strategy in $\mathrm{BM}(X)$. Since at most one player has a winning strategy (see Remarks 2.3(3)), Player I has none, and by Theorem 3.3 the space X is a Baire space. ☺

Remark 4.6. The points x_n named by Player I cannot be dropped from the strong Choquet game in Theorem 4.4: an α-favorable metrizable space need not be completely metrizable. Let $X = \{(a, b) \in \mathbb{R}^2 : b > 0\} \cup (\mathbb{Q} \times \{0\})$, with the topology inherited from $\mathbb{R}^2$. The upper half plane $H = \{(a, b) \in \mathbb{R}^2 : b > 0\}$ is open and dense in X, and Player II wins $\mathrm{BM}(X)$

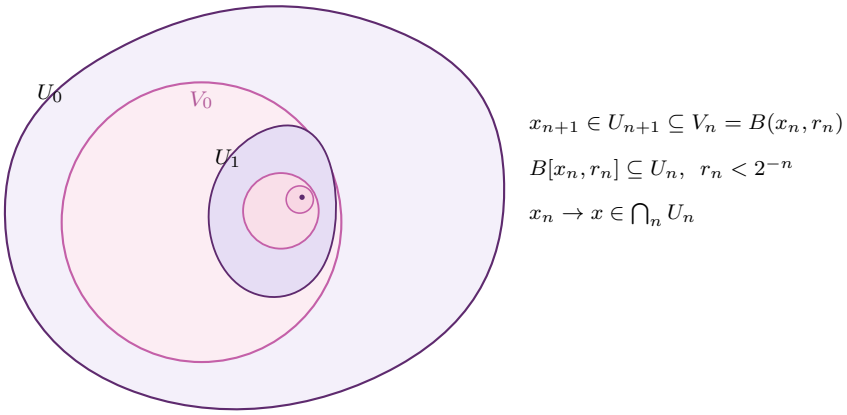


FIGURE 8. The easy half of Theorem 4.4. Player II answers each point x_n of Player I with the open ball $B(x_n, r_n)$, where $r_n < 2^{-n}$ and the closed ball $B[x_n, r_n]$ lies inside the set U_n just played. The points x_n form a Cauchy sequence, and its limit lies in every closed ball $B[x_n, r_n]$, hence in every U_n.

by always answering with an open disk whose closure lies in $U_n \cap H$ and whose radius is less than 2^{-n}: these closed disks are nested and compact, and their common point lies in every U_n. The space X, however, is not completely metrizable. Indeed, a closed subspace of a completely metrizable space is completely metrizable (a Cauchy sequence in the subspace converges in the space, and its limit lies in the subspace), and is therefore a Baire space by Corollary 4.5; yet the closed subspace $\mathbb{Q} \times \{0\}$ of X, a copy of $\mathbb{Q}$ and hence a countable union of nowhere dense singletons, is not a Baire space. By the hard half of Theorem 4.4 (proved in Section 5), it follows that Player II has no winning strategy in $\mathrm{Ch}(X)$. A direct argument, independent of Section 5, leads to the same conclusion: in $\mathrm{Ch}(X)$ Player I can choose his points on the horizontal axis, and Player II must answer with sets that contain them; the strategy by which Player I then wins is shown in Figure 9.

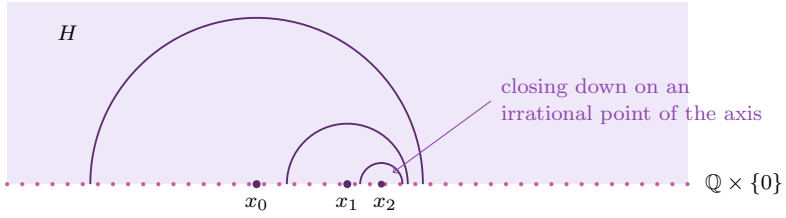


FIGURE 9. Player I wins $\mathrm{Ch}(X)$ for the space X of Remark 4.6. He plays rational points of the horizontal axis, with small balls of X about them, inside the previous answer of Player II, and avoids one more rational each time. The balls close down on a point of the axis that is not rational, hence not in X.

4.1. **Čech-complete spaces.** Because Player I names points, the strong Choquet game detects complete metrizability among metrizable spaces, which the Banach–Mazur game cannot do (see Remark 4.6). Completeness, however, is a metric notion; it has a purely topological counterpart, and the easy half of Theorem 4.4 extends to it.

To make this precise, recall that a *Tychonoff space* is a Hausdorff space in which every closed set and every point outside it are separated by a continuous real-valued function, and that a *compactification* of a Tychonoff space X is a compact Hausdorff space that contains X as a dense subspace. The Stone–Čech compactification βX is the largest one, in the sense that every compactification of X is the image of βX under a continuous map that fixes the points of X.

Definition 4.7 (Čech, 1937). A Tychonoff space X is *Čech-complete* if it is a G_δ subset of its Stone–Čech compactification βX.

A Tychonoff space is Čech-complete if and only if it is a G_δ subset of some compactification, and it is then a G_δ subset of every compactification (see Engelking's book [Eng89, Section 3.9]). In particular, every compact Hausdorff space is Čech-complete, and so is every locally compact Hausdorff space, since such a space, when it is not compact, is open in its one-point compactification. For metrizable spaces nothing new arises: a metrizable space is Čech-complete exactly when it is completely metrizable (Engelking proves this theorem of Čech in his book [Eng89, Theorem 4.3.26]). By Theorem 4.4, Čech-completeness and strong α-favorability therefore coincide on non-empty metrizable spaces. Outside metrizable spaces only one implication survives.

Proposition 4.8. *Every non-empty Čech-complete space is strongly α-favorable.*

Proof. The idea is to play the game in βX, where compactness supplies a common point of the play, while the G_δ condition keeps that point in X. Let X be Čech-complete, and let $G_0, G_1, \dots$ be open subsets of βX with $X = \bigcap_n G_n$; closures below are taken in βX. Along with each answer V_n, Player II records an open set $W_n \subseteq \beta X$ with $V_n = W_n \cap X$, and she chooses the two sets as follows. When Player I plays (x_n, U_n), she takes an open set $\widetilde{U}_n \subseteq \beta X$ with $\widetilde{U}_n \cap X = U_n$; for $n \geq 1$ she may assume that $\widetilde{U}_n \subseteq W_{n-1}$, because $U_n \subseteq V_{n-1} \subseteq W_{n-1}$, and the set $\widetilde{U}_n \cap W_{n-1}$ therefore has the same intersection U_n with X. The point x_n lies in the open set $\widetilde{U}_n \cap G_n$, and the compact Hausdorff space βX is regular; hence there is an open set W_n with $x_n \in W_n$ and $\overline{W_n} \subseteq \widetilde{U}_n \cap G_n$. Player II answers with $V_n = W_n \cap X$, a legal move because $x_n \in V_n \subseteq \widetilde{U}_n \cap X = U_n$.

In every play consistent with this strategy, the inclusions $\overline{W_{n+1}} \subseteq \widetilde{U}_{n+1} \subseteq W_n$ hold for every n, and the sets $\overline{W_n}$ thus form a decreasing sequence of compact sets, which are non-empty because $x_n \in \overline{W_n}$. By compactness, their intersection contains a point y (see Figure 10). Since each $\overline{W_n}$ lies in $\widetilde{U}_n \cap G_n$, the point y lies in $\bigcap_n G_n = X$; consequently it lies in $\widetilde{U}_n \cap X = U_n$ for every n, the play has the common point y, and the strategy is winning. ☺

As in the proof of Corollary 4.5, it follows that every Čech-complete space is a Baire space (see Figure 11). The converse of Proposition 4.8 fails, even among paracompact spaces: the Michael line, obtained from the real line by isolating the irrational points, is strongly α-favorable without being Čech-complete (see Example 4.9 below). Here a Hausdorff space is *paracompact* if every open cover of it has a locally finite open refinement that covers the space; locally finite families and refinements are defined before Lemma 5.3 below.

Example 4.9 (The Michael line). Let M be the real line with the topology in which a set is open if it is the union of a set open in the usual topology and a set of irrational numbers. Every irrational number is then an isolated point, while every rational number keeps its usual neighborhoods (see Figure 12). The space M is Hausdorff and paracompact (see Engelking's book [Eng89, Example 5.1.32]), hence normal, and in particular Tychonoff.

(1) *The space M is strongly α-favorable.* When Player I plays (x_n, U_n) with x_n irrational, Player II answers with $V_n = \{x_n\}$; every later move equals $\{x_n\}$, and the point x_n

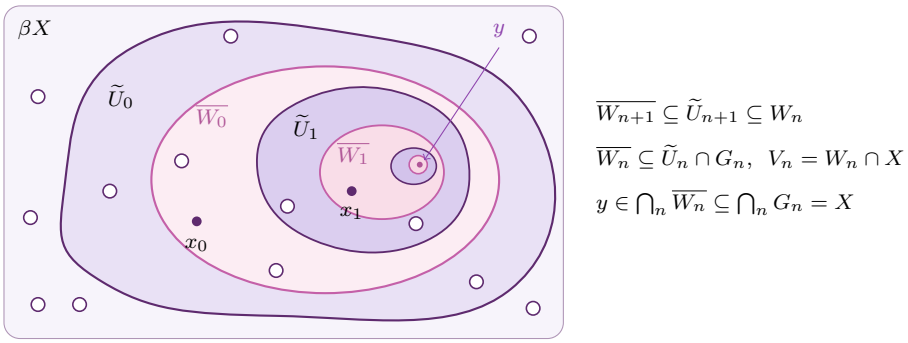


FIGURE 10. The proof of Proposition 4.8, drawn in βX, where $G_0, G_1, \ldots$ are open subsets of βX with $X = \bigcap_n G_n$. The white dots are points of βX that are not in X. The open sets $\widetilde{U}_n$ of βX, whose intersections with X are the moves U_n of Player I (plum), alternate with the closures of the open sets W_n that Player II records with her answers $V_n = W_n \cap X$ (pink), each inside the previous one. Each closure $\overline{W_n}$ lies in G_n and therefore misses the points outside G_n; since every point of βX outside X lies outside some G_n, the white dots are left behind one after another. The nested compact sets $\overline{W_n}$ meet in a point y, which lies in every G_n, hence in X.

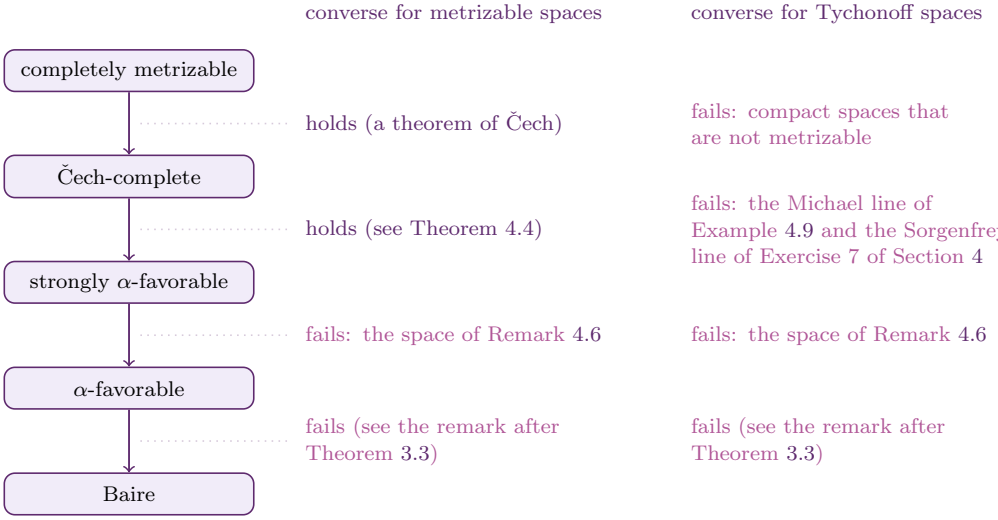


FIGURE 11. Completeness properties of a Tychonoff space, from the strongest to the weakest. From top to bottom, the arrows come from the fact that complete metrizability implies Čech-completeness, from Proposition 4.8, from Lemma 4.2, and from Theorem 3.3 together with Remarks 2.3(3). The two columns on the right tell whether the converse of each arrow holds among metrizable spaces and among all Tychonoff spaces, respectively, and name a space that refutes it when it fails.

lies in every set of the play. When x_n is rational, the set U_n contains a usual open interval about x_n, and Player II answers with an interval $V_n = (x_n - r_n, x_n + r_n)$, where $r_n > 0$ is so small that the closed interval $[x_n - r_n, x_n + r_n]$ lies in U_n. If

Player I plays only rational points, these closed intervals are nested, since each lies in the previous answer; being compact, they have a common point, and that point lies in every U_n.

(2) *The space M is not Čech-complete.* The set $\mathbb{Q}$ is closed in M, because its complement consists of isolated points, and its subspace topology is the usual one. If M were Čech-complete, then its closed subspace $\mathbb{Q}$ would be Čech-complete as well, since closed subspaces of Čech-complete spaces are Čech-complete (see Engelking's book [Eng89, Theorem 3.9.6]), and hence a Baire space; yet $\mathbb{Q}$, the union of countably many nowhere dense singletons, is not a Baire space.

The Michael line separates the two properties in one more respect: unlike Čech-completeness, strong α-favorability does not pass to closed subspaces, since Player I wins $\mathrm{Ch}(\mathbb{Q})$ (see the example after Lemma 4.2).

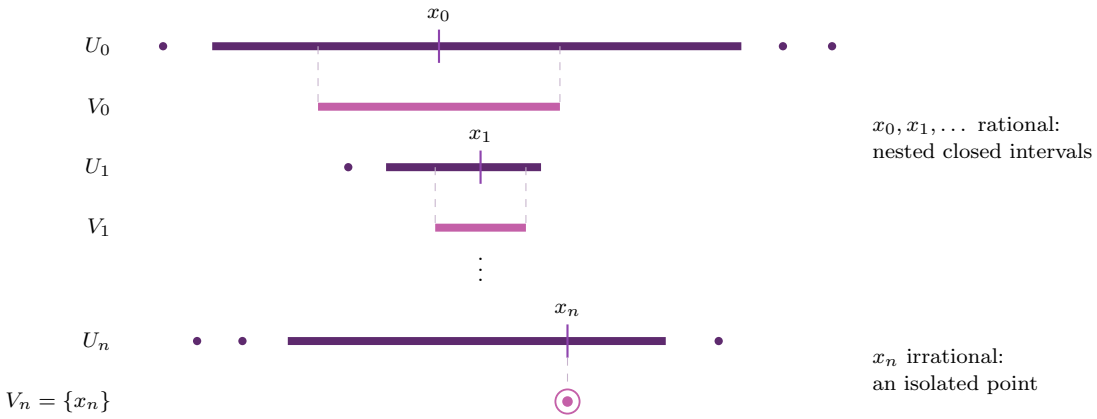


FIGURE 12. Two kinds of answers in the strong Choquet game on the Michael line M of Example 4.9. An open set of M is drawn as a union of open intervals together with isolated irrational points (the small dots). Top: while Player I (plum) plays rational points, Player II (pink) answers with intervals whose closures lie in his sets; these closed intervals are nested and therefore have a common point. Bottom: once Player I plays an irrational point, Player II answers with the singleton of that point, and every later move is the same singleton.

Among paracompact spaces, Čech-completeness also has an exact characterization by a game of another kind, played not on the space but on what the Stone–Čech compactification adds to it: a paracompact space X is Čech-complete if and only if Player II has a winning strategy in the Menger game of Section 15 below, played on the remainder $\beta X \setminus X$. Aurichi and Bella [AB15, Corollary 3.3] deduce this equivalence from a theorem of Telgársky [Tel84].

Exercises.

(1) Let X be a non-empty compact Hausdorff space. Show directly, without Proposition 4.8, that Player II has a winning strategy in $\mathrm{Ch}(X)$, that is, that X is strongly α-favorable.

(2) Let X be a non-empty completely metrizable space with no isolated points. Show that X is uncountable. (Hint: each singleton is closed with empty interior; use Corollary 4.5.)

(3) The interval $(0,1)$ is not complete in the usual metric. Show directly that Player II has a winning strategy in $\mathrm{Ch}((0,1))$. (Hint: imitate the proof of Theorem 4.4, choosing $r_n < 2^{-n}$ so that the closed interval $[x_n - r_n, x_n + r_n]$ lies in U_n.)
(4) Let $X = \{(a,b) \in \mathbb{R}^2 : b > 0\} \cup (\mathbb{Q} \times \{0\})$ be the space of Remark 4.6. Write out in full the winning strategy for Player I in $\mathrm{Ch}(X)$ indicated in Figure 9, and check that it is winning. (Hint: enumerate $\mathbb{Q} \times \{0\}$ as $p_0, p_1, \ldots$, and at step n choose U_n inside a closed disk D_n of radius less than 2^{-n} that misses p_n and, for $n \geq 1$, lies in D_{n-1}.)
(5) $(*)$ Let $P = \mathbb{R} \setminus \mathbb{Q}$ be the space of irrational numbers, with the usual metric, and enumerate $\mathbb{Q}$ as $q_0, q_1, \ldots$. Show directly that Player II has a winning strategy in $\mathrm{Ch}(P)$. (Hint: answer (x_n, U_n) with $V_n = (x_n - r_n, x_n + r_n) \cap P$, where the closed interval $[x_n - r_n, x_n + r_n]$ misses q_n and its points in P lie in U_n.)
(6) Recall that a topological space is *countably compact* if every countable open cover has a finite subcover, or, equivalently, if every decreasing sequence of non-empty closed sets has a non-empty intersection. Show that every non-empty countably compact regular space is strongly α-favorable (this extends Exercise 1). (Hint: answer (x_n, U_n) with an open set V_n such that $x_n \in V_n$ and $\overline{V_n} \subseteq U_n$.)
(7) $(*)$ Let S be the Sorgenfrey line, the set $\mathbb{R}$ with the topology whose basic open sets are the intervals $[a,b)$ with $a < b$ real. Show that S is strongly α-favorable. (Hint: answer (x_n, U_n) with $V_n = [x_n, x_n + r_n)$, where $r_n > 0$ and $[x_n, x_n + 2r_n) \subseteq U_n$; the points x_n form a non-decreasing sequence, bounded above by $x_0 + r_0$, and its limit lies in every U_n.) The space S is, after the Michael line, a second strongly α-favorable space that is not Čech-complete: by the paragraph after Theorem 11.1 below, Čech-complete spaces are strongly Baire, a property defined in Section 11, whereas S is not (see Exercise 4 of Section 11).

5. The converse of Choquet's theorem

In this section we prove the hard half of Theorem 4.4: every non-empty metrizable space on which Player II has a winning strategy in the strong Choquet game is completely metrizable. The proof is long, so we first explain where a complete metric can come from and what the strategy has to do with it.

The completion. Let X be a non-empty metrizable space, let τ be a winning strategy for Player II in the strong Choquet game $\mathrm{Ch}(X)$, fix a compatible metric d on X, and let $\widetilde{X}$ be the completion of (X,d). If d is complete, then $\widetilde{X} = X$; in general the completion contains new points, and the position of X inside it decides whether X is completely metrizable. Two subspaces of $\mathbb{R}$ show the difference. The irrationals form the G_δ set $\bigcap_{q \in \mathbb{Q}} (\mathbb{R} \setminus \{q\})$, and they are completely metrizable: continued fractions make them homeomorphic to the space of sequences of positive integers, which is complete in its usual metric. The rationals do not form a G_δ subset of $\mathbb{R}$: otherwise the rationals and the irrationals would be two disjoint dense G_δ subsets of $\mathbb{R}$, whereas, by the Baire category theorem (see Corollary 4.5), the intersection of two dense G_δ subsets of $\mathbb{R}$ is dense, and in particular non-empty. In Lemma 5.2 below we show that every G_δ subset of a complete metric space is completely metrizable. Our goal is therefore to find open sets $G_1, G_2, \ldots$ of $\widetilde{X}$ with $X = \bigcap_n G_n$. In the proof of Proposition 4.8, a representation of this kind, of a Čech-complete space as a G_δ subset of its Stone–Čech compactification, produced a winning strategy; here the strategy must produce the representation.

The open sets of the strategy. The strategy τ of Player II is a machine that answers each point with an open set that contains it. At the first stage, Player I may play any point x of

X (with the open set X), and the strategy answers with an open set $V \ni x$. These sets cover X; once each of them is enlarged slightly into $\widetilde{X}$, their union is an open subset G_1 of $\widetilde{X}$ that contains X. At the second stage, Player I continues each of these positions with each point of the set just played, and so on. The open set G_n is, roughly, the union of the (enlarged) sets played by Player II at stage n, over all positions, and it therefore contains X. Suppose now that a point y of the completion lies in every G_n. If the sets containing y at the various stages all came from *one* play, then that play would have a common point, because Player II wins it. The sets will be chosen with diameters tending to 0, so this common point would be y; in particular, the point y would belong to X.

The difficulty. At each stage, however, the point y may lie in sets that come from many different positions, and a choice of one position at each stage need not form a single play: for that, the position chosen at stage $n+1$ must extend the one chosen at stage n. If at every stage only *finitely* many positions are involved, then a compactness argument, Kőnig's lemma (see Lemma 5.4 below), makes the choices fit together. The finiteness is supplied by a theorem of A. H. Stone (see Lemma 5.3 below): every open cover of a metric space has an open refinement, still covering the space, in which each point has a neighborhood that meets only finitely many members. The two ingredients combine as shown in Figure 13.

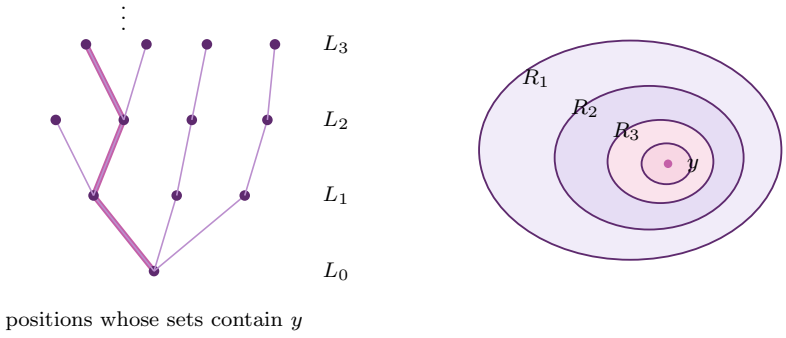


FIGURE 13. The end of the proof of the hard half of Theorem 4.4. Left: let y be a point of the completion $\widetilde{X}$ that lies in every open set G_n built in the proof. The pairs (R, p) of stage n with $y \in R$, each made of a position p and an open set R attached to it, form a finite non-empty set L_n, and each member of L_{n+1} descends from a member of L_n. Kőnig's lemma provides a branch (pink), that is, a single play. Right: along that play the open sets R_n of the chosen pairs, which contain y, shrink to a point; since Player II wins, the play has a common point in X, which must be y.

5.1. **A preliminary lemma and three classical tools.** In this subsection we prove a lemma on extending open sets and the three classical tools of the proof: Alexandrov's theorem, A. H. Stone's theorem and Kőnig's lemma.

Throughout this subsection we fix a metric space (X, d) and a metric space $(\widetilde{X}, \tilde{d})$ that contains X as a dense subset, with $\tilde{d}$ extending d. Closures and diameters of subsets of $\widetilde{X}$ are taken in $\widetilde{X}$. In the preliminary lemma we extend each open set of X to an open set of $\widetilde{X}$ with the same points of X and the same diameter (see Figure 14). For an open set $V \subseteq X$, we write $\widetilde{V} = \widetilde{X} \setminus \overline{X \setminus V}$.

Lemma 5.1. *Let (X, d) and $(\widetilde{X}, \tilde{d})$ be as fixed at the beginning of this subsection, and let V and V' be open subsets of X. Then the sets $\widetilde{V} = \widetilde{X} \setminus \overline{X \setminus V}$ and $\widetilde{V'} = \widetilde{X} \setminus \overline{X \setminus V'}$, with closures taken in $\widetilde{X}$, satisfy the following:*

(1) *the set $\widetilde{V}$ is open in $\widetilde{X}$, and $\widetilde{V} \cap X = V$;*
(2) *if $V \subseteq V'$, then $\widetilde{V} \subseteq \widetilde{V'}$;*
(3) *the diameters satisfy* $\operatorname{diam} \widetilde{V} = \operatorname{diam} V$.

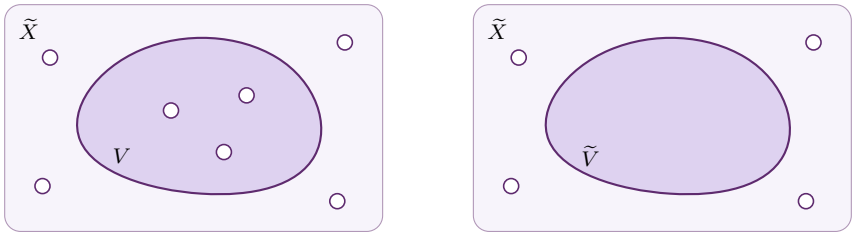


FIGURE 14. The extension of open sets in Lemma 5.1. The white dots are points of $\widetilde{X}$ that are not in X. Left: an open set V of X, which misses the white dots inside it. Right: the open set $\widetilde{V}$ of $\widetilde{X}$, which also contains them. The two sets have the same points of X and the same diameter.

Proof. (1) The set $\widetilde{V}$ is the complement of a closed set, hence open. Let $x \in V$ be given. Since V is open in X, there is $\varepsilon > 0$ such that the open ball $B(x, \varepsilon)$ of X lies in V; the ball of radius ε about x in $\widetilde{X}$ meets X exactly in $B(x, \varepsilon)$, and it therefore misses $X \setminus V$, which gives $x \in \widetilde{V}$. If $x \in X \setminus V$, then $x \in \overline{X \setminus V}$, and $x \notin \widetilde{V}$. Thus $\widetilde{V} \cap X = V$.

(2) If $V \subseteq V'$, then $X \setminus V' \subseteq X \setminus V$, and the complements of the closures reverse the inclusion.

(3) We first show that $\widetilde{V} \subseteq \overline{V}$. Let $y \in \widetilde{V}$ and $\varepsilon > 0$ be given. The points of $\widetilde{V}$ at distance less than ε from y form a non-empty open set, which meets the dense set X; by (1), every point of $\widetilde{V} \cap X$ lies in V. Since ε is arbitrary, every ball about y meets V, that is, $y \in \overline{V}$. It follows that $\operatorname{diam} \widetilde{V} \leq \operatorname{diam} \overline{V} = \operatorname{diam} V$. The reverse inequality holds because $V \subseteq \widetilde{V}$. ☺

The first tool is Alexandrov's theorem, announced at the beginning of this section; it reduces our problem to finding a G_δ set.

Lemma 5.2 (P. S. Alexandrov). *Every G_δ subset of a complete metric space is completely metrizable.*

Proof. Let (Y, ρ_0) be a complete metric space, and let $G = \bigcap_{n \geq 1} G_n$ with each G_n open in Y. The idea is to add to ρ_0 terms that blow up near the complements of the sets G_n, so that a Cauchy sequence cannot approach them. We may assume that each closed set $F_n = Y \setminus G_n$ is non-empty, discarding the indices with $G_n = Y$ (if no index remains, then $G = Y$ and there is nothing to prove). For $x \in G$, the distance $\rho_0(x, F_n) = \inf\{\rho_0(x, z) : z \in F_n\}$ is positive, because F_n is closed and $x \notin F_n$; we put $f_n(x) = 1/\rho_0(x, F_n)$, a continuous function on G, and for $x, y \in G$ we put

$$\rho(x, y) = \rho_0(x, y) + \sum_{n \geq 1} 2^{-n} \min\{1, |f_n(x) - f_n(y)|\}.$$

Each summand of ρ is a pseudometric on G, the series converges, and the first summand ρ_0 is a metric; the sum ρ is therefore a metric on G.

The metric ρ is compatible with the topology of G. Indeed, convergence for ρ implies convergence for ρ_0, because $\rho \geq \rho_0$. Conversely, let $x \in G$ be given, and suppose that a sequence $(x_k)_{k\geq 1}$ in G converges to x with respect to ρ_0. Each term of the series then tends to 0, by the continuity of f_n, and the nth term never exceeds 2^{-n}; dominated convergence for series yields $\rho(x_k, x) \to 0$.

The metric ρ is complete. Let $(x_k)_{k\geq 1}$ be a ρ-Cauchy sequence in G. Since $\rho \geq \rho_0$, the sequence is ρ_0-Cauchy, and it converges to some $y \in Y$ because ρ_0 is complete. For each n, the sequence $(f_n(x_k))_k$ is Cauchy, because $\min\{1, |f_n(x_k) - f_n(x_l)|\} \leq 2^n \rho(x_k, x_l)$; it is therefore bounded by some constant $c_n > 0$. Thus $\rho_0(x_k, F_n) \geq 1/c_n$ for every k, and in the limit $\rho_0(y, F_n) \geq 1/c_n > 0$. It follows that $y \in G_n$ for every n, that is, $y \in G$, and $\rho(x_k, y) \to 0$ by the compatibility just proved. Consequently, the space G carries the complete compatible metric ρ and is completely metrizable. ☺

The second tool supplies the finiteness. A family $\{V_i : i \in I\}$ of subsets of a topological space Y is *locally finite* if every point of Y has a neighborhood that meets V_i for only finitely many indices $i \in I$. A family $\mathcal{V}$ *refines* a family $\mathcal{U}$, and is then called a *refinement* of $\mathcal{U}$, if every member of $\mathcal{V}$ is contained in some member of $\mathcal{U}$.

Lemma 5.3 (A. H. Stone). *Every open cover of a metric space has a locally finite open refinement that covers the space.*

The idea of the proof is to cover the space by small balls, to assign each ball to the first member of the cover that contains its center, to use the ball only when that member contains it with room to spare, and to stop using small balls around points that larger balls already cover.

Proof. Let (Y, ρ) be a metric space, and let $\{U_s : s \in \Lambda\}$ be an open cover of Y; we write $B(c, r)$ for the open ball of radius r about c. By the well-ordering theorem we may assume that Λ is well ordered. For $s \in \Lambda$ and $n \geq 1$, let $V_{s,n}$ be the union of the balls $B(c, 2^{-n})$ over all centers $c \in Y$ such that:

(i) the index s is the least one with $c \in U_s$;
(ii) the center c lies in no set $V_{t,j}$ with $t \in \Lambda$ and $j < n$;
(iii) the ball $B(c, 3 \cdot 2^{-n})$ is contained in U_s.

The definition is a recursion on n, because (ii) involves only the sets with smaller second index. Each $V_{s,n}$ is open, and $V_{s,n} \subseteq U_s$ by (iii).

The sets $V_{s,n}$ cover Y. Let $c \in Y$ be given, let s be the least index with $c \in U_s$, and let n be so large that $B(c, 3 \cdot 2^{-n}) \subseteq U_s$. Either the point c lies in some $V_{t,j}$ with $j < n$, or it satisfies (i)–(iii), and then $c \in B(c, 2^{-n}) \subseteq V_{s,n}$.

The family is locally finite. Let $c \in Y$ be given, and let s and n be indices with $c \in V_{s,n}$; since $V_{s,n}$ is open, there is $j \geq 1$ with $B(c, 2^{-j}) \subseteq V_{s,n}$. We show that the ball $B(c, 2^{-n-j})$ meets no set $V_{t,i}$ with $i \geq n + j$, and at most one set $V_{t,i}$ for each $i < n + j$; the ball then meets at most $n + j - 1$ sets of the family.

Suppose first that $i \geq n + j$, and let c' be a center used for $V_{t,i}$. Since $n < i$, condition (ii) gives $c' \notin V_{s,n}$, so $\rho(c, c') \geq 2^{-j}$. Every point of $B(c', 2^{-i})$ is therefore at distance at least $2^{-j} - 2^{-i} \geq 2^{-j} - 2^{-n-j} \geq 2^{-n-j}$ from c, and the ball $B(c, 2^{-n-j})$ misses $V_{t,i}$.

Suppose next that $i < n + j$, and that $p \in V_{t,i}$ and $q \in V_{t',i}$ with $t < t'$. There are centers c_1 for $V_{t,i}$ and c_2 for $V_{t',i}$ with $\rho(p, c_1) < 2^{-i}$ and $\rho(q, c_2) < 2^{-i}$. Condition (iii) places the ball $B(c_1, 3 \cdot 2^{-i})$ inside U_t, and condition (i) keeps c_2 out of U_t, because t' is the least index with $c_2 \in U_{t'}$ and $t < t'$. Hence $\rho(c_1, c_2) \geq 3 \cdot 2^{-i}$ and

$$\rho(p, q) \geq \rho(c_1, c_2) - \rho(p, c_1) - \rho(q, c_2) > 2^{-i} \geq 2 \cdot 2^{-n-j}.$$

Two points of $B(c, 2^{-n-j})$ are at distance less than $2 \cdot 2^{-n-j}$, so the ball cannot meet both $V_{t,i}$ and $V_{t',i}$. The sets $V_{s,n}$ therefore form a locally finite open refinement of $\{U_s : s \in \Lambda\}$ that covers Y. ☺

The third tool turns finitely many choices at each stage into one coherent choice.

Lemma 5.4 (D. Kőnig). *Let $L_0, L_1, \dots$ be non-empty finite sets, and for each $n \geq 0$ let $\pi_n \colon L_{n+1} \to L_n$ be a function. Then there are $\ell_n \in L_n$, for $n \geq 0$, with $\pi_n(\ell_{n+1}) = \ell_n$ for every n.*

Proof. The idea is to follow, level by level, the elements that are reached from every later level. For $m \geq n$ we write $\pi_{n,m} \colon L_m \to L_n$ for the composite $\pi_n \circ \cdots \circ \pi_{m-1}$ (the identity when $m = n$). We call $\ell \in L_n$ *good* if for every $m \geq n$ some element of L_m is mapped to ℓ by $\pi_{n,m}$. If $\ell = \pi_{n,m}(z)$ for some $z \in L_m$, then $\ell = \pi_{n,k}(\pi_{k,m}(z))$ for every k with $n \leq k \leq m$. Hence an element of L_n is good as soon as it lies in the image of $\pi_{n,m}$ for infinitely many m.

Some element of L_0 is good: every L_m is non-empty, so each m supplies an element of L_0 in the image of $\pi_{0,m}$, and since L_0 is finite, one element is supplied by infinitely many m. If $\ell \in L_n$ is good, then some element of the finite set $\pi_n^{-1}(\ell)$ is good: for each $m > n$, an element of L_m mapped to ℓ passes through some element of $\pi_n^{-1}(\ell)$, and one element of $\pi_n^{-1}(\ell)$ serves infinitely many m. We may therefore choose, recursively, a good $\ell_0 \in L_0$ and, for each n, a good $\ell_{n+1} \in \pi_n^{-1}(\ell_n)$; the resulting sequence satisfies $\pi_n(\ell_{n+1}) = \ell_n$ for every n, as required. ☺

5.2. **The proof.**

Proof of the converse of Theorem 4.4. Let X be a non-empty metrizable space, and let τ be a winning strategy for Player II in $\mathrm{Ch}(X)$. Let d be a compatible metric on X, and let $(\widetilde{X}, \tilde{d})$ be the completion of (X, d), in which X is dense. Following the plan described at the beginning of this section, we build from τ open sets $G_1, G_2, \dots$ of $\widetilde{X}$, each containing X, and we use Kőnig's lemma to show that their intersection is exactly X. As in Lemma 5.1, for an open set $V \subseteq X$ we put $\widetilde{V} = \widetilde{X} \setminus \overline{X \setminus V}$, the closure being taken in $\widetilde{X}$. We write $B(x, r)$ for the open ball of radius r about x in X. For $n \geq 0$, a *position of stage n* is a position of length $2n$ of $\mathrm{Ch}(X)$ consistent with τ; it ends with the move V_{n-1} of Player II when $n \geq 1$, and it is the empty position when $n = 0$.

Step 1: the families $\mathcal{S}_n$. By recursion on $n \geq 0$ we construct families $\mathcal{S}_n$ of pairs (R, p), where R is a non-empty open subset of $\widetilde{X}$ and p is a position of stage n. The union of the sets R with $(R, p) \in \mathcal{S}_n$ is denoted by G_n. We require the following:

- (P1) if $n \geq 1$ and $(R, p) \in \mathcal{S}_n$, then $R \subseteq \widetilde{V}$, where V is the last move of p, and $\operatorname{diam} R \leq 2^{-n}$;
- (P2) the set G_n contains X;
- (P3) the family of sets R, $(R, p) \in \mathcal{S}_n$, is locally finite in G_n;
- (P4) each $(R, p) \in \mathcal{S}_{n+1}$ has a designated *parent* $(R', p') \in \mathcal{S}_n$ such that $R \subseteq R'$ and p extends p' by two moves: Player I plays $(x, R' \cap X)$ for some point $x \in R' \cap X$, and Player II answers as τ prescribes.

We put $\mathcal{S}_0 = \{(\widetilde{X}, \langle\rangle)\}$; conditions (P2) and (P3) hold trivially, and (P1) is vacuous for $n = 0$.

Suppose that $\mathcal{S}_n$ has been constructed. Let $(R', p') \in \mathcal{S}_n$ and $x \in R' \cap X$ be given. The move $(x, R' \cap X)$ is legal for Player I at p': for $n = 0$ it is a first move, and for $n \geq 1$ condition (P1) and Lemma 5.1(1) give $R' \cap X \subseteq \widetilde{V} \cap X = V$, where V is the last move of p'. Let V_x be the answer of τ, which satisfies $x \in V_x \subseteq R' \cap X$, and let q_x be the resulting position of stage

$n+1$. (The sets V_x, the positions q_x and the sets Q_x and O_x below depend also on the pair (R', p'), which we suppress from the notation.) We put

$$Q_x = V_x \cap B(x, 2^{-n-2}) \qquad \text{and} \qquad O_x = R' \cap \widetilde{Q_x}.$$

The set O_x is open in $\widetilde{X}$ and contained in R'. It contains x, because $x \in R'$ and $x \in Q_x \subseteq \widetilde{Q_x}$ by Lemma 5.1(1); it lies in $\widetilde{V_x}$, because $Q_x \subseteq V_x$ and Lemma 5.1(2) applies; and its diameter is at most $\operatorname{diam} \widetilde{Q_x} = \operatorname{diam} Q_x \le 2^{-n-1}$, by Lemma 5.1(3) and because Q_x lies in a ball of radius 2^{-n-2}. By (P2), every point x of X lies in some R' with $(R', p') \in \mathcal{S}_n$, and then $x \in O_x$; hence the sets O_x, over all pairs $(R', p') \in \mathcal{S}_n$ and all $x \in R' \cap X$, cover X. Their union H is open in $\widetilde{X}$, and the sets O_x form an open cover of the metric space H. By Lemma 5.3, this cover has a locally finite open refinement $\mathcal{R}$ that covers H; we discard its empty members. For each $R \in \mathcal{R}$ we choose a set O_x that contains R, coming from a pair (R', p') and a point x, and we put (R, q_x) into $\mathcal{S}_{n+1}$, with designated parent (R', p').

Conditions (P1)–(P4) hold for $n+1$. Indeed, $R \subseteq O_x \subseteq \widetilde{V_x}$, where V_x is the last move of q_x, and $\operatorname{diam} R \le \operatorname{diam} O_x \le 2^{-n-1}$; the union G_{n+1} equals H, which contains X; the family $\mathcal{R}$ is locally finite in $H = G_{n+1}$; and $R \subseteq O_x \subseteq R'$, with q_x extending p' by the moves $(x, R' \cap X)$ and V_x.

Step 2: a G_δ set containing X. The set $G = \bigcap_{n \ge 1} G_n$ is a G_δ subset of $\widetilde{X}$, and $X \subseteq G$ by (P2). It remains to be shown that $G \subseteq X$.

Step 3: every point of G is the common point of a play. Let $y \in G$ be given. For $n \ge 0$, let L_n be the set of pairs $(R, p) \in \mathcal{S}_n$ with $y \in R$. Each L_n is non-empty, because $y \in G_n$ (and L_0 consists of the single pair $(\widetilde{X}, \langle\rangle)$). Each L_n is also finite, by (P3): since $y \in G_n$, some neighborhood of y meets only finitely many of the sets R with $(R, p) \in \mathcal{S}_n$, and every such R that contains y is among them. By (P4), the parent (R', p') of a member (R, p) of L_{n+1} satisfies $y \in R \subseteq R'$, so it lies in L_n; let $\pi_n \colon L_{n+1} \to L_n$ be the parent map. By Lemma 5.4, there are $(R_n, p_n) \in L_n$ such that (R_n, p_n) is the parent of (R_{n+1}, p_{n+1}) for every n.

By (P4), for each $n \ge 0$ the position p_{n+1} extends p_n by a move $(x_n, R_n \cap X)$ of Player I, for some point $x_n \in R_n \cap X$, and by the answer of τ. Hence the positions p_n fit together into a single play of $\mathrm{Ch}(X)$, consistent with τ, in which Player I plays the sets $R_n \cap X$. Since τ is winning, there is a point $z \in \bigcap_n (R_n \cap X)$. For every $n \ge 1$, the points y and z both lie in R_n, which has diameter at most 2^{-n} by (P1). Hence $\tilde{d}(y, z) = 0$, and $y = z \in X$ (see Figure 13).

Step 4: conclusion. By Steps 2 and 3, we have $X = G$, a G_δ subset of the complete metric space $\widetilde{X}$; since $\tilde{d}$ extends d, the topology that X inherits from $\widetilde{X}$ is its own. By Lemma 5.2, the space X is completely metrizable. ☺



Remark 5.5. The points of Player I are what make the proof work: in Step 1, Player I plays every point x of every set $R' \cap X$, and Player II must answer with a set that contains x; this is why each G_n contains all of X. Without the points, the theorem fails (see Remark 4.6). For separable spaces, Kechris gives a different proof [Kec95, Section 8].

Exercises.

(1) Let X be the space $\mathbb{R} \setminus \{0\}$ with the usual metric, whose completion is $\widetilde{X} = \mathbb{R}$. Compute the set $\widetilde{V} = \widetilde{X} \setminus \overline{X \setminus V}$, with the closure taken in $\widetilde{X}$, for $V = (-1, 0) \cup (0, 1)$ and for $V = (0, 1)$, and check the three statements of Lemma 5.1 in both cases.

(2) Show that the open cover $\{(-n, n) : n \ge 1\}$ of $\mathbb{R}$ is not locally finite, and find a locally finite open refinement of it that covers $\mathbb{R}$, as promised by Lemma 5.3.

(3) Apply the proof of Lemma 5.2 to $G = (0,1)$, a single open subset of $Y = \mathbb{R}$ with the usual metric ρ_0, and write down the metric ρ explicitly. Show that the sequence $(1/k)_{k\geq 2}$ is Cauchy for the usual metric but not for ρ.
(4) Show that the conclusion of Lemma 5.4 can fail if the sets L_n are allowed to be infinite. (Hint: let L_0 have one element, let L_n, for $n \geq 1$, consist of the pairs (i, n) of integers with $i \geq n$, let π_0 be constant, and let $\pi_n(i, n+1) = (i, n)$ for $n \geq 1$.)
(5) $(*)$ Let (X, d) be a metric space whose topology is completely metrizable, possibly by a metric other than d. Use Theorem 4.4 and the proof of its converse in Section 5, with the metric d, to show that X is a G_δ subset of the completion of (X, d). Deduce that $\mathbb{Q}$ is not completely metrizable.

6. Summary: from complete metrizability to Baire category

The main results of this part fit into one chain of implications, which runs from complete metrizability, through the two games, down to Baire category.

For a non-empty metrizable space X, with $\mathrm{Ch}(X)$ the strong Choquet game and $\mathrm{BM}(X)$ the Banach–Mazur game on X, the chain is the following:

X is completely metrizable
$\Updownarrow$ Theorem 4.4
Player II has a winning strategy in $\mathrm{Ch}(X)$
$\Downarrow$ Lemma 4.2
Player II has a winning strategy in $\mathrm{BM}(X)$
$\Downarrow$ Remarks 2.3(3)
Player I has no winning strategy in $\mathrm{BM}(X)$
$\Updownarrow$ Theorem 3.3
X is a Baire space
$\Updownarrow$ Theorem 1.3
no non-empty open subset of X is of the first category.

Followed from top to bottom, the chain yields the Baire category theorem (see Corollary 4.5). The implication given by Lemma 4.2 cannot be reversed: the space of Remark 4.6, the open upper half plane together with the rational points of the horizontal axis, is metrizable and Player II wins its Banach–Mazur game, but the space is not completely metrizable, and Player II therefore has no winning strategy in its strong Choquet game. The games $\mathrm{BM}(X, S)$ with a target set $S \subseteq X$, in which Player II wins when the intersection of the moves lies in S, measure the size of sets inside X: Player II has a winning strategy in $\mathrm{BM}(X, S)$ exactly when S is comeager in X, and, when X is a complete metric space, Player I has one exactly when $U \cap S$ is of the first category in X for some non-empty open set U (see Theorem 3.5).

7. Notes and historical remarks

7.1. **Baire category.** The characterization of Baire spaces in Theorem 1.3 is an elementary reformulation of their definition and carries no attribution. It is not to be confused with the *Banach category theorem* [Ban30], which is deeper: in any topological space, the union of an arbitrary family of open sets of the first category is again of the first category. The title of Oxtoby's paper [Oxt57] refers to this theorem.

7.2. **The Banach–Mazur and strong Choquet games.** As told in the Introduction, the Banach–Mazur game is Problem 43 of the Scottish Book, posed by S. Mazur in 1935 [Mau81] (see also Section 26). In Mazur's formulation, two players alternately choose nested intervals of the real line, and the first player wins if their intersection meets a given set E. Mazur observed two facts: the first player wins if E is comeager in some interval, and the second player wins if E is of the first category. He conjectured the converses. On 4 August 1935 Banach recorded in the Scottish Book that Mazur's conjecture is true. Banach's proof was never published. The first published proof is Oxtoby's [Oxt57]. Oxtoby showed that in an arbitrary topological space the second player has a winning strategy exactly when E is of the first category. The other converse, that the first player wins exactly when E is comeager in some non-empty open set, requires further hypotheses on the space; it holds, for instance, in complete metric spaces. Together, Oxtoby's theorem and this converse become the two parts of Theorem 3.5 under the following dictionary:

(1) Mazur's first player is our Player I, and Mazur's second player is our Player II.
(2) Mazur's set E, at which the first player aims, is the complement $X \setminus S$ of the set S at which Player II aims in the game $\mathrm{BM}(X, S)$ on a space X.
(3) "The intersection meets E" becomes "the intersection is not contained in S", so Mazur's first player wins exactly when our Player I wins.

Whether the moves are open or closed intervals makes no difference to the proofs. The game $\mathrm{BM}(X)$, in which Player II needs only a non-empty intersection, differs from $\mathrm{BM}(X, X)$, which Player II wins trivially; the proof of Theorem 3.3 is Oxtoby's argument adapted to $\mathrm{BM}(X)$. Telgársky's survey [Tel87], written for the fiftieth anniversary of the Banach–Mazur game, covers its later history.

The strong Choquet game goes back to Choquet's work on Baire spaces [Cho58]. Choquet expounds the game, together with Theorem 4.4, in his *Lectures on Analysis* [Cho69], where the players are called β and α. Kechris states Theorem 4.4 for separable metrizable spaces, where it characterizes the Polish spaces, that is, the separable completely metrizable spaces [Kec95, Section 8]. The proof in Section 5, for arbitrary metrizable spaces, rests on three classical results. The first, Lemma 5.2, is due to Alexandrov [Ale24]. The second, Lemma 5.3, is A. H. Stone's theorem that metric spaces are paracompact [Sto48]; the short proof given here is M. E. Rudin's [Rud69]. The third, Lemma 5.4, is Kőnig's infinity lemma [Kőn27], first stated for graphs.

7.3. **Čech-complete spaces.** Čech introduced the spaces now named after him in 1937 [Čec37], in the paper in which he constructed the Stone–Čech compactification independently of M. H. Stone; he called them topologically complete, and he proved that a metrizable space is topologically complete exactly when it is completely metrizable. The result of Proposition 4.8, that every Čech-complete space is strongly α-favorable, is folklore. In his survey [Tel87], Telgársky also treats the many variants of the strong Choquet game and the completeness properties that they describe. Michael introduced the space of Example 4.9 in 1963, as a paracompact space whose product with the irrationals is not normal [Mic63].

Part
II

Applications of Banach–Mazur and Choquet games

In Part I we showed that the Banach–Mazur game measures size: a set is large in the sense of Baire category exactly when Player II can steer the play into it (see Theorem 3.5). In this part we put that measure to work outside topology. Each problem in this part asks, in one form or another, that a set be shown to be large, or that an object be shown to exist when no formula for it is in sight; in the last section the question is turned around, and the game explains why an object that the axiom of choice provides cannot be written down. In each case the answer is a game obtained by changing a rule or two, either in the Banach–Mazur game of Section 3 or in a game of Choquet type in which the players shrink sets (see Section 4), and with each new rule the game measures something new. Through all these changes, the two features of games described in the Introduction persist: a winning strategy is built one move at a time, and a player can pursue several goals at once by taking turns between them.

Roadmap

The sections run from the concrete to the set-theoretic. The part opens in number theory, where Schmidt's game, in which the sizes of the moves are prescribed, shows that the badly approximable numbers form a set that is large in dimension and survives countable intersections (see Section 8); read on the space of lattices, the same game becomes a game about bounded orbits of flows (see Section 9). A limit on the computing power of Player II yields a notion of category inside complexity classes (see Section 10), a more demanding winning condition yields a class of spaces, the strongly Baire spaces, on which a separately continuous group multiplication is always jointly continuous (see Section 11), and in a finer topology, Ellentuck's, the game characterizes the families of infinite sets that are Ramsey in a hereditary sense (see Section 12). In Banach spaces one game answers arbitrary sets with relatively weak*-open ones, and another answers block subspaces with vectors; the first detects Asplund spaces, and the second drives Gowers's dichotomy (see Section 13). The part ends with the nonprincipal ultrafilters on the natural numbers (see Section 14).

The sections can be read in any order, with a few dependencies: the section on homogeneous dynamics builds on the proof in Section 8, and the sections on Banach spaces and on ultrafilters both draw on the Ramsey theory of Section 12, which in turn borrows the definition of an ultrafilter from the beginning of Section 14. The sections also differ in what they ask of the reader. In the sections on Diophantine approximation, computational complexity and ultrafilters, which need little beyond Part I, we give complete proofs of the main points. The sections on homogeneous dynamics and Banach spaces reach into Lie groups and functional analysis; each of them opens with a short glossary, and the heart of the first is a computation with 2×2 matrices that we do by hand. In the sections on topological groups and Ramsey theory we sketch the main arguments rather than prove every step. Subsections marked *optional* may be skipped on a first reading. The games of this part are set side by side in Table 1.

Section	**Moves**	**Player II wins if**	**Main result**
8. Diophantine approximation	closed balls whose radii shrink by two fixed factors, one for each player	the limit point lies in the target set	Theorem 8.1
9. Homogeneous dynamics	Schmidt's game on an expanding leaf of a flow on a space of lattices	the orbit of the limit point is bounded	Theorem 9.1
10. Complexity	finite binary strings, each extending the last; Player II must compute her moves quickly	the language produced is not in a given class	Theorem 10.1
11. Topological groups	open sets, as in the Banach–Mazur game	the intersection is non-empty, and every sequence of points of a fixed dense set, one in each of her moves, clusters	Theorem 11.1
12. Ramsey theory	Ellentuck basic sets; Player II adds one element at each move	the set produced lies in a given family of infinite sets	Theorem 12.1
13. Banach spaces	arbitrary sets against relatively weak*-open sets; block subspaces against vectors	the sets shrink to a point in norm, or have empty intersection; the sequence of vectors lies in a given set of sequences	Theorems 13.1 and 13.2
14. Ultrafilters	finite binary strings; sets in a given ultrafilter against points or finite sets	the sequence produced lies in the target set; the set produced lies in the ultrafilter	Theorem 14.1

Table 1. The games of Part II. In every row Player I moves first and the payoff set belongs to Player II.

8. Applications to Diophantine approximation

The problem. Every real number can be approximated by fractions, but some numbers are much harder to approximate than others. Let n be a positive integer, and for $t \in \mathbb{R}$ let $\|t\|$ be the distance from t to the nearest integer. A vector $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ is *badly approximable* if there is a constant $c = c(\mathbf{x}) > 0$ such that

$$\max_{1 \le i \le n} \|q x_i\| \;\ge\; c\, q^{-1/n} \qquad \text{for every integer } q \ge 1.$$

For $n = 1$, writing x for x_1, the condition becomes $|x - p/q| \ge c/q^2$ for every fraction p/q with $p \in \mathbb{Z}$ and $q \ge 1$. Let $\mathbf{Bad}(n)$ be the set of badly approximable vectors of $\mathbb{R}^n$. By Dirichlet's theorem (1842), every $\mathbf{x} \in \mathbb{R}^n$ satisfies $\max_i \|q x_i\| < q^{-1/n}$ for infinitely many integers $q \ge 1$. Badly approximable vectors are those for which this bound cannot be improved by more than a constant factor: they are the most stubborn vectors there are. How big is $\mathbf{Bad}(n)$? In particular, is the intersection of countably many translates of $\mathbf{Bad}(n)$ still big?

8.1. **History of the problem and of the solution using topological games.** For $n = 1$ there is a classical description: an irrational number is badly approximable exactly when the partial quotients in its continued fraction expansion are bounded. As Hurwitz showed in 1891, the golden ratio $(1+\sqrt{5})/2$ is as badly approximable as a number can be; its continued fraction expansion $1 + 1/(1 + 1/(1 + \cdots))$ has every partial quotient equal to 1, the smallest possible value. Badly approximable numbers are at once rare and common. Borel showed in 1909, by an argument that Bernstein completed in 1912, that $\mathbf{Bad}(1)$ has Lebesgue measure zero, so a number chosen at random is almost surely *not* badly approximable. Jarník proved in 1928 that $\mathbf{Bad}(1)$ has Hausdorff dimension 1, the largest possible. The set $\mathbf{Bad}(1)$ is thus tiny for measure and huge for dimension.

Neither notion helps with intersections. Measure sees only that $\mathbf{Bad}(1)$ is a null set, and two sets of full Hausdorff dimension can be disjoint: the set $\mathbf{Bad}(1)$ and its complement are an example, since the complement has full measure and hence full dimension. A question as innocent as whether there is a number x such that $x + \sqrt{p}$ is badly approximable for every prime p was therefore out of reach.

Wolfgang Schmidt solved the problem in 1966 [Sch66]. To do so he invented a new version of the Banach–Mazur game, now called Schmidt's game. He proved that its winning sets have full Hausdorff dimension and survive countable intersections, and that $\mathbf{Bad}(1)$ is one of them. In 1969 he extended the last result to $\mathbf{Bad}(n)$, for every n [Sch69].

The game had more to give. For real weights $i, j > 0$ with $i + j = 1$, let $\mathbf{Bad}(i, j)$ be the set of vectors $(x, y) \in \mathbb{R}^2$ that are badly approximable with weights (i, j), that is, for which there is $c > 0$ with $\max(\|qx\|^{1/i}, \|qy\|^{1/j}) \ge c/q$ for every integer $q \ge 1$. In 1983 Schmidt conjectured that $\mathbf{Bad}(\frac{1}{3}, \frac{2}{3})$ and $\mathbf{Bad}(\frac{2}{3}, \frac{1}{3})$ have a point in common. He observed that a counterexample to his conjecture would prove the Littlewood conjecture, which asserts that $\liminf_{q \to \infty} q\, \|qx\|\, \|qy\| = 0$ for all real numbers x and y. Badziahin, Pollington and Velani proved Schmidt's conjecture in 2011 [BPV11]. In 2016 An showed that the sets $\mathbf{Bad}(i, j)$ are all winning for Schmidt's game, with one and the same parameter for every pair of weights [An16]; the intersection property of winning sets then yields Schmidt's conjecture in a stronger form, namely that every countable intersection of sets $\mathbf{Bad}(i, j)$ has full dimension.

Recognition. Schmidt received the Cole Prize in Number Theory in 1972. He addressed the International Congress of Mathematicians three times: in Nice in 1970, in Vancouver in

1974 (a plenary address) and in Warsaw in 1983. The paper of Badziahin, Pollington and Velani that settled Schmidt's conjecture appeared in the *Annals of Mathematics*.

8.2. **The game.** Schmidt's game is the Banach–Mazur game of Section 3 with one new rule: the sizes of the moves are prescribed in advance. Let (X, d) be a complete metric space, let $S \subseteq X$ be given, and fix two numbers $\alpha, \beta \in (0,1)$, the shrinking factors of Player II and Player I, respectively. The game is played with closed balls $B[x,r] = \{y \in X : d(x,y) \le r\}$ of prescribed radii, where $x \in X$ and $r > 0$. (In a general metric space a ball, as a set, need not determine its center and radius, and Schmidt formulates the game with the pairs (x,r); in $\mathbb{R}^n$, the only space used below, the distinction disappears.) Schmidt called the players Black and White: Black, who moves first, is our Player I, and White is our Player II. In Choquet's terminology our Player I is β and our Player II is α; in Choquet's games, as in Schmidt's, the letter α thus belongs to Player II. The resemblance ends there: here α and β are shrinking factors, and the α-winning sets defined below have nothing to do with the α-favorable spaces of Section 4.

(1) Player I chooses a closed ball $U_0 = B[x_0, r_0]$ with any radius $r_0 > 0$ he likes.
(2) Player II chooses a closed ball $V_0 \subseteq U_0$ of radius αr_0.
(3) Player I chooses a closed ball $U_1 \subseteq V_0$ of radius $\beta\alpha r_0$.
(4) The players continue in this way, and the play produces nested balls $U_0 \supseteq V_0 \supseteq U_1 \supseteq V_1 \supseteq \cdots$, where, for every $k \ge 0$, the ball U_k has radius $r_k = (\alpha\beta)^k r_0$ and the ball V_k has radius αr_k.

Since the radii tend to 0 and the space is complete, the balls close down on a single point x_∞ (as in the proof of Theorem 4.4). Player II wins if $x_\infty \in S$. We call this game the (α,β)-*game for* S. The set S is α-*winning* if Player II has a winning strategy in the (α,β)-game for S for every $\beta \in (0,1)$. The order of the quantifiers matters. Player II declares her handicap α first, promising to shrink by the factor α every time, and only then does Player I pick β, as aggressive as he likes. Her strategy may depend on β, but the number α is fixed once and for all. This order of quantifiers is what lets α-winning sets survive countable intersections (see the proof of Theorem 8.1(2) below). On the real line the balls are just closed intervals, and we speak of lengths instead of radii (see Figure 15).

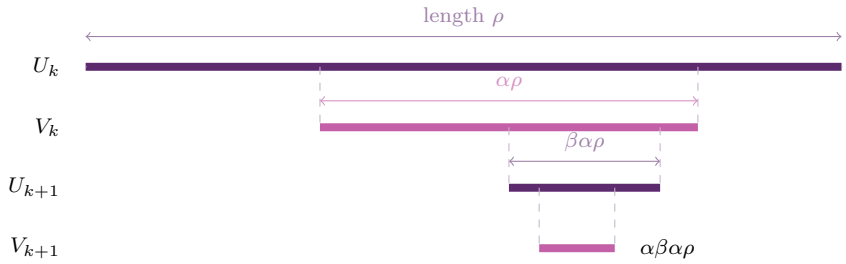


FIGURE 15. Rounds k and $k+1$ of Schmidt's (α,β)-game on the line, for some $k \ge 0$, drawn with $\alpha = \frac{1}{2}$ and $\beta = \frac{2}{5}$; Player I's interval U_k has length ρ. Player I (plum) and Player II (pink) take turns choosing closed intervals, each inside the previous one, and the lengths are prescribed: Player II shrinks by the factor α, Player I by the factor β. Only the positions of the intervals are up to the players.

Because the sizes of the moves are prescribed, neither player can make the balls arbitrarily small in a single move, and this small change lets the game detect Hausdorff dimension (see

the proof of Theorem 8.1(1) below). The game also parts company with category. The set **Bad**(1) is $\frac{1}{2}$-winning (see Theorem 8.1(3) below), and yet it is meager: for each $c > 0$, the numbers x with $|x - p/q| \geq c/q^2$ for every fraction p/q form a closed set that contains no rational number, hence a nowhere dense set, and the union of these sets over $c = 1, \frac{1}{2}, \frac{1}{3}, \ldots$ is **Bad**(1).

8.3. **The Main Result of this section.** Schmidt proved three things about α-winning sets: they are large, they survive countable intersections, and the badly approximable vectors form one of them.

Theorem 8.1 (Schmidt, 1966 and 1969). *Let n be a positive integer.*

(1) *If $0 < \alpha < 1$ and $S \subseteq \mathbb{R}^n$ is α-winning, then the intersection of S with every non-empty open set $U \subseteq \mathbb{R}^n$ has Hausdorff dimension n.*
(2) *For every $\alpha \in (0, 1)$, the intersection of countably many α-winning subsets of $\mathbb{R}^n$ is α-winning.*
(3) *The set **Bad**(1) is $\frac{1}{2}$-winning, and there is $\alpha \in (0, 1)$ such that **Bad**(n) is α-winning.*

Together, the three parts answer both questions of the problem. Let $\alpha \in (0, 1)$ be a number for which **Bad**(n) is α-winning, as in part (3). A translate of an α-winning set is α-winning (see Exercise 2 at the end of this section). By part (2), the set $\bigcap_k (\mathbf{Bad}(n) + \mathbf{t}_k)$ is therefore α-winning for every sequence $(\mathbf{t}_k)_{k \geq 1}$ of vectors of $\mathbb{R}^n$, and by part (1) this intersection has full dimension in every non-empty open set. In particular, every non-empty open interval contains a set of Hausdorff dimension 1 of numbers x such that $x + \sqrt{p}$ is badly approximable for every prime p.

8.4. **Solving the problem with the game.** We go through the three parts of Theorem 8.1, starting with the one that is pure number theory. For simplicity we prove that **Bad**(1) is α-winning for every $\alpha < \frac{1}{2}$; Schmidt's sharper bookkeeping also handles $\alpha = \frac{1}{2}$ [Sch66].

Badly approximable numbers: the plan. Fix $\alpha \in (0, \frac{1}{2})$ and $\beta \in (0, 1)$, and consider the (α, β)-game for **Bad**(1) on the real line, in which Player I opens with an interval U_0 of length ρ_0. We put $R = 1/(\alpha\beta) > 1$. Every round shrinks the intervals by the factor R, and for every $k \geq 0$ the interval U_k has length $\rho_k = \rho_0 R^{-k}$. We will find a constant $c > 0$ and a strategy for Player II such that the final point x satisfies

$$\left| x - \frac{p}{q} \right| \;\geq\; \frac{c}{q^2} \qquad \text{for every fraction } p/q \text{ with } p \in \mathbb{Z} \text{ and } q \geq 1.$$

It suffices to consider fractions in lowest terms: if $p/q = p'/q'$ with $q \leq q'$, the inequality for p/q implies the one for p'/q', because $c/q^2 \geq c/q'^2$. From now on every fraction is in lowest terms. We call the interval $\left(\frac{p}{q} - \frac{c}{q^2}, \frac{p}{q} + \frac{c}{q^2}\right)$ the *no-go zone* of p/q. Player II's job is to keep the final point out of every no-go zone. There are infinitely many zones, but she has infinitely many moves, and the proof consists in giving each move a manageable amount of work.

Idea 1: fractions with small denominators keep their distance. If $p/q \neq p'/q'$, then

$$\left| \frac{p}{q} - \frac{p'}{q'} \right| = \frac{|pq' - p'q|}{qq'} \;\geq\; \frac{1}{qq'},$$

because the numerator $pq' - p'q$ is a non-zero integer. Two fractions whose denominators are both small therefore cannot be close together, and this little inequality drives the argument.

Idea 2: sort the fractions into generations. Let $A = 1/(3\rho_0)$. We say that p/q belongs to *generation* 0 if $q^2 < A$, and to *generation* $k \geq 1$ if

$$AR^{k-1} \;\leq\; q^2 \;<\; AR^k.$$

Every fraction belongs to exactly one generation, and generation k is the one that Player II will deal with in round k. Grouping the denominators by size in this way keeps the fractions of a single generation far apart: by Idea 1, two different fractions p/q and p'/q' of generation k satisfy $qq' < AR^k$, and they therefore lie at distance more than $1/(AR^k) = 3\rho_0 R^{-k} = 3\rho_k$. If we fatten Player I's interval U_k by ρ_k on each side, we obtain an interval of length $3\rho_k$, which can contain *at most one* fraction of generation k. That fraction, if it exists, is the *dangerous fraction* of round k.

Idea 3: the no-go zones are small. A fraction p/q of generation $k \geq 1$ has $q^2 \geq AR^{k-1}$, so the no-go zone of p/q has length

$$\frac{2c}{q^2} \;\leq\; \frac{2c}{AR^{k-1}} \;=\; 6cR\,\rho_k.$$

We choose $c > 0$ so small that $6cR \leq 1 - 2\alpha$ and $2c \leq (1-2\alpha)\rho_0$ (the second condition takes care of generation 0, whose zones have length at most $2c$). Then every zone of generation k has length at most $(1-2\alpha)\rho_k$, which is less than ρ_k. In particular, a zone of generation k that reaches into U_k has its center within $\rho_k/2$ of U_k; it therefore belongs to a fraction in the fattened interval, that is, to the dangerous fraction.

The strategy. In round k, Player II looks at Player I's interval U_k and at the no-go zone of the dangerous fraction of round k, if there is one. Removing the zone from U_k leaves at most two closed intervals, of total length at least $\rho_k - (1-2\alpha)\rho_k = 2\alpha\rho_k$, and one of them therefore has length at least $\alpha\rho_k$. Player II plays her interval V_k inside that one; if there is no dangerous fraction, she plays any interval of the prescribed length inside U_k. That is the whole strategy.

Proof that the strategy wins. Let p/q be a fraction, and let k be its generation. The final point x lies in $V_k \subseteq U_k$. If the zone of p/q reaches into U_k, the fraction is the dangerous one of round k; the interval V_k avoids its zone, and so does the point x. Otherwise the zone misses U_k altogether, and with it the point x. Either way $|x - p/q| \geq c/q^2$, whence $x \in \mathbf{Bad}(1)$. The constant c depends on α, β and ρ_0, and this dependence is harmless, since the definition of $\mathbf{Bad}(1)$ allows each number a constant of its own.

The game has done exactly what we promised in the Introduction: it has turned a global question into a local one. The condition "$|x - p/q| \geq c/q^2$ for all fractions" involves infinitely many fractions at once; the game splits it into infinitely many small jobs, each of them easy (dodge one interval).

Higher dimensions. For $\mathbf{Bad}(n)$ the same plan works, with Idea 1 replaced by the *simplex lemma*: if rational points $\mathbf{p}_0/q_0, \ldots, \mathbf{p}_n/q_n$ of $\mathbb{R}^n$ (with $\mathbf{p}_i \in \mathbb{Z}^n$ and $q_i \geq 1$) do not lie on a common hyperplane, then the simplex they span has volume at least $1/(n!\,q_0 \cdots q_n)$. (The volume is $1/n!$ times the absolute value of the determinant of the $(n+1) \times (n+1)$ matrix whose i-th row is $(1, \mathbf{p}_i/q_i)$. Multiplying the i-th row by q_i produces an integer matrix, and a non-zero determinant of this kind is therefore at least $1/(q_0 \cdots q_n)$ in absolute value, exactly as in Idea 1.) Consequently, with the generations chosen as in Idea 2 (by the size of $q^{1+1/n}$ instead of q^2, since the no-go zone of $\mathbf{p}/q$ now has radius $c\,q^{-1-1/n}$ in the maximum norm), all the rational points of one generation that lie near Player I's ball of the corresponding round sit on a single hyperplane, and in each round Player II has to dodge a thin slab around one hyperplane instead of one interval.

Full dimension. Suppose that Player II has a winning strategy in the (α, β)-game for a set $S \subseteq \mathbb{R}^n$. Inside Player II's ball V_k, of radius αr_k, Player I has room for roughly β^{-n} disjoint balls of radius $\beta\alpha r_k$, and he may play any of them. Let him try them all, while Player II keeps following her strategy. The result is a tree of plays: every ball has about β^{-n} children, and the radii shrink by the factor $\alpha\beta$ from one level of the tree to the next. The final points of all these plays form a Cantor set, and every point of this Cantor set lies in S, because Player II won every one of those plays. Since the children of each ball can be chosen well separated, a Cantor set built this way has Hausdorff dimension at least roughly

$$\frac{\log(\beta^{-n})}{\log(1/(\alpha\beta))} = \frac{n\log(1/\beta)}{\log(1/\alpha)+\log(1/\beta)}$$

(see Schmidt [Sch66] for the precise bound), and this quantity tends to n as $\beta \to 0$. Since an α-winning set is winning for *every* β, letting $\beta \to 0$ shows that such a set has dimension n. If Player I opens with a ball inside a non-empty open set U, the same construction yields the local version: the set $S \cap U$ has dimension n.

Countable intersections. Let $S_1, S_2, \ldots$ be α-winning subsets of $\mathbb{R}^n$. Player II now shares her time among them. She splits the rounds into infinitely many infinite arithmetic progressions (for instance, the k-th progression may consist of the rounds $2^{k-1}(2m+1)-1$, $m \geq 0$), and in the rounds of the k-th progression she plays her strategy for S_k, ignoring everything else. From the point of view of S_k, everything that happens between two consecutive rounds of the k-th progression (her moves for the other sets and Player I's moves) looks like one very big shrinking move by Player I. If the progression has difference d_k, the game for S_k is thus the (α, β_k)-game with $\beta_k = \beta(\alpha\beta)^{d_k-1}$. Since S_k is α-winning, she has a winning strategy in that game, and the final point, which is the same for every k, lies in every S_k. Her time-sharing strategy is a fugue: each voice enters at a time of its own and follows a line of its own, and in the end all of them sound together. This is the second feature of games described in the Introduction, and it is exactly what the problem needed: Hausdorff dimension has no such device, and measure, which has one, sees only that $\mathbf{Bad}(n)$ is null.

Exercises.

(1) Let n be a positive integer, let $0 < \alpha < 1$ be given, and let $S \subseteq \mathbb{R}^n$ be α-winning. Show that S meets every non-empty open subset of $\mathbb{R}^n$, that is, that S is dense in $\mathbb{R}^n$.

(2) Let n be a positive integer, let $\alpha \in (0,1)$ be given, let $S \subseteq \mathbb{R}^n$ be α-winning, and let a real number $\lambda \neq 0$ and a vector $\mathbf{t} \in \mathbb{R}^n$ be given. Show that the set $\lambda S + \mathbf{t}$ is α-winning. (Hint: the map $\mathbf{x} \mapsto \lambda\mathbf{x} + \mathbf{t}$ carries $B[\mathbf{x}, r]$ onto $B[\lambda\mathbf{x}+\mathbf{t}, |\lambda| r]$; transport the strategy.)

(3) Let $\alpha \in (0,1)$ and $\beta \in (0, \frac{1}{2})$ be given, and let $S \subseteq \mathbb{R}$ be countable. Show, without using Theorem 8.1, that Player I has a winning strategy in the (α, β)-game for S. Conclude that $\mathbb{Q}$ is not α-winning for any $\alpha \in (0,1)$. (Hint: since $2\beta < 1$, a closed interval contains two disjoint closed subintervals of β times its length.)

(4) Let $0 < \alpha < \frac{1}{2}$ be given. Show directly that the set of irrational numbers is α-winning, and more generally that the complement of every countable subset of $\mathbb{R}$ is α-winning.

(5) $(*)$ Let n be a positive integer, let $0 < \alpha' < \alpha < 1$ be given, and let $S \subseteq \mathbb{R}^n$ be α-winning. Show that S is α'-winning. (Hint: given $\beta' \in (0,1)$, put $\beta = \alpha'\beta'/\alpha$, so that $\alpha\beta = \alpha'\beta'$; in the (α', β')-game, Player II answers Player I's ball of radius r_k with a ball of radius $\alpha' r_k$ inside the ball of radius αr_k prescribed by her strategy for the (α, β)-game.)

9. Applications to homogeneous dynamics

The problem. A *lattice* in $\mathbb{R}^m$, for a positive integer m, is a set of the form $g\mathbb{Z}^m$ with g an invertible real $m \times m$ matrix, and the lattice $g\mathbb{Z}^m$ is *unimodular* if $\det g = \pm 1$. The unimodular lattices form a space, which can be identified with the quotient $\mathrm{SL}(m,\mathbb{R})/\mathrm{SL}(m,\mathbb{Z})$ of the group of real $m \times m$ matrices of determinant 1 by its subgroup of integer matrices. Flows on this space move lattices around by stretching some directions and squeezing others, for example the flow

$$g_t = \mathrm{diag}\big(e^t, \dots, e^t, e^{-nt}\big), \qquad t \geq 0,$$

on the unimodular lattices in $\mathbb{R}^{n+1}$, where n is a positive integer (here $m = n+1$) and g_t carries a lattice Λ to the lattice $g_t\Lambda$. For a flow of this kind almost every lattice has a dense orbit $\{g_t\Lambda : t \geq 0\}$, which therefore wanders all over the space, while the lattices whose orbits stay in a bounded region form a set of measure zero. How big is this set of measure zero?

In this section we show that the game of Section 8 is a game about lattices being stretched and squeezed, so a reader of that section already knows the main idea. The words "Lie group", "lattice" and "horospherical subgroup" will appear, but the heart of the matter is a computation with 2×2 matrices acting on the plane, which we do by hand below; the Lie group language is just the coordinate system that lets the same computation run in higher dimensions.

A small glossary. The reader may keep the example $G = \mathrm{SL}(m,\mathbb{R})$, $\Gamma = \mathrm{SL}(m,\mathbb{Z})$ in mind throughout.

- A *Lie group* is a group that is also a smooth manifold on which multiplication and inversion are smooth, such as $\mathrm{SL}(m,\mathbb{R})$.
- A *lattice* Γ in a Lie group G is a discrete subgroup such that the quotient G/Γ has finite volume. The word is used in two senses here: a lattice in $\mathbb{R}^m$ is a grid of points (a lattice, in the sense just defined, in the group $\mathbb{R}^m$), and a lattice in G is a subgroup of matrices. The two are linked by the identification of the coset $g\,\mathrm{SL}(m,\mathbb{Z})$ with the grid $g\mathbb{Z}^m$, under which $\mathrm{SL}(m,\mathbb{R})/\mathrm{SL}(m,\mathbb{Z})$ becomes the space of unimodular grids.
- "Almost every" and "measure zero" refer to the natural measure on G/Γ induced by the Haar measure of G, which the flow preserves.
- A set of points of G/Γ is *bounded* if its closure is compact; for grids, boundedness is made concrete by Mahler's criterion in the history below.
- A flow g_t is *quasi-unipotent* if conjugation by g_t stretches no direction of G exponentially, that is, if all the eigenvalues of the linear maps that this conjugation induces on the tangent space of G at the identity (the Lie algebra of G) have absolute value 1; for $G = \mathrm{SL}(m,\mathbb{R})$ this happens exactly when all the eigenvalues of the matrices g_t have absolute value 1.
- A lattice Γ in G is *irreducible* if no subgroup of finite index in Γ is a product of lattices in two non-trivial factors of G; in a simple group such as $\mathrm{SL}(m,\mathbb{R})$, every lattice is irreducible.

Other terms (nilpotent, semisimple, mixing) appear only where we describe the general case, and a reader may pass over them.

9.1. **History of the problem and of the solution using topological games.** The space of lattices is not compact: one can squash a lattice to create a very short non-zero vector, and then squash further. "Going to infinity" in this space is usually called "going into the cusp",

and a lattice deep in the cusp is a very thin needle-like grid with a vector of length nearly zero. By Mahler's compactness criterion (1946), this is the only way to escape to infinity: a set of unimodular lattices is bounded exactly when the lengths of their shortest non-zero vectors are bounded away from zero.

The link with the previous section was found by Dani. He showed in 1985 [Dan85] that badly approximable vectors correspond exactly to bounded orbits of the flow above. In 1986 he used Schmidt's games to show that bounded orbits are abundant for several families of flows [Dan86]. In his plenary address at the International Congress of Mathematicians in Kyoto in 1990 [Mar91], Margulis conjectured that this abundance is universal: for every one-parameter flow that is not quasi-unipotent (roughly, every flow that really stretches something), the set of points with bounded orbits is *thick*, that is, it meets every non-empty open set in a set of full Hausdorff dimension.

Kleinbock and Margulis proved the conjecture in 1996 [KM96]. They worked on the expanding leaves of the flow, but their argument rests on the mixing of the flow, and it produces sets of full dimension rather than winning strategies. Kleinbock and Weiss introduced *modified Schmidt games*, adapted to flows that stretch different directions at different rates. In 2013 they proved that sets of bounded orbits are winning for these games [KW13], a result that adds the countable intersection property to thickness. In 2018 An, Beresnevich and Velani turned to the sets $\mathbf{Bad}(i,j)$ of Section 8, of vectors badly approximable with weights (i,j). They established that on every nondegenerate planar curve, that is, every twice continuously differentiable curve in the plane whose curvature vanishes nowhere, the points of $\mathbf{Bad}(i,j)$ form a winning subset of the curve for Schmidt's game [ABV18].

Recognition. Margulis received the Fields Medal in 1978, the Wolf Prize in 2005 and the Abel Prize in 2020. Dani received the Shanti Swarup Bhatnagar Prize in 1990. He was an invited speaker at the International Congress of Mathematicians in Zürich in 1994. The dynamical approach to Diophantine approximation that grew out of this work led Kleinbock and Margulis to a proof of the Baker–Sprindžuk conjectures, published in the *Annals of Mathematics* in 1998 [KM98]; that proof uses non-divergence estimates, not games.

9.2. **The game.** The game is Schmidt's game of Section 8, played not on the whole space but on an *expanding leaf* of a flow g_t on G/Γ. The expanding horospherical subgroup H consists of the elements h of G with $g_{-t}hg_t \to e$ as $t \to \infty$, where e is the identity of G; these are the directions that the flow stretches. Distances in these directions grow exponentially, and a tiny ball soon becomes a ball of unit size; this growth lets Player II deal with the problem one scale at a time (see the dictionary between scales and times below). The group H is a nilpotent Lie group with a natural metric, and the expanding leaf through a point z of G/Γ is the orbit Hz, the image of H under the map $h \mapsto hz$, which need not be injective. The players play Schmidt's game with closed balls in H, and Player II wins if the final point h is such that hz has a bounded forward orbit. For the diagonal flow above, the subgroup H consists of the matrices $\left(\begin{smallmatrix} I_n & \mathbf{y} \\ 0 & 1 \end{smallmatrix}\right)$, and the leaf through $\mathbb{Z}^{n+1}$ is the set of lattices $\Lambda_{\mathbf{y}} = \left(\begin{smallmatrix} I_n & \mathbf{y} \\ 0 & 1 \end{smallmatrix}\right)\mathbb{Z}^{n+1}$ with $\mathbf{y} \in \mathbb{R}^n$; the game on this leaf is therefore literally Schmidt's game on $\mathbb{R}^n$.

9.3. **The Main Result of this section.** The main result has two parts: a dictionary between numbers and orbits, due to Dani, and the theorem of Kleinbock and Margulis.

Theorem 9.1 (Dani, 1985; Kleinbock–Margulis, 1996).

(1) Dani's correspondence. *Let n be a positive integer, let $g_t = \operatorname{diag}(e^t, \dots, e^t, e^{-nt})$ for $t \geq 0$, and for $\mathbf{x} \in \mathbb{R}^n$ let $\Lambda_{\mathbf{x}} = \left(\begin{smallmatrix} I_n & \mathbf{x} \\ 0 & 1 \end{smallmatrix}\right) \mathbb{Z}^{n+1}$. A vector $\mathbf{x} \in \mathbb{R}^n$ lies in the set $\mathbf{Bad}(n)$ of badly approximable vectors if and only if the forward orbit $\{g_t \Lambda_{\mathbf{x}} : t \geq 0\}$ is bounded in the space of unimodular lattices in $\mathbb{R}^{n+1}$.*

(2) Thickness of bounded orbits. *For every non-quasi-unipotent one-parameter subgroup of a connected semisimple Lie group G without compact factors, and every irreducible lattice Γ in G, the set of points of G/Γ with bounded orbits is thick, that is, it meets every non-empty open subset of G/Γ in a set of full Hausdorff dimension.*

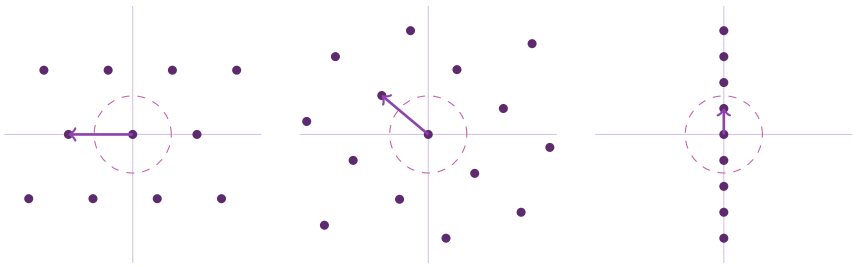
By part (1), Schmidt's theorem of Section 8 is a theorem about one particular flow: the vectors $\mathbf{x}$ whose lattices $\Lambda_{\mathbf{x}}$ have bounded forward orbits form the set $\mathbf{Bad}(n)$, which is α-winning for some $\alpha \in (0, 1)$. Part (2) extends the abundance of bounded orbits to every flow that stretches; for it the game serves as a plan rather than as a proof, as we explain in the paragraph on the general case below.

9.4. **Solving the problem with the game.** We first follow the whole mechanism in the simplest case $n = 1$, where every step can be computed by hand, and then explain what changes in general.

Lattices in the plane. For $n = 1$ we have $g_t = \operatorname{diag}(e^t, e^{-t})$, and for $x \in \mathbb{R}$ we write $u_x = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)$; then

$$\Lambda_x = u_x \mathbb{Z}^2 = \{(p + qx,\ q) : p, q \in \mathbb{Z}\}, \qquad g_t \Lambda_x = \{(e^t(p + qx),\ e^{-t} q) : p, q \in \mathbb{Z}\}.$$

The flow stretches the first coordinate and squeezes the second, and the *product* of the two coordinates of a vector, namely $q(p + qx)$, does not change along the orbit. The whole argument rests on this invariant product (see Figure 16).



(a) $x = (\sqrt{5} - 1)/2$, $t = 0$ (b) $x = (\sqrt{5} - 1)/2$, $t = 1.6$ (c) $x = 1/2$, $t = 1.6$

FIGURE 16. The grid $g_t \Lambda_x = \{(e^t(p + qx),\ e^{-t} q) : p, q \in \mathbb{Z}\}$ near the origin, with a shortest non-zero vector drawn as an arrow and the circle of radius 0.6 about the origin drawn dashed. The flow $g_t = \operatorname{diag}(e^t, e^{-t})$ stretches the grid horizontally by e^t and squeezes it vertically by e^{-t}. For the badly approximable number $x = (\sqrt{5} - 1)/2$, in (a) and (b), the grid is sheared but no non-zero point ever comes close to the origin, so the orbit stays bounded. For the rational number $x = 1/2$, in (c), the points $(0, 2ke^{-t})$, which come from $(p, q) = (-k, 2k)$ with $k \in \mathbb{Z}$, crowd toward the origin, and the orbit goes into the cusp.

Dani's correspondence, by hand. Suppose first that $x \in \mathbf{Bad}(1)$: there is a constant $c > 0$ with $|p + qx| \geq c/|q|$ for all integers p and q with $q \neq 0$. Let $(e^t(p + qx), e^{-t}q)$ be a non-zero vector of $g_t\Lambda_x$, with $t \geq 0$. If $q \neq 0$, the product of the coordinates of this vector has absolute value $|q|\,|p + qx| \geq c$, so the two coordinates cannot both be smaller than $\sqrt{c}$, and the vector has length at least $\sqrt{c}$. If $q = 0$, the vector is $(e^t p, 0)$ with $p \neq 0$, of length at least $e^t \geq 1$. Hence no lattice on the forward orbit has a non-zero vector shorter than $\min(1, \sqrt{c})$, and by Mahler's criterion the orbit is bounded.

Conversely, suppose that $x \notin \mathbf{Bad}(1)$. Then for every $\varepsilon \in (0, 1)$ there are integers p and $q \geq 1$ with $|p + qx| < \varepsilon/q$. Let t be the time with $e^{-t}q = \sqrt{\varepsilon}$, which is positive because $q \geq 1 > \sqrt{\varepsilon}$. At that time the vector coming from (p, q) has second coordinate $\sqrt{\varepsilon}$ and first coordinate of absolute value $e^t|p + qx| < (q/\sqrt{\varepsilon})(\varepsilon/q) = \sqrt{\varepsilon}$, and it is therefore shorter than $\sqrt{2\varepsilon}$. Since ε is arbitrary, the orbit dives deeper and deeper into the cusp, and it is unbounded by Mahler's criterion.

A good rational approximation $-p/q$ of x is thus *the same thing* as an excursion of the orbit into the cusp at a time $t \approx \log q$.

The dictionary between scales and times. A one-line computation gives

$$g_t\, u_x\, g_{-t} = u_{e^{2t}x},$$

so for every unimodular lattice Λ the flow carries the expanding leaf $\{u_x\Lambda : x \in \mathbb{R}\}$ onto the leaf through $g_t\Lambda$, stretching it by the factor e^{2t}. An interval of length ρ in the leaf is stretched to length about 1 at the time t with $e^{2t} \approx 1/\rho$. We now return to the proof in Section 8 that $\mathbf{Bad}(1)$ is α-winning, in which every round shrinks the intervals by the factor $R = 1/(\alpha\beta)$ and the fractions are sorted into generations by the size of their denominators. In round k, Player I's interval has length $\rho_k = \rho_0 R^{-k}$, where ρ_0 is the length of his first interval, and the fractions p/q Player II worries about are those of generation k, with q^2 within a constant factor of R^k; by Dani's correspondence as worked out above, these are, up to constant factors, the fractions that send the orbit into the cusp at the time t with $e^{2t} \approx R^k \approx 1/\rho_k$. In other words, *round k of the game corresponds to the time at which the current interval has been stretched to unit size, and in that round Player II dodges the points whose orbits visit the cusp at that time.* Read in this dictionary, Schmidt's proof is a proof about orbits, and the formula $g_t u_x g_{-t} = u_{e^{2t}x}$ translates each step of one proof into the corresponding step of the other.

The general case. Kleinbock and Margulis follow the same plan for an arbitrary flow that stretches, working on an expanding leaf Hz. At each scale they look at a ball in the leaf, wait (in their heads) until the flow has stretched that ball to unit size, and determine the points of the ball that are deep in the cusp at that time. For the diagonal flow on the plane this bad set was a single short interval, the no-go zone of one fraction. In general there is no single dangerous fraction, and here lies the real work: using the mixing of the flow, Kleinbock and Margulis show [KM96] that the bad set occupies only a small proportion of the ball. Removing the bad set at every scale leaves a Cantor-like set whose points never go far into the cusp, and therefore have bounded orbits, and whose dimension is close to that of the leaf. The set of points with bounded orbits thus has full dimension on every expanding leaf. Their argument does not produce a winning strategy, since a set that occupies a small proportion of a ball may still meet every ball of the prescribed size inside it; the winning property came later, with the modified games of Kleinbock and Weiss (see the history above).

From leaves to the whole space. Roughly, near any point the space looks like a product of a piece of an expanding leaf with the directions that the flow does not stretch, and whether an orbit stays bounded depends only mildly on those other directions. Full dimension on

every expanding leaf therefore yields full dimension in the whole space, that is, thickness (see Kleinbock and Margulis [KM96]).

Exercises.

(1) Let $x \in \mathbb{R}$ and $t \geq 0$ be given, and let g_t, u_x and Λ_x be as in the case $n = 1$ above. Check by multiplying matrices that $g_t u_x g_{-t} = u_{e^{2t}x}$. Show that Λ_x is unimodular, and that the product of the two coordinates of a vector of $g_t\Lambda_x$ coming from (p, q) does not depend on t.

(2) Let $x \in \mathbb{R}$ be given. Show that $\Lambda_{x+1} = \Lambda_x$. Explain why this is consistent with Theorem 9.1(1), by checking directly that $x \in \mathbf{Bad}(1)$ if and only if $x + 1 \in \mathbf{Bad}(1)$.

(3) Let $x = a/b$, with integers a and $b \geq 1$. Find a non-zero vector of $g_t\Lambda_x$ of length be^{-t}, and deduce that the orbit $\{g_t\Lambda_x : t \geq 0\}$ is unbounded. Check that this agrees with Theorem 9.1(1), by showing that no rational number is badly approximable.

(4) $(*)$ Let $\varphi = (1+\sqrt{5})/2$ and $\varphi' = (1-\sqrt{5})/2$. Show that $(p+q\varphi)(p+q\varphi') = p^2+pq-q^2$ for all integers p, q, and deduce that $|q|\,|p+q\varphi| \geq 1/(1+\sqrt{5})$ whenever $q \neq 0$. Conclude that, for every $t \geq 0$, every non-zero vector of $g_t\Lambda_\varphi$ has length at least $(1 + \sqrt{5})^{-1/2}$.

10. Applications to computational complexity

The problem. A *language* is a set L of finite binary strings, and a complexity class is a set of languages, such as the class P of languages decidable in polynomial time, the class E = DTIME($2^{O(n)}$) of languages decidable by a deterministic Turing machine in time $2^{O(n)}$ on inputs of length n, or the class ESPACE = DSPACE($2^{O(n)}$) of languages decidable by such a machine in space $2^{O(n)}$. We would like to say that "most" languages in E are hard, or that "most" languages in ESPACE need big circuits. Ordinary Baire category and Lebesgue measure on the space of all languages are useless for this: every complexity class is countable, hence meager and of measure zero, and therefore P and E look equally small. The problem is to find a notion of "small" that works *inside* a complexity class.

10.1. **History of the problem and of the solution using topological games.** The problem was solved by Kurt Mehlhorn in 1973 [Meh73], who introduced resource-bounded Baire category, in which the descriptions of meager sets must themselves be computable with limited resources. Jack Lutz reformulated the theory in terms of the Banach–Mazur game played with resource-bounded strategies, in his thesis of 1987 and in a paper published in 1990 [Lut90]. Among other results, he proved that languages with high circuit complexity are typical in ESPACE in this sense. He also introduced the companion notion of resource-bounded measure, which he developed in a paper of 1992 [Lut92]. Both notions became standard tools for quantitative questions about complexity classes.

Recognition. Mehlhorn received the Leibniz Prize of the Deutsche Forschungsgemeinschaft in 1986.

10.2. **The game.** We list all binary strings in the standard order, namely $\lambda, 0, 1, 00, 01, 10, 11, 000, \ldots$, where λ is the empty string (shorter strings first, and strings of the same length in dictionary order). A language L is then the same thing as an infinite 0–1 sequence, the *characteristic sequence* of L, whose N-th bit says whether the N-th string belongs to L. A finite 0–1 string w is a partial description of a language, since w decides membership for the first $|w|$ strings, where $|w|$ is the length of w. The 2^n strings of length n come after the $2^n - 1$ shorter ones (see Exercise 1 below), so a description that decides all strings of length less than n has

length at least $2^n - 1$. This exponential gap is what turns time polynomial in the length of a description into time $2^{O(n)}$ on inputs of length n.

Let $\mathcal{C}$ be a class of languages. The game for $\mathcal{C}$ is played as follows.

(1) The players alternately extend finite binary strings, $w_0 \sqsubset w_1 \sqsubset w_2 \sqsubset \cdots$, where $w \sqsubset w'$ means that w is a proper initial segment of w', with Player I playing w_{2k} and Player II playing w_{2k+1} for $k \geq 0$.
(2) The play produces the language L whose characteristic sequence is $\bigcup_k w_k$.
(3) Player II wins if $L \notin \mathcal{C}$.

This is the Banach–Mazur game on the Cantor space of infinite 0–1 sequences, in which the move w stands for the basic open set N_w of languages whose characteristic sequence begins with w. By the theorem of Banach and Oxtoby, Player II has a winning strategy exactly when $\mathcal{C}$ is meager (see Theorem 3.5; its proof works just as well when the moves are restricted to basic open sets). The resource-bounded version adds one rule: *Player II is not allowed to think for too long.* Her strategy must compute each move in time polynomial in the length of the current string (or in space polynomial in that length, when we work in ESPACE). Player I has no restrictions at all. A class is *meager in* E if Player II has a winning strategy that obeys the time bound, and *meager in ESPACE* if she has one that obeys the space bound.

10.3. **The Main Result of this section.** A notion of smallness is useful only if it does not declare everything small. The theory rests on a resource-bounded version of the Baire category theorem, part (1) of Theorem 10.1 below, which we prove in the next subsection. Since E is not meager in itself, its meager subclasses are genuinely small, and the notion can tell its subclasses apart; the same holds for ESPACE.

Theorem 10.1 (Mehlhorn, 1973; Lutz, 1990). *Let* P, E *and* ESPACE *be the classes of languages decidable in polynomial time, in time* $2^{O(n)}$ *and in space* $2^{O(n)}$ *on inputs of length* n*, respectively, and let meagerness be understood in the resource-bounded sense of the game above.*

(1) Baire category theorem. *The class* E *is not meager in itself.*
(2) *The class* P *is meager in* E.
(3) *The class* SIZE$(2^n/n)$ *of languages that, for all sufficiently large* n*, are decided on inputs of length* n *by Boolean circuits with at most* $2^n/n$ *gates is meager in* ESPACE.

By part (3), big circuit complexity is typical in ESPACE. Lutz also proved the measure version: in resource-bounded measure, almost every language in ESPACE has circuit-size complexity above $2^n/n$ [Lut92]. Here category and measure agree, although, as on the real line, they can disagree in general (see Remark 1.2(2)).

10.4. **Solving the problem with the game.** The game formulation makes the definition of category effective in the most natural way imaginable: one simply limits the resources of the winning player. The proofs then become constructions of efficient strategies, and they are pleasantly concrete.

The Baire category theorem: the class E **is not meager in itself.** The idea is that Player I can play so predictably that the whole play is computable from Player II's strategy. Suppose that Player II has a polynomial-time strategy. We show that Player I can make the play produce a language in E, whence the strategy does not win the game for the class E.

His strategy is the simplest possible: at each of his moves he appends a single 0. The whole play is then computable: to decide whether the N-th string is in L, we run the game until the current string has length at least N and read off bit N. Each round adds at least one bit, and therefore at most N rounds are needed. Each of Player II's moves is made from a string of length at most N, so it takes time polynomial in N; hence bit N is computed in time polynomial in N. Since the strings of length n sit at positions $N \approx 2^n$, membership of a string of length n is decided in time polynomial in 2^n, that is, in time $2^{O(n)}$. Thus $L \in \mathrm{E}$, and the class E is not meager in itself.

The class P is meager in E: diagonalization, one move at a time. Let $M_1, M_2, \ldots$ be a list of Turing machines such that M_i halts within $n^i + i$ steps on inputs of length n and every language in P is decided by some M_i. Player II wants to make sure that L disagrees with every M_i on at least one string, and each of her moves takes care of a few more machines. Let w be the current string, and let s be the least length such that w decides no string of length s; then the string w decides only strings of length less than s, so $|w| < 2^s$, and all 2^s strings of length s are still free. For each positive integer $i \le \log s$ (here and below, log is the logarithm to base 2), Player II picks the i-th string x_i of length s, runs M_i on x_i, and extends w so that

$$x_i \in L \iff M_i \text{ rejects } x_i,$$

filling all the other new positions with 0. Each simulation takes at most $s^{\log s} + \log s$ steps, which is less than 2^s once s is large, and therefore the whole move takes time polynomial in 2^s. (Here $|w| \ge 2^{s-1} - 1$, because w decides all strings of length less than $s - 1$; a bound polynomial in 2^s is thus polynomial in $|w|$, as the rules require.) Player I may then extend the string as much as he likes, but he cannot undo what she wrote. Once $s \ge 2$, each of her moves decides a string of length s, whence s grows from each of her moves to the next. Since Player II moves infinitely often, every M_i is eventually defeated on some string, and $L \notin \mathrm{P}$ follows.

The one clever point is the restriction $i \le \log s$. Player II cannot afford to simulate M_i on short strings when i is large: the s^i steps of the simulation could exceed her time budget, which is a polynomial in 2^s fixed once and for all. To defeat M_{1000} on strings of length 10, for instance, she might need 10^{1000} steps while writing about 2^{10} bits, and a strategy that did this for every i at lengths that grow too slowly would obey no polynomial bound at all. She waits instead until $s \ge 2^{1000}$, when the strings are long enough that simulating M_{1000} costs little relative to $|w|$. This "diagonalization with delay" is a standard trick of complexity theory, and the game makes it look natural.

Circuit size: a counting argument, one length at a time. Here the resource bound is space instead of time, and the key fact is Shannon's counting argument (1949): there are 2^{2^n} Boolean functions of n variables, one for each truth table. A circuit, whose size is its number of gates, can be described by listing the operation and the two inputs of each gate, and a careful count (taking into account that renaming the gates does not change the circuit) shows that for large n there are fewer circuits of size at most $2^n/n$ than there are truth tables. Hence some Boolean function of n variables has no circuit of size at most $2^n/n$.

Player II turns this count into a strategy. At each of her moves she takes the smallest length n such that no string of length n has been decided yet. If n is too small for the count above, she extends the current string with 0s until it decides every string of length n. Otherwise she finds, by brute force, the first truth table on n variables that has no circuit of size at most $2^n/n$: she goes through the truth tables one by one, and for each of them she checks every circuit of size at most $2^n/n$ to see whether it computes that truth table.

She then fills the remaining positions before the strings of length n with 0, and defines L on the 2^n strings of length n to be that truth table. The search takes an astronomical amount of *time*, but very little *space*: at any moment she needs to remember only one truth table and one circuit, which takes space polynomial in 2^n, hence polynomial in the length of the current string. Space can be reused and time cannot, and Player II is charged only for space. Since she does this for infinitely many n, the final language has no circuits of size at most $2^n/n$ at infinitely many lengths, and therefore $L \notin \mathrm{SIZE}(2^n/n)$.

Exercises.

(1) Number the binary strings in the standard order starting from 0, with the empty string λ as string number 0. Show that, for every integer $n \geq 0$, the strings of length n are those numbered $2^n - 1$ through $2^{n+1} - 2$, and that a string w of length n has number $m - 1$, where m is the number whose binary expansion is the string obtained by writing 1 in front of w. Deduce that a finite 0–1 string decides all strings of length less than n exactly when its length is at least $2^n - 1$.

(2) Consider the game of this section without resource bounds. Show that Player II has a winning strategy for every countable class $\mathcal{C}$, and that Player I has a winning strategy for the class of languages containing the empty string λ.

(3) Fill in the proof that E is not meager in itself. Let p be an increasing polynomial such that Player II computes her answer to a string w in time at most $p(|w|)$, and let Player I always append a single 0. Show that bit N of the resulting language can be computed in time polynomial in N. (Hint: every move that has to be simulated is made from a string of length at most N.)

(4) Show that $s^{\log_2 s} + \log_2 s < 2^s$ for every integer $s \geq 17$, but not for $s = 16$. Explain why the proof that P is meager in E needs such an inequality only for large s.

(5) $(*)$ A circuit with s gates on n variables is a list of s gates, each labeled AND, OR or NOT and given two inputs among the variables and the earlier gates (a NOT gate ignores its second input), the last gate being the output. Show that there are at most $s\,(3(n+s)^2)^s$ circuits with between 1 and s gates, and deduce that, for all sufficiently large n, some Boolean function of n variables has no circuit with at most $2^n/(10n)$ gates.

11. Applications to topological groups

The problem. A *semitopological group* is a group with a topology in which multiplication is *separately* continuous: for every element a of the group, the translations $x \mapsto xa$ and $x \mapsto ax$ are continuous. A semitopological group is a *topological group* if multiplication is jointly continuous and inversion is continuous. Separate continuity is much weaker than joint continuity (the function $f(x, y) = xy/(x^2 + y^2)$, with $f(0, 0) = 0$, is continuous in each variable separately but not jointly continuous at the origin; see Figure 17). When does separate continuity of multiplication force a semitopological group to be a topological group? Behind this lies a more general question: when is a separately continuous real-valued function on a product $X \times Y$ of topological spaces jointly continuous at many points?

In this section we show that the answer is a matter of completeness, and that the right amount of completeness is measured by a game of Banach–Mazur type in which Player II must meet a more demanding winning condition.

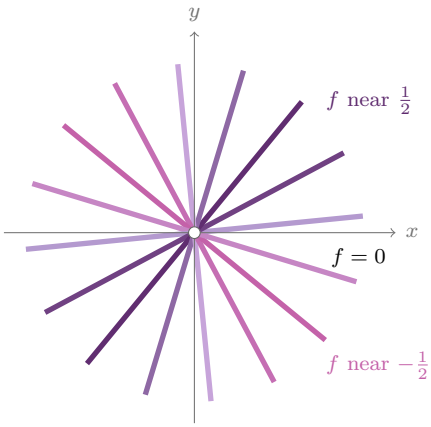


FIGURE 17. The function $f(x,y) = xy/(x^2+y^2)$, with $f(0,0) = 0$, is constant on each ray from the origin: on the ray at angle θ to the positive x-axis it equals $\frac{1}{2}\sin 2\theta$; the color shows the value. The function is continuous away from the origin and vanishes on the axes; hence it is continuous in each variable separately. In every neighborhood of the origin, the function takes all the values between $-\frac{1}{2}$ and $\frac{1}{2}$; hence it is not jointly continuous there.

11.1. **History of the problem and of the solution using topological games.** Baire proved in his thesis of 1899 that a separately continuous function on $\mathbb{R}^2$ is jointly continuous at the points of a dense G_δ set. In 1957 Ellis proved that every locally compact Hausdorff semitopological group is a topological group [Ell57], and in 1974 Namioka proved a general joint continuity theorem for functions on $X \times K$, with X complete in a suitable sense and K compact, from which he deduced Ellis's theorem [Nam74].

The game-theoretic solution came in three steps. In 1981 Christensen replaced the completeness hypothesis on X in Namioka's theorem by the existence of a winning strategy in a topological game [Chr81]. In 1983 Saint-Raymond showed that the weaker hypothesis that the opposing player has no winning strategy suffices [SR83]. He deduced that the conclusion of Namioka's theorem holds whenever X is a metrizable Baire space. The third step concerns groups. Bouziad proved in 1996 [Bou96] that every Čech-analytic Baire semitopological group is a topological group; the Čech-analytic spaces form a class that contains the Čech-complete spaces of Subsection 4.1. In 2001 Kenderov, Kortezov and Moors introduced a game that describes a class of spaces, the strongly Baire spaces, on which every semitopological group is a topological group [KKM01].

11.2. **The game.** Let X be a topological space, and let $D \subseteq X$ be a dense set. The game $\mathrm{SB}(D)$ (for "strongly Baire") is due to Kenderov, Kortezov and Moors, who call the players β and α; here β becomes Player I and α becomes Player II.

(1) Player I chooses a non-empty open set $U_1 \subseteq X$.
(2) Player II chooses a non-empty open set $V_1 \subseteq U_1$.
(3) In general, for $k \geq 1$, Player I chooses a non-empty open $U_{k+1} \subseteq V_k$ and Player II a non-empty open $V_{k+1} \subseteq U_{k+1}$.
(4) Player II wins if $\bigcap_k V_k \neq \emptyset$ and every sequence $(a_k)_{k\geq 1}$ with $a_k \in V_k \cap D$ has a cluster point in X.

A regular space is *strongly Baire* if, for some dense D, Player I has no winning strategy in $\mathrm{SB}(D)$. This game is the Banach–Mazur game of Section 3 with a more demanding winning condition for Player II; the games of Christensen and Saint-Raymond are variants in which Player II also picks points as she goes.

The motivating examples are the Čech-complete spaces of Subsection 4.1: on such a space Player II wins $\mathrm{SB}(X)$ with the strategy of the proof of Proposition 4.8, stripped of the points, and only the condition on cluster points needs a new argument. Let X be Čech-complete, say with $X = \bigcap_k G_k$ for sets G_k open in a compactification bX, and for each move U_k of Player I let $\widetilde{U}_k$ be an open subset of bX with $\widetilde{U}_k \cap X = U_k$. Player II wins with $D = X$ by choosing V_k whose closure in bX lies inside $G_k \cap \widetilde{U}_k$; such a set exists because bX is regular. The closures form a decreasing sequence of non-empty compact sets. Their intersection lies in $\bigcap_k G_k = X$, and it equals $\bigcap_k V_k$ because the closure of V_{k+1} meets X inside $U_{k+1} \subseteq V_k$. Finally, a sequence with $a_k \in V_k$ has a cluster point in bX; since $a_j \in V_k$ for $j \geq k$, this point lies in every closure, hence in $\bigcap_k V_k$.

11.3. **The Main Result of this section.** The main result goes from functions to groups: Namioka's joint continuity theorem, Ellis's theorem on locally compact groups, and the theorem of Kenderov, Kortezov and Moors, which contains Ellis's.

Theorem 11.1 (Ellis, 1957; Namioka, 1974; Kenderov–Kortezov–Moors, 2001).

(1) Namioka's theorem. *Let X be a Čech-complete space, let K be a compact space, and let $f\colon X \times K \to \mathbb{R}$ be separately continuous. Then there is a dense G_δ set $A \subseteq X$ such that f is jointly continuous at every point of $A \times K$.*

(2) Ellis's theorem. *Every locally compact Hausdorff semitopological group is a topological group.*

(3) *Every semitopological group whose underlying space is strongly Baire is a topological group.*

By the argument of the preceding subsection, every Čech-complete space, in particular every locally compact Hausdorff space and every complete metric space, is strongly Baire (a Čech-complete space is Tychonoff by definition, hence regular); hence part (3) contains part (2). Part (1) has a useful reformulation (see Exercise 5 below). Let X, K and f be as in part (1), and let $C(K)$ be the space of continuous real-valued functions on K. For $x \in X$, the function f is jointly continuous at every point of $\{x\} \times K$ exactly when the map $x' \mapsto f(x', \cdot)$ from X to $C(K)$, which is continuous for the topology of pointwise convergence, is continuous at x for the topology of the supremum norm. Namioka's theorem is therefore about upgrading pointwise convergence to uniform convergence, at least at many points.

11.4. **Solving the problem with the game: a sketch.** The arguments of Christensen, of Saint-Raymond and of Kenderov, Kortezov and Moors share a pattern worth learning, of which we give only a sketch; for the details, see the original papers [Chr81, SR83, KKM01]. Suppose, for instance, that a semitopological group with identity e has a strongly Baire underlying space X, and that multiplication is not jointly continuous at (e, e). Then there is a neighborhood W of e such that, for all neighborhoods O_1 and O_2 of e, there are $u \in O_1$ and $v \in O_2$ with $uv \notin W$. Player I uses these pairs to build a strategy: at each move he chooses an open set inside Player II's last move that records a new pair of witnesses, chosen so that separate continuity at any cluster point of the witnesses would be violated. Since X

is strongly Baire, there is a dense set $D \subseteq X$ such that Player I has no winning strategy in $\mathrm{SB}(D)$, and some play of $\mathrm{SB}(D)$ consistent with this strategy is therefore won by Player II. That play provides a cluster point of the witnesses, and separate continuity at that point leads to a contradiction. (The witnesses must be taken from the dense set D, so that the winning condition of the game, which only speaks of sequences from D, applies to them; arranging this is the technical heart of the proofs, and we omit it.) In short, the failure of joint continuity is turned into a strategy for Player I, and the topological hypothesis says exactly that such strategies can be beaten. Continuity of inversion is handled separately. For locally compact groups the argument yields Ellis's theorem, and Namioka's original deduction of that theorem, which uses translations to spread a single point of joint continuity over the whole group, follows the same idea.

Exercises.

(1) Let $f(x,y) = xy/(x^2+y^2)$ for $(x,y) \neq (0,0)$, and $f(0,0) = 0$. Show that f is continuous in each variable separately, that f is not jointly continuous at $(0,0)$, and that f is jointly continuous at every other point of $\mathbb{R}^2$.

(2) Let X be a strongly Baire space, and show that X is a Baire space. (Hint: by Theorem 3.3, a non-empty space X is a Baire space if and only if Player I has no winning strategy in the Banach–Mazur game $\mathrm{BM}(X)$, and a winning strategy for Player I in $\mathrm{BM}(X)$ is also winning in $\mathrm{SB}(D)$ for every dense set D.) Deduce that $\mathbb{Q}$ is not strongly Baire.

(3) Let X be a complete metric space. Show directly that Player II has a winning strategy in $\mathrm{SB}(X)$, by answering U_k with a non-empty open set V_k such that $\overline{V_k} \subseteq U_k$ and $\operatorname{diam} V_k < 2^{-k}$.

(4) Let $\mathbb{R}$ carry the Sorgenfrey topology, whose basic open sets are the intervals $[a,b)$ with $a < b$ real. Show that addition is jointly continuous, that the translations are homeomorphisms, and that $x \mapsto -x$ is not continuous. Assuming that the Sorgenfrey line is regular, deduce from Theorem 11.1(3) that it is not strongly Baire.

(5) $(*)$ Let X be a topological space, let K be a compact space, let $f\colon X \times K \to \mathbb{R}$ be separately continuous, and let $x \in X$ be given. Show that f is jointly continuous at every point of $\{x\} \times K$ if and only if the map $x' \mapsto f(x', \cdot)$ from X to $C(K)$ is continuous at x for the supremum norm.

12. Applications to infinite-dimensional Ramsey theory

The problem. Ramsey's theorem (1930) says that, for every positive integer n, whenever the n-element subsets of $\omega = \{0,1,2,\dots\}$ are colored with two colors, there is an infinite $H \subseteq \omega$ all of whose n-element subsets have the same color. We now color the *infinite* subsets instead. For $H \subseteq \omega$, let $[H]^\omega$ be the set of infinite subsets of H. Call $\mathcal{A} \subseteq [\omega]^\omega$ *Ramsey* if there is an infinite $H \subseteq \omega$ such that $[H]^\omega \subseteq \mathcal{A}$ or $[H]^\omega \cap \mathcal{A} = \emptyset$. Using the axiom of choice, one can build sets that are not Ramsey: if $\mathcal{U}$ is a nonprincipal ultrafilter on ω (see the definitions at the beginning of Section 14), the set of all $X = \{x_0 < x_1 < \cdots\}$ in $[\omega]^\omega$ for which the union of the intervals $[x_{2k}, x_{2k+1}) = \{m \in \omega : x_{2k} \leq m < x_{2k+1}\}$, for $k \in \omega$, belongs to $\mathcal{U}$ is such a set (see Exercise 6 below). Which sets are Ramsey?

There is a Rothko hiding in every Pollock: however wildly two colors are splashed on the pairs of natural numbers, some infinite set has all of its pairs painted in a single color. In this section we show that the picture survives the passage from pairs to infinite sets whenever the coloring is regular enough: the sets that are Ramsey in a hereditary sense are exactly those

with the Baire property in a suitable topology, and every analytic set, one that is empty or a continuous image of the space of all sequences of natural numbers, is among them. The tool is a game of Banach–Mazur type.

12.1. **History of the problem and of the solution using topological games.** Ramsey's theorem appeared in 1930 [Ram30]. In 1952 Erdős and Rado showed, using the axiom of choice, that the infinite-dimensional analogue of Ramsey's theorem fails [ER52]. Nash-Williams proved in 1965 that clopen sets are Ramsey, for the topology that $[\omega]^\omega$ inherits from the Cantor space 2^ω when every subset of ω is identified with its characteristic function [NW65]. Galvin and Prikry extended this result to Borel sets [GP73], and Silver extended it to analytic sets [Sil70], by a proof that used metamathematical methods. Silver's paper appeared in 1970, three years before that of Galvin and Prikry. Mathias developed the theory further, with applications to forcing, in his work on happy families [Mat77].

Erik Ellentuck solved the problem in 1974 [Ell74]. He introduced a new topology on $[\omega]^\omega$, finer than the usual one, and proved that the sets that are Ramsey in a hereditary sense are exactly those with the Baire property in this topology. Silver's theorem follows from Ellentuck's in a few lines. In 1983 Kastanas [Kas83] characterized the same sets by the determinacy of a game, which is a Banach–Mazur game in the Ellentuck topology. In Section 13 we show that Gowers's game is the analogue of the Kastanas game for block sequences in Banach spaces.

12.2. **The game.** The game is played in a new topology whose open sets look strange at first, so we begin with the idea behind them. In the usual topology on $[\omega]^\omega$, which it inherits from the Cantor space 2^ω (see the history above), two infinite sets are close if they begin in the same way, that is, if they have the same elements below some large integer. Ellentuck's open sets ask for more: they fix the beginning s *and* restrict the rest of the set to come from a given reservoir M. For a finite $s \subseteq \omega$ and an infinite $M \subseteq \omega$ with $\max s < \min M$ (a condition that holds vacuously when $s = \emptyset$), let

$$[s, M] = \{X \in [\omega]^\omega : s \text{ is an initial segment of } X \text{ and } X \subseteq s \cup M\},$$

where s is an *initial segment* of X if $s \subseteq X$ and every element of X that is smaller than some element of s lies in s. Think of s as a stem that has already been decided and of M as the pool of candidates for the rest. The sets $[s, M]$ form a basis of a topology, the *Ellentuck topology* (see Exercise 1 below). A set $\mathcal{A} \subseteq [\omega]^\omega$ is *completely Ramsey* if for every $[s, M]$ there is an infinite $N \subseteq M$ with $[s, N] \subseteq \mathcal{A}$ or $[s, N] \cap \mathcal{A} = \emptyset$.

Let $\mathcal{A} \subseteq [\omega]^\omega$ be given, let $s \subseteq \omega$ be finite, and let $N \subseteq \omega$ be infinite with $\max s < \min N$. The Kastanas game $K(\mathcal{A}, s, N)$ is played as follows.

(1) Player I chooses an infinite $M_0 \subseteq N$.
(2) Player II chooses $n_0 \in M_0$ and an infinite $N_0 \subseteq M_0$ with $n_0 < \min N_0$.
(3) In round $k \geq 1$, Player I chooses an infinite $M_k \subseteq N_{k-1}$, and Player II chooses $n_k \in M_k$ and an infinite $N_k \subseteq M_k$ with $n_k < \min N_k$.
(4) Player II wins if $s \cup \{n_0, n_1, \dots\} \in \mathcal{A}$.

In Kastanas's paper the payoff set is Player I's; since the main result of the next subsection (see Theorem 12.1) is about determinacy, and $\mathcal{A}$ is completely Ramsey exactly when its complement $[\omega]^\omega \setminus \mathcal{A}$ is (see Exercise 4 below), the switch does not matter. The moves are basic open sets in disguise: Player I's move M_k stands for $[s \cup \{n_0, \dots, n_{k-1}\}, M_k]$, and Player II answers with the smaller basic open set $[s \cup \{n_0, \dots, n_k\}, N_k]$. The Kastanas game

is thus a Banach–Mazur game in the Ellentuck topology, in which Player II must commit to one more element at each move (see Figure 18).

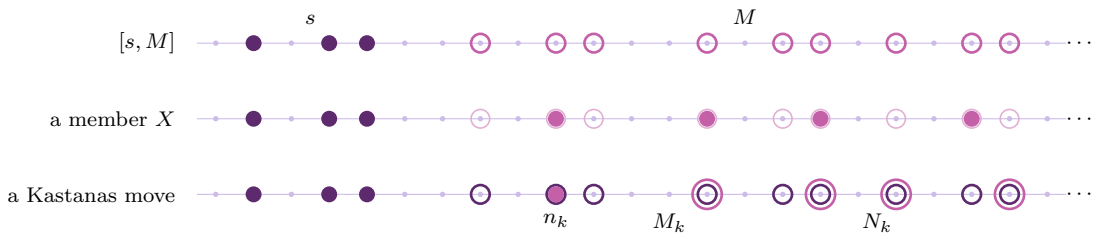


FIGURE 18. The Ellentuck topology on $[\omega]^\omega$. Top: a basic open set $[s, M]$ is given by a finite stem s (filled plum) and an infinite set M above it (circles). Middle: a member of $[s, M]$ is the stem together with an infinite subset of M (filled pink). Bottom: round k of the Kastanas game, for some $k \in \omega$, played above the stem: Player I offers an infinite set M_k (plum circles), and Player II commits to one element $n_k \in M_k$ and keeps an infinite set $N_k \subseteq M_k$ above it (pink circles) for the rest of the play.

12.3. **The Main Result of this section.** The main result is Ellentuck's theorem, in the game form due to Kastanas.

Theorem 12.1 (Ellentuck, 1974; Kastanas, 1983). *For $\mathcal{A} \subseteq [\omega]^\omega$ the following are equivalent:*

(1) *the set $\mathcal{A}$ is completely Ramsey;*
(2) *the set $\mathcal{A}$ has the Baire property in the Ellentuck topology;*
(3) *for every finite $s \subseteq \omega$ and every infinite $N \subseteq \omega$ with $\max s < \min N$, the Kastanas game $K(\mathcal{A}, s, N)$ is determined.*

In any topological space, the sets with the Baire property are closed under the Souslin operation (see [Kec95]). This operation assigns to a family of sets F_u, indexed by the non-empty finite sequences u of natural numbers, the union, over all sequences $x = (x_0, x_1, \dots)$ of natural numbers, of the sets $\bigcap_{n\geq 1} F_{(x_0,\dots,x_{n-1})}$. In $[\omega]^\omega$ with the usual topology, as in every separable completely metrizable space, the analytic sets are exactly the sets obtained from closed sets by this operation (see [Kec95]). Every closed set of the usual topology is closed in the finer Ellentuck topology. Hence every analytic set, and in particular every Borel set, has the Baire property in the Ellentuck topology, and is therefore completely Ramsey by Theorem 12.1. Since completely Ramsey sets are Ramsey (see Exercise 4 below), the argument recovers in one stroke the Galvin–Prikry theorem and Silver's theorem, by which Borel sets and analytic sets, respectively, are Ramsey.

12.4. **Solving the problem with the game.** The proofs of the Galvin–Prikry theorem and of Ellentuck's theorem rest on one device, the *combinatorial forcing* of Galvin and Prikry [GP73]. Fix $\mathcal{A} \subseteq [\omega]^\omega$. For a finite $s \subseteq \omega$ and an infinite $M \subseteq \omega$ with $\max s < \min M$, we say that M *accepts* s if $[s, M] \subseteq \mathcal{A}$, and that M *rejects* s if no infinite $N \subseteq M$ accepts s. In plain words, the set M accepts the stem s if every completion of s by infinitely many elements of M lies in $\mathcal{A}$, and M rejects s if no thinning out of M produces a set that accepts s. By a diagonal argument, one finds, inside any infinite M, an infinite N that *decides* every

finite $s \subseteq N$: for each such s, the set of elements of N above $\max s$ accepts or rejects s. Ellentuck's insight was that these diagonal arguments are exactly the proof that $[\omega]^\omega$, with the Ellentuck topology, is a Baire space, and that the nowhere dense sets of the Ellentuck topology are exactly the sets $\mathcal{A}$ such that every $[s, M]$ contains some $[s, N]$ disjoint from $\mathcal{A}$ [Ell74]. The Kastanas game turns the diagonalization into a play: a winning strategy for Player II in $K(\mathcal{A}, s, N)$, run against all possible moves of Player I and glued together by a diagonal fusion, produces an infinite $M \subseteq N$ with $[s, M] \subseteq \mathcal{A}$, and a winning strategy for Player I produces one with $[s, M] \cap \mathcal{A} = \emptyset$ [Kas83]. The property of being completely Ramsey is thus to the Kastanas game what the Baire property is to the Banach–Mazur game.

12.5. **The notation $[\omega]^\omega$ and the partition calculus.** The games of this section are played on the set $[\omega]^\omega$ of infinite subsets of ω, with the Ellentuck topology, and the notation for that set differs from the notation of Section 2. It comes from the partition calculus, as we now explain. For a set S and a cardinal κ, one writes $[S]^\kappa = \{X \subseteq S : |X| = \kappa\}$ and $[S]^{<\kappa} = \{X \subseteq S : |X| < \kappa\}$; thus $[\omega]^\omega$ and $[\omega]^{<\omega}$ are the families of infinite and of finite subsets of ω, as opposed to the spaces A^ω and $A^{<\omega}$ of infinite and of finite sequences of elements of a set A (see Section 2). This is the notation of the partition calculus of P. Erdős and R. Rado [ER56], in which, for cardinals κ, λ, μ and r, the relation $\kappa \to (\lambda)^r_\mu$ means that, whenever $|S| = \kappa$ and $[S]^r$ is partitioned into μ pieces, there is a set $H \subseteq S$ with $|H| = \lambda$ such that $[H]^r$ lies inside a single piece; such a set H is called *homogeneous*. Ramsey's theorem becomes $\omega \to (\omega)^r_k$ for all positive integers r and k. For the infinite exponent the relation fails: using the axiom of choice, Erdős and Rado partitioned $[\omega]^\omega$ into two pieces with no infinite homogeneous set [ER52]. Positive results recover $\omega \to (\omega)^\omega_2$ for partitions that are regular enough: Galvin and Prikry for Borel partitions [GP73], Silver for analytic ones [Sil70], and Ellentuck for partitions with the Baire property in the Ellentuck topology [Ell74]; in this section, the Kastanas game is the bridge to these theorems. The partition calculus, and the school of infinite combinatorics that Erdős and his collaborators built around it, are set out in the monograph of Erdős, Hajnal, Máté and Rado [EHMR84].

Exercises.

(1) Show that the sets $[s, M]$ form a basis of a topology on $[\omega]^\omega$. (Hint: if $[s, M] \cap [t, N]$ is non-empty, then one of s and t, say s, is an initial segment of the other, and the intersection is $[t, P]$, where P is the set of elements of $M \cap N$ greater than $\max t$.)

(2) Show that the Ellentuck topology is finer than the usual (Cantor space) topology: for a positive integer n and a set $t \subseteq \{0, \dots, n-1\}$, the set of $X \in [\omega]^\omega$ with $X \cap \{0, \dots, n-1\} = t$ is a basic set $[s, M]$. Show also that, if $M \subseteq \omega$ is infinite and $\omega \setminus M$ is infinite, the set $[\emptyset, M]$ has empty interior in the usual topology.

(3) Let $\mathcal{A}$ be the set of $X \in [\omega]^\omega$ containing infinitely many even numbers, and let s and N be as in the definition of the Kastanas game. Show that Player II has a winning strategy in $K(\mathcal{A}, s, N)$ if N contains only finitely many odd numbers, and that Player I has one otherwise. Find in each case an infinite $N' \subseteq N$ with $[s, N'] \subseteq \mathcal{A}$ or $[s, N'] \cap \mathcal{A} = \emptyset$.

(4) Show that a set $\mathcal{A} \subseteq [\omega]^\omega$ is completely Ramsey if and only if $[\omega]^\omega \setminus \mathcal{A}$ is, and that every completely Ramsey set is Ramsey.

(5) (∗) Color the two-element subsets of ω red or blue, and let $\mathcal{A}$ be the set of $X \in [\omega]^\omega$ whose two least elements form a red pair. Show that $\mathcal{A}$ is completely Ramsey. (Hint: treat separately the cases in which the stem s of a basic set $[s, M]$ has at least two elements, one element, or none; in the last case use Ramsey's theorem.)

(6) Let $\mathcal{U}$ be a nonprincipal ultrafilter on ω, viewed as a subset of $[\omega]^\omega$. Show that $\mathcal{U}$ itself is Ramsey, by finding an infinite $H \notin \mathcal{U}$ and checking that $[H]^\omega \cap \mathcal{U} = \emptyset$. For $X = \{x_0 < x_1 < \cdots\}$ in $[\omega]^\omega$, let Y_X be the union of the intervals $[x_{2k}, x_{2k+1}) = \{m \in \omega : x_{2k} \leq m < x_{2k+1}\}$ for $k \in \omega$, and let $\mathcal{A}$ be the set of X with $Y_X \in \mathcal{U}$. Show that Y_X and $Y_{X\setminus\{x_0\}}$ are disjoint and that their union is $\{m \in \omega : m \geq x_0\}$. Deduce that $\mathcal{A}$ is not Ramsey.

13. Applications to Banach space theory

The problems. (1) *Differentiability of convex functions.* In calculus, a convex function on an open interval is differentiable at all but countably many points. In infinite dimensions things are much less tidy. A Banach space X is an *Asplund space* if every continuous convex function on an open convex subset of X is Fréchet differentiable (see the glossary below) on a dense G_δ subset of its domain. Which spaces are Asplund spaces?

(2) *The unconditional basic sequence problem.* Does every infinite-dimensional Banach space contain an unconditional basic sequence, that is, a sequence $(e_n)_{n\geq 1}$ that is a basis (see the glossary below) of the closed subspace spanned by $e_1, e_2, \ldots$ and for which every convergent expansion $\sum_n a_n e_n$ keeps converging when the signs of its terms are changed or its terms are rearranged?

In this section we put two games to work in Banach spaces. The fragmentability game of Kenderov and Moors detects the Asplund property, and Gowers's game drives his dichotomy theorem, which explains the answer to the unconditional basic sequence problem. Functional analysis lies outside the background we assume, so we begin with a short glossary.

A small glossary. Let X be a Banach space.

- The *dual* X^* is the Banach space of bounded linear functionals on X, and its closed unit ball is denoted by B_{X^*}. The *weak* topology* on X^* is the topology of pointwise convergence on the elements of X.
- A function f on an open subset O of X is *Gâteaux differentiable* at a point $x \in O$ if there is $\varphi \in X^*$ with $f(x + th) = f(x) + t\varphi(h) + o(t)$ as the real number t tends to 0, for each fixed $h \in X$, and *Fréchet differentiable* at x if there is $\varphi \in X^*$ with $f(x + h) = f(x) + \varphi(h) + o(\|h\|)$ as $\|h\| \to 0$.
- A *basis* of X is a sequence $(e_n)_{n\geq 1}$ in X such that every $x \in X$ has a unique expansion $x = \sum_n a_n e_n$ with scalars a_n, and a *basic sequence* is a sequence that is a basis of its closed linear span. For a constant $C \geq 1$, a basic sequence $(e_n)_{n\geq 1}$ is *C-unconditional* if $\|\sum_n \varepsilon_n a_n e_n\| \leq C\|\sum_n a_n e_n\|$ for all finitely supported sequences of scalars (a_n) and all signs $\varepsilon_n = \pm 1$, and *unconditional* if it is C-unconditional for some C.
- The space X is the *direct sum* of closed subspaces Y_1 and Y_2 if $Y_1 \cap Y_2 = \{0\}$ and $Y_1 + Y_2 = X$.
- Let K be a set with a topology $\mathcal{T}$ and a metric d. The set K is *fragmented* by d if every non-empty subset of K has non-empty relatively $\mathcal{T}$-open subsets of arbitrarily small d-diameter.

13.1. **History of the problems and their solutions using topological games.** Mazur proved in 1933 that continuous convex functions on separable Banach spaces are Gâteaux differentiable on dense G_δ sets, and Asplund introduced the spaces now named after him in 1968, in a paper in *Acta Mathematica* [Asp68]. The characterization problem was solved in 1975 by Namioka and Phelps [NP75] and by Stegall [Ste75]. The property behind the

Asplund spaces is fragmentability, and Kenderov and Moors described it by a game, first in 1996 and then, in the form used below, in a paper published in 1999 [KM99].

The unconditional basic sequence problem was one of the central open problems of Banach space theory for decades. The problem was solved, negatively, by Gowers and Maurey in 1991 (published in 1993) [GM93]. They built a *hereditarily indecomposable* space: an infinite-dimensional Banach space so tangled that none of its infinite-dimensional closed subspaces is the direct sum of two infinite-dimensional closed subspaces. Gowers then showed that such strange spaces are no accident. In papers of 1996 and 2002, by means of an infinite game played with subspaces and vectors, he proved that every infinite-dimensional Banach space contains an infinite-dimensional subspace that either has an unconditional basis or is hereditarily indecomposable [Gow96, Gow02]. The second paper, which contains the Ramsey theorem behind the game, appeared in the *Annals of Mathematics*.

Recognition. Gowers received the Fields Medal in 1998 for research connecting functional analysis and combinatorics; this research includes the solution of the unconditional basic sequence problem and the dichotomy theorem. He was an invited speaker at the International Congress of Mathematicians in Zürich in 1994, and he received the Prize of the European Mathematical Society in 1996.

13.2. **The games.**

13.2.1. *The fragmentability game.* Let K be a set with two topologies $\mathcal{T}_1$ and $\mathcal{T}_2$. The case to keep in mind is the dual ball B_{X^*} of a Banach space X, with the weak* topology as $\mathcal{T}_1$ and the norm topology as $\mathcal{T}_2$. The game is played as follows.

(1) Player I chooses a non-empty subset $U_0 \subseteq K$.
(2) Player II chooses a non-empty relatively $\mathcal{T}_1$-open subset $V_0 \subseteq U_0$.
(3) In round $k \geq 1$, Player I chooses a non-empty subset $U_k \subseteq V_{k-1}$, and Player II a non-empty relatively $\mathcal{T}_1$-open subset $V_k \subseteq U_k$.
(4) Player II wins if $\bigcap_k U_k$ is empty, or if $\bigcap_k U_k$ is a single point x and every $\mathcal{T}_2$-neighborhood of x contains some U_k.

This is a Banach–Mazur game with an asymmetry: Player I may play any non-empty subset of the previous move, while Player II must play relatively $\mathcal{T}_1$-open ones; she wins if the sets close in on at most one point and, when they close in on a point, they do so in the topology $\mathcal{T}_2$.

13.2.2. *Gowers's game.* Let X be a Banach space with a basis $(e_n)_{n\geq 1}$. A *block vector* is a non-zero finite linear combination of the e_n, and a *block subspace* is the closed span of a sequence of block vectors $x_1, x_2, \dots$ with *successive supports*: every basis vector used in x_k comes before every basis vector used in x_{k+1}. A sequence of block vectors with successive supports is a *block sequence*, and a vector is *normalized* if it has norm 1. Let Σ be a set of sequences of normalized block vectors. In round $k \geq 1$, Player I chooses an infinite-dimensional block subspace Y_k, and Player II chooses a normalized block vector $x_k \in Y_k$. Player II wins if $(x_1, x_2, \dots) \in \Sigma$. Player I's subspaces play the role of open sets. In the Banach–Mazur game a small neighborhood limits where the other player may look, and a block subspace does the same. Player I may choose it to live as far out along the basis as he likes; he thereby forces Player II's next vector to be supported beyond any finite stage, and he can also restrict the kind of vectors available to her.

13.3. **The Main Results of this section.** Differentiability is tied to the dual space for a simple reason: the derivative of a function on X at a point is a linear functional, that is, an element of X^*. Differentiability is therefore a question about how the candidate derivatives sit inside X^*; for a convex function f on an open convex subset O of X and a point $x \in O$, these are the *subgradients*, the functionals $\varphi \in X^*$ with $f(y) \geq f(x) + \varphi(y - x)$ for all $y \in O$. Roughly, Fréchet differentiability at a point means that subgradients at nearby points are close to the derivative *in norm*, and the norm is what fragmentability controls. In the first result below we collect several characterizations of Asplund spaces, one of which is exactly what the game of Kenderov and Moors detects; the second result is Gowers's dichotomy.

Theorem 13.1 (Asplund, 1968; Namioka–Phelps, 1975; Stegall, 1975). *For a Banach space X the following are equivalent:*

(1) *the space X is an Asplund space;*
(2) *every separable subspace of X has a separable dual;*
(3) *the dual X^* has the Radon–Nikodým property, a property from the theory of vector-valued measures that we do not use below;*
(4) *the dual ball B_{X^*}, with the weak* topology, is fragmented by the norm (see the glossary at the beginning of this section); that is, every non-empty subset of B_{X^*} has non-empty relatively weak*-open subsets of arbitrarily small norm diameter.*

Kenderov and Moors [KM99] proved that the fragmentability game detects exactly this property: for a set K with two topologies $\mathcal{T}_1$ and $\mathcal{T}_2$, the space K, with the topology $\mathcal{T}_1$, is fragmented by a metric on K whose topology contains $\mathcal{T}_2$ if and only if Player II has a winning strategy in the fragmentability game on K. In particular, by condition (4), Player II wins the game on B_{X^*}, with the weak* and norm topologies, whenever X is an Asplund space.

Theorem 13.2 (Gowers's dichotomy, 1996 and 2002). *Every infinite-dimensional Banach space contains an infinite-dimensional subspace that either has an unconditional basis or is hereditarily indecomposable.*

The two alternatives exclude each other, because a hereditarily indecomposable space contains no unconditional basic sequence [GM93]: the closed spans of the even-numbered and of the odd-numbered terms of such a sequence would be two infinite-dimensional closed subspaces whose direct sum is the closed span of the whole sequence. Every infinite-dimensional Banach space, however large, thus contains a subspace that is either perfectly civilized or thoroughly rebellious; there is no third option.

13.4. **Solving the problems with the games.**

13.4.1. *Asplund spaces.* Problem (1) was solved in 1975 without games, by Namioka and Phelps and by Stegall. What the game of Kenderov and Moors adds is a new way to establish fragmentability, and with it the Asplund property: instead of producing a fragmenting metric, one describes how Player II should play. We give the easy half of their theorem.

Let X be a Banach space, suppose that B_{X^*} is fragmented by the norm, and consider the fragmentability game on B_{X^*}, with the weak* topology as $\mathcal{T}_1$ and the norm topology as $\mathcal{T}_2$. Then Player II has a simple strategy: she answers each move U_k with a relatively weak*-open subset $V_k \subseteq U_k$ of norm diameter less than 2^{-k}. Since $U_{k+1} \subseteq V_k$ for every k, the intersection $\bigcap_k U_k$ has norm diameter 0, and it is either empty or a single point x. In the second case, the

set U_{k+1} lies in the norm ball of radius 2^{-k} about x, whence every norm neighborhood of x contains some U_k, and Player II wins. Conversely, Kenderov and Moors build a fragmenting metric out of a winning strategy for Player II; roughly, two points are close when the strategy takes a long time to separate them. The advantage of the game is that a strategy is a local object, built one move at a time, while a fragmenting metric is a global one, and describing how to play is often much easier than writing down a metric.

13.4.2. *Gowers's dichotomy.* Let X be a Banach space with a basis. Gowers's infinite Ramsey theorem [Gow02] says that, for every set Σ of sequences of normalized block vectors that is analytic in the product of countably many copies of X, and for every sequence of positive tolerances, there is a block subspace Y of X such that either no block sequence of Y lies in Σ, or Player II has a winning strategy in Gowers's game played below Y (Player I choosing block subspaces of Y) for a slight enlargement of Σ, obtained by wiggling the members of Σ within the given tolerances. The theorem is the counterpart, for block sequences, of the Galvin–Prikry theorem and Ellentuck's theorem on Ramsey sets (see Section 12), and Gowers's game plays the part of the Kastanas game. An exact Ramsey statement is false for vectors; what survives is the game, played up to the given tolerances. Gowers's proof uses a "combinatorial forcing" argument, in which subspaces are said to accept or reject finite block sequences.

Roughly, the dichotomy follows from an application of Gowers's infinite Ramsey theorem to the set of block sequences that fail to be C-unconditional, for a constant $C \geq 1$ [Gow96, Gow02]. If, for some constant C, some block subspace avoids these sequences, then every normalized block sequence in that subspace is C-unconditional, and the subspace has an unconditional basis. Otherwise, for every C and in every block subspace, Player II can produce block sequences that fail to be C-unconditional, and from her strategies Gowers extracts a hereditarily indecomposable subspace: none of its infinite-dimensional closed subspaces is the direct sum of two infinite-dimensional closed subspaces. The game locates this subspace; that such spaces exist at all is the theorem of Gowers and Maurey.

Exercises.

(1) Let c_0 be the space of real sequences $(a_n)_{n\geq 1}$ converging to 0, with the supremum norm, and, for $1 \leq p < \infty$, let ℓ_p be the space of real sequences $(a_n)_{n\geq 1}$ for which the norm $\|(a_n)\|_p = (\sum_n |a_n|^p)^{1/p}$ is finite. Show that the standard unit vector bases of these spaces are 1-unconditional.

(2) Let $f(x,y) = x^3y/(x^4+y^2)$ for $(x,y) \neq (0,0)$, and $f(0,0) = 0$. Show that f is Gâteaux differentiable at the origin, with $\varphi = 0$, but not Fréchet differentiable there. (Hint: look at $f(x,x^2)$.)

(3) Let (K,d) be a metric space, and let $\mathcal{T}_1$ and $\mathcal{T}_2$ both be the topology of d. Show that Player II has a winning strategy in the fragmentability game, by answering U_k with the intersection of U_k and a ball of radius 2^{-k-1} about a point of U_k.

(4) Let X be a Banach space with a 1-unconditional basis $(e_n)_{n\geq 1}$, and let $x_1, x_2, \ldots$ be block vectors with successive supports. Show that $\|\sum_k \varepsilon_k b_k x_k\| \leq \|\sum_k b_k x_k\|$ for all finite sequences of scalars b_k and signs $\varepsilon_k = \pm 1$.

(5) (∗) Let X be a Banach space with a basis $(e_n)_{n\geq 1}$, and consider Gowers's game in X. Show that Player II has a winning strategy when Σ is the set of all sequences of normalized block vectors with successive supports, and that Player I has a winning strategy when Σ is the set of sequences whose first term is $e_1/\|e_1\|$. (Hint: the coefficient functionals of a basis are continuous.)

14. Applications to ultrafilters

In the last section of this part the Banach–Mazur game is turned on the axiom of choice itself, through the ultrafilters on the natural numbers. We begin with the vocabulary needed to state the problem.

Definitions. A *filter* on ω is a non-empty family $\mathcal{F}$ of subsets of ω that does not contain $\emptyset$, is closed under finite intersections, and contains every superset of each of its members. An *ultrafilter* is a filter $\mathcal{U}$ such that for every $x \subseteq \omega$ either x or $\omega \setminus x$ belongs to $\mathcal{U}$. For each $n \in \omega$, the ultrafilter of all subsets of ω containing n is called *principal*; every other ultrafilter is *nonprincipal*. A nonprincipal ultrafilter contains every cofinite set (see Exercise 1 at the end of this section). Among the nonprincipal ultrafilters, an ultrafilter $\mathcal{U}$ is a *P-point* if, for every partition of ω into pieces that do not belong to $\mathcal{U}$, some element of $\mathcal{U}$ meets every piece in a finite set, and it is *selective* if for every such partition some element of $\mathcal{U}$ meets every piece in at most one point. The set of all ultrafilters on ω, with the topology whose basic open sets are the sets of ultrafilters containing a given subset of ω, is a compact space, denoted by $\beta\omega$. When each $n \in \omega$ is identified with the principal ultrafilter of the sets containing n, the subspace $\omega^* = \beta\omega \setminus \omega$ consists of the nonprincipal ultrafilters; these spaces appear only in passing.

The problem. Identify every subset of ω with its characteristic function; the power set $\mathcal{P}(\omega)$ then becomes the Cantor space 2^ω of infinite binary sequences, with the product topology, and a filter on ω becomes a set of reals. (In set theory the elements of 2^ω are called reals, and its subsets sets of reals.) The axiom of choice provides nonprincipal ultrafilters on ω in abundance (there are $2^{2^{\aleph_0}}$ of them, the points of ω^*), and yet nobody has ever exhibited one. The problem is to explain this: to determine how irregular a nonprincipal ultrafilter must be as a set of reals, how much choice its existence requires, and which ultrafilters are tame enough to carry the Ramsey theory of Section 12.

A principal ultrafilter is a point of ω in disguise: it collects the sets that contain the point, and the two-valued measure that it defines (measure 1 for its members and 0 for all other subsets of ω) is the Dirac measure at the point. A nonprincipal ultrafilter is a point at infinity that passes judgment on every subset of ω, consistently and irrevocably, and always against the finite ones. Seen through the Banach–Mazur game on the Cantor space, played with finite binary strings (Ulam's game below), such a judge is invisible to topology: in Theorem 14.1(1) below we show that in the game it defines neither player has a winning strategy. Seen through Galvin's games, the well-mannered judges are the selective ultrafilters and the P-points, and in Theorem 14.1(4) below we show that they are exactly the judges that cannot be outplayed.

14.1. **History of the problem and of the solution using topological games.** Ultrafilters came into mathematics through two doors, the algebra of sets and general topology. Tarski proved in 1930, using the axiom of choice, that every filter of subsets of a set extends to an ultrafilter [Tar30]; Stone made the ultrafilters of a Boolean algebra into the points of a compact space in 1936 and 1937 [Sto36]; and Cartan introduced filters and ultrafilters in 1937 to describe convergence in general topology [Car37]. In 1938 Sierpiński observed that a nonprincipal ultrafilter on ω, regarded as a set of reals, is not Lebesgue measurable [Sie38]; the symmetry behind his argument, the flip $x \mapsto \omega \setminus x$, also shows that it does not have the Baire property (we prove both facts in Theorem 14.1(2) below). In 1962 Mycielski and Steinhaus proposed the Axiom of Determinacy, by which every game in which the players choose natural numbers is determined [MS62]. The axiom makes every set of reals Lebesgue

measurable and gives it the Baire property, and it therefore rules out nonprincipal ultrafilters on ω altogether (see Theorem 14.1(3) below). Feferman showed in 1965, by forcing, that the existence of a definable nonprincipal ultrafilter on ω cannot be proved in ZFC (Zermelo–Fraenkel set theory with the axiom of choice) [Fef65], and Shelah showed in 1984 that ZF (the same theory without the axiom of choice) together with the axiom of dependent choices is consistent with the statement that every set of reals has the Baire property, relative to ZF alone [She84]. Some choice beyond dependent choice is therefore needed to produce a single nonprincipal ultrafilter on ω.

Meanwhile the ultrafilters themselves were being sorted. Walter Rudin proved in 1956 that under the continuum hypothesis there are P-points, and deduced that ω^* is not homogeneous, that is, that some point of ω^* cannot be carried to some other point by a homeomorphism of ω^* [Rud56]; Frolík proved in 1967 that ω^* is not homogeneous in ZFC [Fro67]. Booth and Kunen studied the selective ultrafilters [Boo70], and Kunen proved that a nonprincipal ultrafilter on ω is selective exactly when, for every coloring of the two-element subsets of ω in two colors, the ultrafilter contains a set all of whose two-element subsets have the same color. Mathias showed in 1977 that selective ultrafilters carry the Ramsey theory of analytic sets [Mat77]. Kunen proved in 1976 that it is consistent with ZFC that there are no selective ultrafilters [Kun76], and Shelah proved shortly afterwards that it is consistent that there are no P-points at all (see the account of Wimmers [Wim82]).

The topological side of the problem was settled by Jalali-Naini in his thesis of 1976 [JN76] and, independently, by Talagrand in 1980 [Tal80]. Both characterized the meager filters by a combinatorial condition on partitions of ω into intervals. Talagrand's motivation came from pointwise compact sets of measurable functions, a subject closely tied to the theory of Rosenthal compacta, begun by Rosenthal and developed by Bourgain, Fremlin and Talagrand. The games enter twice. The irregularity of a nonprincipal ultrafilter follows from a strategy-stealing argument in the Banach–Mazur game on the Cantor space, which rests on the same symmetry $x \mapsto \omega \setminus x$ as Sierpiński's proof (we give the argument in the proof of Theorem 14.1(1) below); and Galvin and McKenzie characterized the selective ultrafilters and the P-points by the games described below (see the book of Bartoszyński and Judah [BJ95]).

Recognition. Talagrand received the Abel Prize in 2024, and Shelah the Wolf Prize in 2001.

14.2. **The games.**

Ulam's game. Let $Z \subseteq 2^\omega$ be given. In the game $\mathrm{Ul}(Z)$, Player I chooses a non-empty finite binary string s_0, Player II answers with a non-empty finite binary string t_0, Player I continues with a non-empty string s_1, and so on, forever. Player II wins if the concatenation

$$x = s_0{}^\frown t_0{}^\frown s_1{}^\frown t_1{}^\frown \cdots$$

belongs to Z, where $u{}^\frown v$ denotes the finite binary string u followed by the binary sequence v, finite or infinite. For a finite binary string u, let $N_u = \{x \in 2^\omega : u \subseteq x\}$ be the basic clopen set of the sequences that begin with u. Each move produces a position, the concatenation of the moves made so far, and the basic clopen set of that position lies inside the basic clopen set of the previous one; the game is therefore the Banach–Mazur game $\mathrm{BM}(2^\omega, Z)$ of Section 3, with the moves restricted to basic clopen sets. We name it after Ulam, who proposed, right after Problem 43 of the Scottish Book, a variant in which each move is a single binary digit (see Section 26). Since the Cantor space is a compact metric space, both parts of the theorem of Banach and Oxtoby apply (see Theorem 3.5; its proof works equally well with moves restricted to basic open sets), and here they read as follows: Player II has

a winning strategy in $\mathrm{Ul}(Z)$ if and only if Z is comeager, and Player I has one if and only if $Z \cap N_u$ is meager for some finite binary string u.

Galvin's games. Let $\mathcal{U}$ be an ultrafilter on ω. In the *selective game* $S(\mathcal{U})$, Player I chooses $A_0 \in \mathcal{U}$, Player II chooses $n_0 \in A_0$, Player I chooses $A_1 \in \mathcal{U}$, Player II chooses $n_1 \in A_1$, and so on; Player II wins if $\{n_0, n_1, \dots\} \in \mathcal{U}$. In the *P-point game* $P(\mathcal{U})$, Player I again chooses sets $A_k \in \mathcal{U}$, for $k \in \omega$, but Player II answers each A_k with a finite set $F_k \subseteq A_k$ instead of a single element, and she wins if $\bigcup_k F_k \in \mathcal{U}$. Player I plays the skeptic: at each turn he names a set that $\mathcal{U}$ approves of, and he challenges Player II to assemble, one element (or one finite batch) at a time, a set that $\mathcal{U}$ approves of as well.

The tame ultrafilters of the main result are the P-points and the selective ultrafilters defined at the beginning of this section. Every selective ultrafilter is a P-point, since meeting each piece of a partition in at most one point is a special case of meeting it in a finite set; the counterpart of this inclusion for the games is the content of Exercise 4 below.

14.3. **The Main Result of this section.** In the main result we gather the behavior of nonprincipal ultrafilters in the two kinds of games, together with its consequences for category, measure and determinacy: in Ulam's game they are invisible, and in Galvin's games the tame ones are exactly those that cannot be outplayed.

Theorem 14.1 (Sierpiński, 1938; Mycielski–Steinhaus, 1962; Galvin–McKenzie). *Let $\mathcal{U}$ be a nonprincipal ultrafilter on ω, regarded as a subset of 2^ω. Then the following hold.*

(1) *The game $\mathrm{Ul}(\mathcal{U})$ is not determined.*
(2) *The set $\mathcal{U}$ is neither meager nor comeager, it does not have the Baire property, and it is not measurable for the coin-tossing measure on 2^ω, the product of countably many copies of the measure on $\{0,1\}$ that gives measure $\frac{1}{2}$ to each of 0 and 1.*
(3) *If the game $\mathrm{Ul}(Z)$ is determined for every $Z \subseteq 2^\omega$, then every subset of 2^ω has the Baire property, and there is no nonprincipal ultrafilter on ω.*
(4) *The ultrafilter $\mathcal{U}$ is selective if and only if Player I has no winning strategy in $S(\mathcal{U})$, and it is a P-point if and only if Player I has no winning strategy in $P(\mathcal{U})$.*

By part (2), it is no accident that nobody has written down a nonprincipal ultrafilter on ω: analytic subsets of 2^ω (the empty set and the continuous images of the space ω^ω of all sequences of natural numbers) and coanalytic ones (their complements) have the Baire property (see [Kec95]), so no nonprincipal ultrafilter is analytic or coanalytic. The proofs of (1) and (2), given below, use only two properties of $\mathcal{U}$:

- membership in $\mathcal{U}$ does not depend on finitely many digits, because $\mathcal{U}$ is nonprincipal;
- for every $x \subseteq \omega$, the complement $\bar{x} = \omega \setminus x$ belongs to $\mathcal{U}$ if and only if x does not, because $\mathcal{U}$ is an ultrafilter.

14.4. **Solving the problem with the games.**

Stealing a strategy. The idea is to let Player I play two games against a strategy of Player II at once, the second always the mirror image of the first, so that the strategy is forced to put both a set and its complement into $\mathcal{U}$. For a finite binary string s, we write $\bar{s}$ for the string obtained by flipping every digit of s. Suppose that τ is a winning strategy for Player II in $\mathrm{Ul}(\mathcal{U})$. Player I opens the first game with the string 0, and the strategy τ answers t_0. He opens the second game with $\overline{0^\frown t_0}$ and receives the answer u_0. Back in the first game, he plays $\bar{u}_0$ and receives t_1; in the second, he plays $\bar{t}_1$ and receives u_1; and so on (Figure 19). By

construction, each block of either play is the flip of the corresponding stretch of the other, and therefore the play x of the first game and the play y of the second satisfy $y = \bar{x}$. Both plays are won by τ, whence $x \in \mathcal{U}$ and $\bar{x} \in \mathcal{U}$, which is impossible. Hence Player II has no winning strategy.

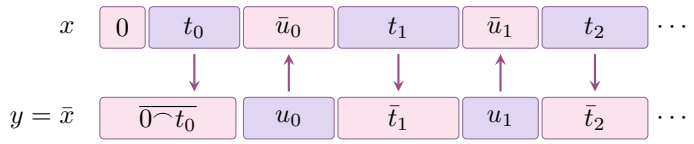


FIGURE 19. Stealing Player II's strategy in $\mathrm{Ul}(\mathcal{U})$. The top row is the play x of the first game and the bottom row the play $y = \bar{x}$ of the second. Lavender blocks are answers of the strategy τ (t_k in the first game, u_k in the second), blush blocks are moves of Player I, and each arrow copies an answer of τ in one game, flipped, as a move of Player I in the other.

Suppose now that Player I has a winning strategy σ, and let s be its first move. From the position s on, the strategy σ answers the moves of Player II and forces $x \notin \mathcal{U}$, that is, $\bar{x} \in \mathcal{U}$; flipping every move, we obtain a strategy that, from the position $\bar{s}$, answers its opponent and forces the play into $\mathcal{U}$. Player II can transplant this strategy into $\mathrm{Ul}(\mathcal{U})$. She makes her first move at least as long as s, so that at her next turn the position is longer than s. She then pretends that the position is $\bar{s}$ followed by a single move of her opponent, namely the digits of the actual position beyond the length of s, and from then on she answers as the flipped strategy does. The resulting play differs in finitely many digits from a play that the flipped strategy wins, and membership in $\mathcal{U}$ does not depend on finitely many digits; the transplanted strategy is therefore a winning strategy for Player II in $\mathrm{Ul}(\mathcal{U})$, which we have just seen to be impossible. This proves (1).

Baire property and measure. If Z is comeager, then Player II has a winning strategy in $\mathrm{Ul}(Z)$: for each $k \in \omega$, she plays her k-th move into the k-th of countably many dense open sets whose intersection lies in Z; if Z is meager, then Player I has a winning strategy of the same kind. By (1), the set $\mathcal{U}$ is therefore neither meager nor comeager. Next, a set $Z \subseteq 2^\omega$ that is invariant under finite changes (that is, that contains every sequence differing in finitely many digits from a member of Z) and has the Baire property is meager or comeager. Indeed, suppose that Z is not meager. Since Z has the Baire property, it differs by a meager set from an open set, which must be non-empty; hence the difference $N_s \setminus Z$ is meager for some string s. For every string t of the same length as s, the homeomorphism of 2^ω that replaces the initial segment s by t carries N_s onto N_t and preserves Z; it follows that $N_t \setminus Z$ is meager as well. The finitely many sets N_t cover 2^ω, whence the complement of Z is meager. As $\mathcal{U}$ is invariant under finite changes and is neither meager nor comeager, it does not have the Baire property. Measure is no better. The flip $x \mapsto \bar{x}$ preserves the coin-tossing measure λ and carries $\mathcal{U}$ onto its complement, so, if $\mathcal{U}$ were measurable, we would have $\lambda(\mathcal{U}) = 1 - \lambda(\mathcal{U}) = \frac{1}{2}$, while Kolmogorov's zero–one law, applied to a measurable set that is invariant under finite changes, gives $\lambda(\mathcal{U}) \in \{0, 1\}$. This proves (2).

Determinacy kills ultrafilters. We show that, if all of Ulam's games are determined, then every subset of 2^ω has the Baire property; part (2) then leaves no room for a nonprincipal ultrafilter. Assume that every game $\mathrm{Ul}(Z)$ is determined, and let $Z \subseteq 2^\omega$ and a finite binary string s be given. Applying determinacy to the set $\{z \in 2^\omega : s^\frown z \in Z\}$ and the theorem of Banach and Oxtoby (see Theorem 3.5), we find a string $s' \supseteq s$ such that Z is comeager in

$N_{s'}$ or meager in $N_{s'}$. Let O be the union of the sets N_u, for finite binary strings u, in which Z is comeager, and let O' be the union of those in which Z is meager. The union $O \cup O'$ is then a dense open set, and the symmetric difference $Z \triangle O$ is covered by $O \setminus Z$, by $O' \cap Z$, and by the nowhere dense set $2^\omega \setminus (O \cup O')$, all of which are meager. Since Z was arbitrary, every subset of 2^ω has the Baire property, and by (2) there is no nonprincipal ultrafilter on ω. Applied to the sets of a pointclass, a family of sets of reals that is closed under continuous preimages (such as the class of analytic sets), the same argument leads to the remarks on definable ultrafilters at the end of this section.

Selective ultrafilters. The proof of part (4) uses a theorem of Kunen, recorded in the history above, which we quote without proof.

Lemma 14.2 (Kunen). *If $\mathcal{U}$ is a selective ultrafilter on ω, then for every coloring of the two-element subsets of ω in two colors there is a set $H \in \mathcal{U}$ all of whose two-element subsets have the same color.*

The idea of the first half of part (4) is to color each pair according to whether the strategy of Player I still allows its larger element after its smaller one; a homogeneous set in $\mathcal{U}$, that is, a set in $\mathcal{U}$ all of whose pairs have the same color, then exists by Lemma 14.2, and Player II can play such a set in full. Suppose that $\mathcal{U}$ is selective, and let σ be a strategy for Player I in $S(\mathcal{U})$. For a finite set $s \subseteq \omega$, listed increasingly, let $A_s \in \mathcal{U}$ be the move of σ after Player II has played the elements of s (we put $A_s = \omega$ if this sequence of moves is not legal), and let $B_n = \bigcap\{A_s : s \subseteq \{0, \dots, n-1\}\}$. As a finite intersection of members of $\mathcal{U}$, the set B_n belongs to $\mathcal{U}$. We color a pair $m < n$ pink if $n \in B_{m+1}$, and purple otherwise. By Lemma 14.2 there is a homogeneous set $H \in \mathcal{U}$; replacing H by $H \cap A_\emptyset$, which is still homogeneous and belongs to $\mathcal{U}$, we may assume that $H \subseteq A_\emptyset$. The color of H is not purple: for $m = \min H$, the sets $H \setminus \{m\}$ and B_{m+1} both belong to $\mathcal{U}$ and therefore meet, and a common point n yields a pink pair $\{m, n\}$ inside H. The color of H is thus pink. We list H increasingly as $h_0 < h_1 < \cdots$. Since $H \subseteq A_\emptyset$ and every pair in H is pink, we have $h_0 \in A_\emptyset$, and $h_{k+1} \in B_{h_k+1} \subseteq A_{\{h_0,\dots,h_k\}}$ for every k. Hence Player II may answer σ with $h_0, h_1, h_2, \dots$ in turn, and the set she produces is $H \in \mathcal{U}$. Consequently, the strategy σ is not winning.

Conversely, suppose that $\mathcal{U}$ is not selective, and fix a partition $\{P_j : j \in J\}$ of ω, indexed by a set J, into pieces outside $\mathcal{U}$ such that no element of $\mathcal{U}$ meets every piece in at most one point. Player I wins by always removing the pieces already visited. When Player II has played elements of $P_{j_0}, \dots, P_{j_{k-1}}$, Player I plays $\omega \setminus (P_{j_0} \cup \cdots \cup P_{j_{k-1}})$, which belongs to $\mathcal{U}$ because a finite union of sets outside an ultrafilter lies outside it. The set produced by Player II then meets each piece in at most one point, and it is not in $\mathcal{U}$. The P-point game is handled by the same two arguments, with finite sets in place of single points and the P-point property in place of homogeneity; for the details we refer to the book of Bartoszyński and Judah [BJ95]. This proves (4).

Selectivity is exactly what the Ramsey theory of Section 12 needs. By the theorem of Mathias recalled in the history above, if $\mathcal{U}$ is selective, then every analytic $\mathcal{A} \subseteq [\omega]^\omega$ has a homogeneous set in $\mathcal{U}$: there is $H \in \mathcal{U}$ with $[H]^\omega \subseteq \mathcal{A}$ or $[H]^\omega \cap \mathcal{A} = \emptyset$ [Mat77]. In the other direction, every nonprincipal ultrafilter, selective or not, yields a subset of $[\omega]^\omega$ that is not Ramsey (see the problem at the beginning of Section 12 and Exercise 6 at the end of that section).

14.5. **Further remarks (optional).**

Definable ultrafilters. The proof of part (3) of Theorem 14.1 works one pointclass at a time. The *projective* sets are the sets of reals obtained from the Borel sets by finitely many applications of the operations of taking continuous images and complements, and *projective*

determinacy is the statement that every game in which the players choose natural numbers and whose payoff set is projective is determined. Under projective determinacy (a consequence of the existence of infinitely many Woodin cardinals, by the theorem of Martin and Steel mentioned in Section 16 below), every projective set has the Baire property, so no nonprincipal ultrafilter is projective either. Without an extra hypothesis of this kind, the bound recorded after the main result, that no nonprincipal ultrafilter is analytic or coanalytic, cannot be extended to the next class of projective sets, the Δ^1_2 sets, which we now define. A set of reals is Σ^1_2 if it is a continuous image of a coanalytic subset of ω^ω, and Δ^1_2 if both the set and its complement are Σ^1_2. In Gödel's constructible universe, a transfinite recursion along the canonical well-ordering of the reals, which is Σ^1_2, produces a Σ^1_2 nonprincipal ultrafilter $\mathcal{U}$, which is even Δ^1_2, since $x \notin \mathcal{U}$ if and only if $\omega \setminus x \in \mathcal{U}$.

Meager filters. The game argument shows that a nonprincipal ultrafilter is not meager; in the following theorem we characterize the meager filters, and thereby isolate the part of the ultrafilter property that is responsible.

Theorem 14.3 (Jalali-Naini, 1976; Talagrand, 1980). *A filter $\mathcal{F}$ on ω that contains every cofinite set is meager, as a subset of 2^ω, if and only if there is a partition of ω into consecutive finite intervals $I_0 < I_1 < \cdots$ (every element of I_n being smaller than every element of I_{n+1}, for $n \in \omega$) such that every element of $\mathcal{F}$ meets all but finitely many of the I_n.*

Proof. One direction is a direct check. For the other, the idea is to build a single set $e \subseteq \omega$ such that the points of the k-th of a sequence of closed nowhere dense sets covering $\mathcal{F}$ disagree with e on every later block of a partition into intervals, and these blocks are the intervals we need. If such a partition exists, then the filter is contained in the union over $m \in \omega$ of the sets $\{x : x \cap I_n \neq \emptyset \text{ for all } n \geq m\}$. These sets are closed, and nowhere dense, because any finite string can be extended by zeros across a whole interval I_n. Suppose now that $\mathcal{F}$ is meager, and let $C_0 \subseteq C_1 \subseteq \cdots$ be closed nowhere dense sets whose union contains $\mathcal{F}$. We choose $0 = n_0 < n_1 < \cdots$ and binary strings w_k of length $n_{k+1} - n_k$ such that $N_{u^\frown w_k} \cap C_k = \emptyset$ for every string u of length n_k. This is possible because there are only finitely many such u, and we may extend a candidate for w_k once for each of them, each time avoiding C_k. Let $I_k = [n_k, n_{k+1})$ and let $e \subseteq \omega$ be the set whose characteristic function is $w_0{}^\frown w_1{}^\frown \cdots$. Since $C_m \subseteq C_k$ for $k \geq m$, every point of C_m disagrees with e somewhere on I_k for each $k \geq m$. Now let $x \in \mathcal{F}$ be given. The set $x \cup e$ then belongs to $\mathcal{F}$ and therefore lies in C_m for some m; for every $k \geq m$, the sets $x \cup e$ and e then differ somewhere in I_k, which means that x meets I_k. Hence the intervals I_k witness the condition of the theorem. ☺

A nonprincipal ultrafilter $\mathcal{U}$ fails the condition of Theorem 14.3 for every partition: the union of the even intervals and the union of the odd intervals are complementary, so one of them belongs to $\mathcal{U}$, and it misses infinitely many intervals. This is a second proof that $\mathcal{U}$ is not meager. By Theorem 14.3, nonmeagerness also survives finite-to-one images. Let $\mathcal{F}$ be a filter on ω that contains every cofinite set, let $f\colon \omega \to \omega$ be finite-to-one, and let $f(\mathcal{F}) = \{y \subseteq \omega : f^{-1}(y) \in \mathcal{F}\}$ be the image filter. If $f(\mathcal{F})$ is meager, witnessed by a partition $\{I_n\}$, then every $x \in \mathcal{F}$ meets $f^{-1}(I_n)$ for all but finitely many n, because the image $f[x] = \{f(m) : m \in x\}$ belongs to $f(\mathcal{F})$. The sets $f^{-1}(I_n)$ are finite and pairwise disjoint, and infinitely many of them are non-empty; hence any partition of ω into consecutive intervals each of which contains a non-empty set of the form $f^{-1}(I_n)$ witnesses that $\mathcal{F}$ is meager.

Points that nobody can point at. The space ω^* is the Stone space of the Boolean algebra $\mathcal{P}(\omega)/\mathrm{fin}$, the quotient of $\mathcal{P}(\omega)$ by the ideal of finite sets, and its points, the ultrafilters of this algebra, correspond to the nonprincipal ultrafilters on ω. Pospíšil showed in 1937 that

there are $2^{2^{\aleph_0}}$ of them, by means of an independent family of $2^{\aleph_0}$ subsets of ω in the sense of Fichtenholz, Kantorovich and Hausdorff: every intersection of finitely many members of the family and of the complements of finitely many other members is infinite. Hence every choice, for each member, of either the member or its complement extends to a nonprincipal ultrafilter, and distinct choices give distinct ultrafilters. A P-point is exactly a point $p \in \omega^*$ such that every G_δ subset of ω^* containing p is a neighborhood of p. If $D \subseteq \omega^*$ is countable and p lies in the closure of D but not in D, then the complement $\omega^* \setminus D$ is a G_δ set containing p that is not a neighborhood of p; no homeomorphism of ω^* can therefore move a P-point to such a point. Points of this kind exist in ZFC, and P-points exist under the continuum hypothesis; this is Rudin's proof that ω^* is not homogeneous [Rud56]. In 1978 Kunen proved in ZFC that there are *weak P-points*, points of ω^* that are not limits of any countable subset of the rest of ω^* [Kun80]; they give a proof that ω^* is not homogeneous without the continuum hypothesis. All of these points are produced by choice, and by the remarks on definable ultrafilters above, none of them can be produced by a definition of low complexity. The large-cardinal picture is sharper still: Todorčević proved that, in the presence of large cardinals, the selective ultrafilters are exactly the filters that are generic over $L(\mathbb{R})$, the smallest transitive model of ZF that contains all the ordinals and all the reals, for the partial order $\mathcal{P}(\omega)/\mathrm{fin}$ (see Farah [Far98]). A selective ultrafilter $\mathcal{U}$ is thus a generic object for the very Boolean algebra whose Stone space it inhabits, and every statement about it that holds in $L(\mathbb{R})[\mathcal{U}]$, the smallest such model that also contains $\mathcal{U}$, is forced by one of its own members, as every true statement about a Cohen real is forced by a finite initial segment of that real.

Exercises.

(1) Let $\mathcal{U}$ be an ultrafilter on ω that contains a finite set. Show that $\mathcal{U}$ is principal. Deduce that every nonprincipal ultrafilter $\mathcal{V}$ on ω contains every cofinite set, and that if $x \in \mathcal{V}$ and $y \subseteq \omega$ differs from x in finitely many points, then $y \in \mathcal{V}$.

(2) For each of the following sets $Z \subseteq 2^\omega$, decide which player has a winning strategy in $\mathrm{Ul}(Z)$, and check the answer against Theorem 3.5: (a) the sequences with infinitely many terms equal to 1; (b) the sequences that are eventually 0; (c) the sequences x with $x(n) = 1$ for infinitely many even n; (d) the sequences x with $x(0) = 0$.

(3) Let $\mathcal{F}$ be the filter of cofinite subsets of ω. Show that $\mathcal{F}$ is meager in 2^ω, first directly and then by means of Theorem 14.3.

(4) Let $\mathcal{U}$ be a nonprincipal ultrafilter on ω such that Player I has no winning strategy in $S(\mathcal{U})$. Show, without using Theorem 14.1, that Player I has no winning strategy in $P(\mathcal{U})$ either. (Hint: Player II may play one-element sets in $P(\mathcal{U})$.)

(5) $(*)$ Let $\mathcal{U}$ be a nonprincipal ultrafilter on ω with the property in the conclusion of Lemma 14.2. Show that $\mathcal{U}$ is selective. (Hint: given a partition of ω into pieces outside $\mathcal{U}$, color a pair according to whether its two elements lie in the same piece.)

Part

III

Applications of other topological games

The last section of Part II ended with a game that is not determined, the game defined by a nonprincipal ultrafilter on ω (see Theorem 14.1). In this part the question of determinacy, whether one of the players has a winning strategy, becomes the theme. The players no longer shrink a set: in the Menger game they choose open covers and finite subfamilies of them, and in the games of descriptive set theory they call out natural numbers, one at a time. What survives from Part I is the framework of Section 2 (see Definition 2.2), in which a game is a tree of positions together with a set of plays won by Player II.

Roadmap

A property of a space can be turned into a statement about one player of a game, and a stronger property into a statement about the other player. When this happens, the two properties coincide on a space exactly when the game on that space is determined. In Section 15 we follow this idea through Menger's conjecture: the Menger property and, for separable metrizable spaces, σ-compactness are equivalent to statements about the two players of the Menger game (see Theorem 15.1), and the theorem of Fremlin and Miller shows that the game can be undetermined on a set of reals. In Section 16 determinacy changes its role: it becomes a tool rather than a question. There the games are played with natural numbers, and Martin's theorem says that every game with a Borel payoff set is determined (see Theorem 16.1). Wadge's comparison of two Borel sets then follows from the determinacy of a single game, in which a winning strategy for either player yields a continuous reduction: of the first set to the second, or of the second to the complement of the first. The two sections are independent of each other.

15. Applications to set-theoretic topology

The problem. Compact sets are the "small" sets of topology, and countable unions of compact sets (σ-compact sets) are almost as tame. In 1924 Karl Menger proposed a covering property that looked like a clean characterization of σ-compactness. A topological space X has the *Menger property* if for every sequence $\mathcal{U}_1, \mathcal{U}_2, \dots$ of open covers of X there are finite sets $\mathcal{F}_n \subseteq \mathcal{U}_n$ such that $\bigcup_{n\geq 1} \mathcal{F}_n$ covers X. Every σ-compact space has the Menger property: if $X = \bigcup_{n\geq 1} K_n$ with each K_n compact, it suffices to take for $\mathcal{F}_n$ a finite subset of $\mathcal{U}_n$ that covers K_n. Menger conjectured the converse for metric spaces. The problem is to decide Menger's conjecture, in particular for sets of real numbers.

15.1. **History of the problem and of the solution using topological games.** Menger introduced the property in 1924 [Men24], in a different but equivalent form, while studying a basis property of metric spaces. He conjectured that, among metric spaces, the property characterizes σ-compactness. Hurewicz gave the property the form above in a paper of 1926 [Hur26]. In the same paper he proved a theorem that is now stated as a characterization of the property by an infinite game; the game form is due to later authors, notably Telgársky and Scheepers. Hurewicz also introduced a stronger property, for which he made the analogous conjecture. Under the continuum hypothesis Menger's conjecture fails: a Luzin set, an uncountable set of reals meeting every meager set in a countable set, has the Menger property but is not σ-compact. In 1984 Telgársky characterized σ-compactness of metrizable spaces by the same game [Tel84]. Menger's conjecture thereby became the statement that the Menger game is determined on every set of reals.

In 1988 David Fremlin and Arnold Miller solved the problem [FM88]: they proved, without any hypothesis beyond ZFC (Zermelo–Fraenkel set theory with the axiom of choice), that there is a set of reals with the Menger property that is not σ-compact. The closely related Rothberger property (1938), in which a single member of each cover is chosen, has a parallel story in terms of games. In 1978 Galvin introduced the point-open game [Gal78], in which one player chooses points and the other chooses open sets containing them (see the next subsection). In 1994 Pawlikowski proved that the point-open game on a set of reals is undetermined if and only if the set is uncountable and has the Rothberger property [Paw94]. A set of reals has *strong measure zero* if, for every sequence of positive lengths, it can be covered by a sequence of intervals with those lengths. Borel conjectured in 1919 that every set of reals of strong measure zero is countable; Laver proved in 1976 that the conjecture is consistent [Lav76]. Since the Rothberger property implies strong measure zero, every set of reals with the Rothberger property is countable in the model of set theory that Laver constructed, and by the characterization of 1994 the point-open game is determined there on every set of reals. Under the continuum hypothesis, on the other hand, there are Luzin sets, and the point-open game on every Luzin set is undetermined, by a theorem of Recław [Rec94].

15.2. **The games.** The *Menger game* on a topological space X is played in rounds $n = 1, 2, \dots$ (see Figure 20). In round n, Player I chooses an open cover $\mathcal{U}_n$ of X, and Player II chooses a finite set $\mathcal{F}_n \subseteq \mathcal{U}_n$. Player II wins if $\bigcup_n \mathcal{F}_n$ covers X. In the *Rothberger game* Player II is more frugal still: she chooses a single member of each cover, and she wins if the chosen sets cover X. Galvin's *point-open game* reverses the winning condition. In round n, Player I chooses a point $x_n \in X$ and Player II an open set $U_n \ni x_n$; Player II still chooses the open sets, as in the Menger and Rothberger games, but she now wins if the sets U_n do not cover X. The point-open game is dual to the Rothberger game, in the sense that Player I

has a winning strategy in either of the two games exactly when Player II has one in the other [Gal78]. In the literature Player I is called ONE and Player II is called TWO.

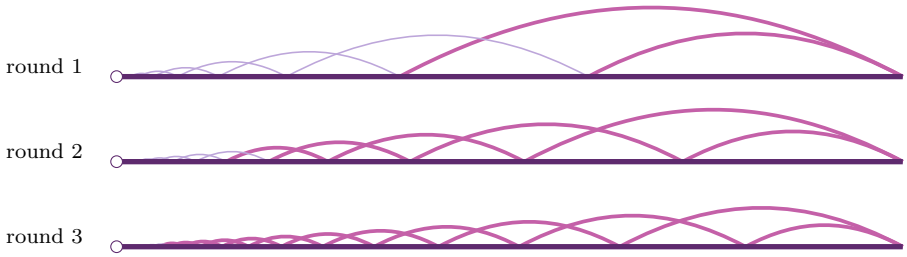


FIGURE 20. The Menger game on the half-open interval $X = (0, 4]$. In each round Player I chooses an open cover of X by intervals that accumulate at 0, drawn as arcs; Player II keeps finitely many of them (pink). Her choice in a single round never covers X, since the union of finitely many intervals of the cover misses an interval $(0, \varepsilon)$ with $\varepsilon > 0$. Player II nevertheless has a winning strategy, because X is σ-compact: in each round she covers a larger compact piece of X, and her choices together cover all of X.

15.3. **The Main Result of this section.** Hurewicz and Telgársky turned the Menger property and σ-compactness into statements about the two players of the Menger game, and Fremlin and Miller proved that the gap between these two statements is real.

Theorem 15.1 (Hurewicz, 1926; Telgársky, 1984; Fremlin–Miller, 1988).

(1) *A topological space X has the Menger property if and only if Player I has no winning strategy in the Menger game on X.*
(2) *A separable metrizable space X is σ-compact if and only if Player II has a winning strategy in the Menger game on X.*
(3) *There is a set of reals with the Menger property that is not σ-compact. Consequently, the Menger game on this set is not determined, and Menger's conjecture is false.*

15.4. **Solving the problem with the games.**

The Menger property. One direction is immediate. If Player I has no winning strategy and open covers $\mathcal{U}_1, \mathcal{U}_2, \ldots$ of X are given, then the strategy "play $\mathcal{U}_n$ in round n" is not winning: some play against it is won by Player II, and her moves witness the Menger property. The converse is the substantial direction, and we give only an outline. Applied to a constant sequence of covers, the Menger property shows that X is Lindelöf, that is, that every open cover of X has a countable subcover (see Exercise 2 at the end of this section). Player II may therefore restrict herself, at each position, to a fixed countable subcover of the cover that Player I's strategy plays. Only countably many finite positions can then arise, and the strategy plays only countably many covers. The Menger property, applied to a suitable arrangement of these countably many covers, yields a play that defeats the strategy. We omit the details, which go back to Hurewicz [Hur26].

σ-compactness. If $X = \bigcup_{n\geq 1} K_n$ with K_n compact and increasing, Player II has a winning strategy: in round n, she chooses a finite subfamily of $\mathcal{U}_n$ that covers K_n. Conversely,

Telgársky unravels a winning strategy of Player II, position by position, into countably many compact sets that cover X [Tel84].

Menger's conjecture. By parts (1) and (2) of Theorem 15.1, a separable metrizable space with the Menger property that is not σ-compact is exactly one on which neither player has a winning strategy in the Menger game. Fremlin and Miller obtain such a set of reals from combinatorial properties of families of functions from ω to ω under eventual domination, in an argument that splits into cases [FM88]. A function g *eventually dominates* a function f if $f(n) \leq g(n)$ for all but finitely many $n \in \omega$, and a family of such functions is *dominating* if every function from ω to ω is eventually dominated by a member of the family. The link with the Menger property is a later theorem of Hurewicz, from 1927: a subset of the space ω^ω of these functions, with the product topology (see Section 16), has the Menger property exactly when no continuous image of it in ω^ω is a dominating family. Seen as a statement about games, Menger's conjecture is attractive and false for one and the same reason: it asserts a form of determinacy. On analytic sets of reals, those that are empty or continuous images of ω^ω, the Menger game is determined, since an analytic set with the Menger property is σ-compact; on wild sets it need not be, and Fremlin and Miller proved that a wild enough set exists without any extra axioms.

Exercises.

(1) Let X be a countable space. Show that Player II has a winning strategy in the Rothberger game on X, and that Player I has a winning strategy in the point-open game on X.
(2) Show that a space with the Menger property is Lindelöf. Deduce that an uncountable discrete space does not have the Menger property, and describe a winning strategy for Player I in the Menger game on such a space.
(3) Show that the Menger property is inherited by closed subspaces and preserved by continuous surjections.
(4) Show that Player II has a winning strategy in the Menger game on $[0,1]$, but that Player I has a winning strategy in the Rothberger game on $[0,1]$. (Hint: in round n, let Player I cover $[0,1]$ by relatively open intervals of length at most 2^{-n-2}; a finite family of intervals covering $[0,1]$ has total length at least 1.)
(5) $(*)$ Show that the space ω^ω of infinite sequences of natural numbers, with the product topology, does not have the Menger property, and hence is not σ-compact. (Hint: let $\mathcal{U}_n$ consist of the sets $\{x \in \omega^\omega : x(n) = k\}$ for $k \in \omega$.) Deduce from Theorem 15.1(1) that Player I has a winning strategy in the Menger game on ω^ω, and find one directly.

16. Applications to descriptive set theory

The games of this section are played with natural numbers, and their payoff sets are sets of infinite sequences of natural numbers, which descriptive set theory sorts by their complexity; we begin by recalling the space in which these sets live and the classes into which they are sorted.

Definitions. The space ω^ω of infinite sequences of natural numbers carries the product topology: for a finite sequence s of natural numbers, the basic open set $N_s = \{x \in \omega^\omega : s \subseteq x\}$ consists of the sequences that begin with s, and a set is open if it is a union of such sets. (By tradition the space ω^ω is called "the Baire space". The name is not to be confused with the notion of a Baire space in Definition 1.1, although the space ω^ω, being completely metrizable, is a Baire space in that sense as well.) The *Borel sets* of a topological space form the smallest family of subsets of the space that contains the open sets and is closed under complements

and countable unions. They are arranged in levels: the open sets, the closed sets, then the F_σ sets (countable unions of closed sets) and the G_δ sets (countable intersections of open sets), and so on through the countable ordinals; the level at which a Borel set first appears is its *Borel rank*. A subset of a separable completely metrizable space, such as ω^ω, the Cantor space 2^ω of infinite binary sequences, or $\mathbb{R}$, is *analytic* if it is empty or is a continuous image of ω^ω.

The problem. Here is the simplest infinite game imaginable: fix a set $E \subseteq \omega^\omega$ of infinite sequences of natural numbers. In the *Gale–Stewart game* $G(E)$, Player I and Player II take turns calling out natural numbers, Player I playing $x(0), x(2), \dots$ and Player II playing $x(1), x(3), \dots$, and Player II wins if the resulting sequence x belongs to E. In the language of Definition 2.2, the game $G(E)$ is the pair $(\omega^{<\omega}, E)$, with moves in $A = \omega$. Recall that a game is determined if one of the players has a winning strategy (see Remarks 2.3(4)). For finite games this is always the case, by Zermelo's theorem; in chess, where draws are possible, the same argument shows that one of the players can force a win or both can force a draw, although nobody knows which. For infinite games determinacy can fail: using the axiom of choice one can build sets E for which neither player has a winning strategy in $G(E)$. A nonprincipal ultrafilter $\mathcal{U}$ on ω gives such a set. In Ulam's game $\mathrm{Ul}(\mathcal{U})$ of Section 14, the players alternately choose non-empty finite binary strings, and Player II wins if the concatenation of the strings is the characteristic function of a set in $\mathcal{U}$; this game is not determined (see Theorem 14.1), and it becomes a Gale–Stewart game once the countably many non-empty finite binary strings are numbered, since calling out a number then means playing a string. The problem is to determine which sets, and in particular which Borel sets, give determined games, and what the answer reveals about the structure of the Borel sets.

16.1. **History of the problem and of the solution using infinite games.** The story of this problem is unusual: the game came first, and the applications came later. Gale and Stewart introduced these games in 1953 [GS53]. They proved that $G(E)$ is determined when E is open or closed, constructed an undetermined game from a well-ordering of the reals, and asked how far determinacy extends. Wolfe proved in 1955 that games with F_σ or G_δ payoff sets are determined [Wol55], and Davis extended the result in 1964 to the next level of the Borel hierarchy [Dav64]. In 1970 Martin proved that analytic games are determined if there is a measurable cardinal [Mar70]. In 1971 Friedman showed that any proof of Borel determinacy must use uncountably many iterations of the power set operation [Fri71]. The problem therefore lay beyond the methods of ordinary analysis. That a statement about sets of real numbers should need, for its proof, sets of sets of sets of ... of reals, iterated uncountably many times, was something new.

In 1975 Donald A. Martin solved the problem [Mar75]: every Borel game is determined. His proof uses iterated power sets of the reals of every countable length, and by Friedman's theorem no proof can do with less. Even before Martin's proof, Wadge had compared sets of reals through a game of his own. In work of 1972, written up in his thesis of 1983 [Wad83], he showed that determinacy makes the Borel sets almost linearly ordered under continuous reducibility; Martin and Monk showed that this ordering is well-founded. Determinacy was later extended far beyond the Borel sets: Martin and Steel proved in 1989 that projective games, those whose payoff sets are obtained from the Borel sets by finitely many applications of the operations of taking continuous images and complements, are determined if there are infinitely many Woodin cardinals [MS89].

Recognition. Martin's proof appeared in the *Annals of Mathematics*. Martin, Steel and Woodin received the Karp Prize of the Association for Symbolic Logic in 1988 for their work on determinacy, and the proof of projective determinacy by Martin and Steel appeared in the *Journal of the American Mathematical Society*.

16.2. **The games.** Besides the Gale–Stewart game $G(E)$ of the problem, we need a game for comparing two sets $E, F \subseteq \omega^\omega$. In the *Wadge game* $G_{\mathrm{W}}(E, F)$, Player I plays $x(0), x(1), \dots$ and Player II plays $y(0), y(1), \dots$, alternately, Player I first, and Player II wins if

$$x \in E \iff y \in F.$$

Player II thus tries to imitate Player I: whenever his sequence lands in E, hers should land in F, and vice versa. The play is the single sequence $(x(0), y(0), x(1), y(1), \dots)$; the Wadge game is therefore a Gale–Stewart game, and its payoff set is Borel when E and F are. A strategy of Player II turns the sequence of Player I into her own, one term at a time (see Figure 21).

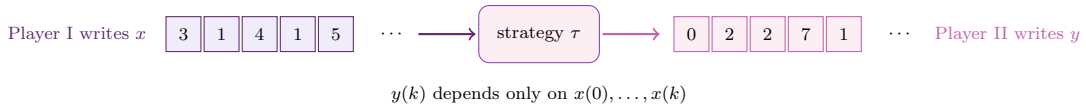


FIGURE 21. A strategy τ of Player II in the Wadge game $G_{\mathrm{W}}(E, F)$, for sets $E, F \subseteq \omega^\omega$, works like a machine that reads the sequence x of Player I and writes her sequence y, one term at a time; each term depends only on what she has seen so far. Such a machine defines a continuous map $x \mapsto y$, and if τ is winning, then the sequence x lies in E exactly when y lies in F, and E is therefore the preimage of F under the map.

16.3. **The Main Result of this section.** The main result is Martin's theorem, together with Wadge's consequence for the comparison of Borel sets. For $E, F \subseteq \omega^\omega$ write $E \leq_W F$, and say that E is *Wadge reducible* to F, if there is a continuous $f\colon \omega^\omega \to \omega^\omega$ with $E = f^{-1}(F)$.

Theorem 16.1 (Martin, 1975; Wadge, 1972).

(1) Borel determinacy. *For every Borel set $E \subseteq \omega^\omega$, the Gale–Stewart game $G(E)$ is determined.*
(2) Wadge's lemma. *For all Borel sets $E, F \subseteq \omega^\omega$, either $E \leq_W F$ or $F \leq_W \omega^\omega \setminus E$.*

By part (2), any two Borel sets are comparable under $\leq_W$, up to complements. Martin and Monk showed that $\leq_W$ is also well-founded on the Borel sets. Hence, if two sets $E, F \subseteq \omega^\omega$ are called *Wadge equivalent* when $E \leq_W F$ and $F \leq_W E$, the Borel sets, up to Wadge equivalence, form a well-founded hierarchy that refines the classical Borel hierarchy and is almost linear: two Borel sets that are not comparable under $\leq_W$ are, up to Wadge equivalence, a set and its complement, whence every antichain has at most two elements. The position of a Borel set in this hierarchy measures its complexity far more finely than its Borel rank does.

16.4. **Solving the problem with the games.**

Open games. The idea is the maxim "never step into a position from which you are already lost", and the openness of E guarantees that this cautious play is enough. Suppose that E is open and that Player II has no winning strategy in $G(E)$. We shall call a finite position *good for Player I* if Player II has no winning strategy from that position; by hypothesis, the empty position is good. If a position with Player I to move is good for him, some move leads to a good position, for otherwise Player II would have a winning strategy from every successor position and hence from the position itself. If a position with Player II to move is good for Player I, so is every position reached in one move, since a move of Player II to a position that is not good would give her a winning strategy from the position she moved from. Player I can therefore play so that every position of the play is good; let x be a play in which he does so. If x were in E, the openness of E would provide an initial segment s of x with $N_s \subseteq E$. Player II would then win from the position s whatever the players did, and this position would not be good for Player I. Consequently, the play x lies outside E: the rule "move to a good position" is a winning strategy for Player I, and the game is determined.

Borel games. Martin's proof [Mar75] reduces every Borel game to a clopen game on a larger tree. In outline, for each Borel set E he builds a game G' whose moves are more complicated objects than natural numbers, together with a continuous map from plays of G' to plays of $G(E)$ that sends strategies to strategies; the payoff set of G', the preimage of E under this map, is clopen. The argument for open games works on any tree, and the game G' is therefore determined; the transfer of strategies makes $G(E)$ determined as well. The construction proceeds by induction on the Borel rank of E, and at each step the moves of the new game are sets of positions of the old one; this is why Martin's proof climbs through the iterated power sets of $\mathbb{R}$, in agreement with Friedman's theorem.

Wadge's lemma. Let $E, F \subseteq \omega^\omega$ be Borel sets. The payoff set of the Wadge game $G_{\mathrm{W}}(E, F)$ is then Borel, and by Borel determinacy one of the players has a winning strategy in this game. If Player II has one, her strategy is a blueprint for a continuous function: Player I feeds in an input sequence x one term at a time, and Player II's step-by-step answers are the output y. Each term of the output depends only on finitely many terms of the input, so the map $x \mapsto y$ is continuous; since the strategy is winning, the input x lies in E if and only if the output y lies in F, whence $E \leq_W F$. If Player I has one, his strategy turns each sequence y of moves of Player II continuously into a sequence x, with $y \in F$ if and only if $x \notin E$; this map witnesses $F \leq_W \omega^\omega \setminus E$. The game thus produces the reduction itself, and the comparability of Borel sets is literally the determinacy of a game.

Exercises.

(1) For each of the following sets $E \subseteq \omega^\omega$, decide which player has a winning strategy in $G(E)$, and describe one: (a) the sequences with $x(1) > x(0)$; (b) the sequences with $x(1) < x(0)$; (c) the sequences for which $\{x(2n) : n \in \omega\}$ is finite; (d) the sequences with $x(2n+1) = x(0)$ for infinitely many n.

(2) Show that $\leq_W$ is reflexive and transitive, and that $E \leq_W F$ implies $\omega^\omega \setminus E \leq_W \omega^\omega \setminus F$.

(3) Let $E \subseteq \omega^\omega$ be closed. Show that $G(E)$ is determined, by adapting the argument for open games with the roles of the players exchanged. (Hint: call a position good for Player II if Player I has no winning strategy from it.)

(4) Let E be the set of $x \in \omega^\omega$ with $x(n) = 0$ for some n, and let $F = N_{(0)}$ be the set of sequences beginning with 0. Show that Player I has a winning strategy in $G_{\mathrm{W}}(E, F)$, and that E is not Wadge reducible to F. Find a continuous f with $F = f^{-1}(\omega^\omega \setminus E)$, as Theorem 16.1(2) predicts.

(5) $(*)$ Let $E \subseteq \omega^\omega$ be open but not closed. Show that $F \leq_W E$ for every open set $F \subseteq \omega^\omega$. (Hint: fix $z \in \overline{E} \setminus E$; in $G_{\mathrm{W}}(F, E)$, let Player II copy z until an initial segment p of Player I's sequence satisfies $N_p \subseteq F$.)

Part

IV

Applications of non-topological games

The games of Part III are infinite, and whether they are determined is a deep question: Martin's theorem settles it for Borel payoff sets, and the axiom of choice provides games that are not determined. The games of this part are finite, played on boards and on families of finite sets, and for them determinacy is Zermelo's elementary theorem, recalled in the problem at the beginning of Section 16. The spirit of the earlier parts is nevertheless at work here, far from topology: a global question becomes a local one, what to play next.

Roadmap

In this part we prove two classical theorems by games. In each of them, a statement about a large family of objects (all two-colorings of a set, all points of a square) is reduced to a statement about a finite game. In Section 17 a random coloring shows that a good coloring exists; the potential function of Erdős and Selfridge takes the place of the random choice and builds such a coloring one element at a time (see Theorem 17.1). In Section 18 we show that the board game Hex cannot end in a draw, and we derive from this fact about finite boards the Brouwer fixed point theorem in dimension two (see Theorem 18.1). The two sections can be read in any order, and neither needs more than Section 2 from Part I.

17. Applications to the probabilistic method

The problem. The probabilistic method proves that combinatorial objects exist by showing that a random object has the required property with positive probability. For example, Erdős proved in 1963 that if n is a positive integer and a family $\mathcal{F}$ of n-element sets has fewer than 2^{n-1} members, then the elements of these sets can be colored red and blue so that no member of $\mathcal{F}$ is monochromatic: a random coloring makes a given member of $\mathcal{F}$ monochromatic with probability 2^{1-n}, so the expected number of monochromatic members is less than 1, and some coloring has none. Such proofs give no construction. The problem is to turn them into explicit, deterministic, efficient constructions.

17.1. **History of the problem and of the solution using combinatorial games.** The probabilistic method goes back to a 1943 paper of Szele on tournaments. Erdős made it

famous in 1947 with his lower bound for diagonal Ramsey numbers [Erd47]. The coloring theorem above is from 1963 [Erd63]. In 1973 Erdős and Selfridge studied the games now called Maker–Breaker games, in which the players alternately claim elements of a board and Maker tries to claim all the elements of some winning set [ES73]. They gave a strategy for Breaker based on a potential function: the expected number of winning sets that Maker would complete if each unclaimed element were given to Maker or to Breaker by the toss of a fair coin. Breaker, they showed, can play so that this expectation never grows over a move of Breaker followed by a move of Maker. With one player instead of two, the same potential argument solves the problem for the coloring theorem and related results: it colors the elements one at a time and never lets the expected number of monochromatic sets reach 1.

The argument was recognized as a general derandomization technique, the method of conditional probabilities. Spencer systematized the method in 1987 [Spe87], and Raghavan extended it in 1988 [Rag88] with pessimistic estimators: efficiently computable upper bounds on the conditional probabilities of failure, which take the place of those probabilities when they are too costly to compute. Beck built a theory of positional games on the Erdős–Selfridge theorem and presented it in his book of 2008 [Bec08].

Recognition. Erdős received the Wolf Prize in 1983/84.

17.2. **The game.** Let $\mathcal{F}$ be a family of subsets of a finite set V. In the *Maker–Breaker game* on $(V, \mathcal{F})$, Maker is Player I and Breaker is Player II. The players alternately claim unclaimed elements of V, Player I first, until all elements are claimed. Player I wins if he has claimed all the elements of some $A \in \mathcal{F}$, and Player II wins otherwise. Tic-tac-toe is the example to keep in mind: the set V is the board and the family $\mathcal{F}$ is the set of lines, with the difference that Breaker wins merely by preventing Maker from completing a line. On the board $\{1, \dots, n\}^d$, for integers $n \geq 2$ and $d \geq 1$, a *line* is a set $\{c_1, \dots, c_n\}$ of cells in which each of the d coordinates is either the same in all the cells, or equal to m in c_m for $1 \leq m \leq n$, or equal to $n + 1 - m$ in c_m for $1 \leq m \leq n$, and at least one coordinate is not the same in all the cells; ordinary tic-tac-toe is the case $n = 3$, $d = 2$.

17.3. **The Main Result of this section.** The main result is the Erdős–Selfridge theorem, together with two classical consequences: a derandomized coloring and a result on multidimensional tic-tac-toe.

Theorem 17.1 (Erdős–Selfridge, 1973). *Let $\mathcal{F}$ be a family of subsets of a finite set V.*

(1) *If $\sum_{A \in \mathcal{F}} 2^{-|A|} < \frac{1}{2}$, then in the Maker–Breaker game on $(V, \mathcal{F})$ in which Maker moves first, Breaker has a winning strategy.*
(2) *If $\sum_{A \in \mathcal{F}} 2^{1-|A|} < 1$, then a coloring of V in two colors with no monochromatic member of $\mathcal{F}$ can be computed in time polynomial in $|V|$ and $|\mathcal{F}|$.*
(3) *If $n \geq 2$ and $d \geq 1$ are integers with $\frac{1}{2}\big((n+2)^d - n^d\big) < 2^{n-1}$, then the first player cannot force a win in tic-tac-toe on the board $\{1, \dots, n\}^d$, the game in which the players alternately claim cells and the first to claim all the cells of a line wins.*

Part (3) follows from part (1) because the board (see Figure 22 for the case $n = 3$, $d = 2$) has $\frac{1}{2}\big((n+2)^d - n^d\big)$ lines of n cells each: the second player uses Breaker's strategy and occupies a cell of every line. By the Hales–Jewett theorem (1963), on the other hand, the first player wins when d is large compared with n [HJ63].

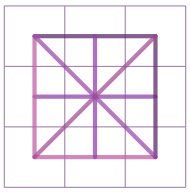


FIGURE 22. The eight winning lines of ordinary tic-tac-toe, each of three cells. Here the sum of $2^{-|A|}$ over the eight lines A is $8 \cdot 2^{-3} = 1$, which is not less than $\frac{1}{2}$, so Theorem 17.1(1) does not apply; in fact Maker wins the Maker–Breaker game on this board (see Exercise 3 at the end of this section).

17.4. **Solving the problem with the game.** The proof of Theorem 17.1(1) turns the expectation of the problem into a potential function that Player II never lets grow. At any stage of a play of the Maker–Breaker game on $(V, \mathcal{F})$, we call $A \in \mathcal{F}$ *alive* if Player II has claimed none of the elements of A, we denote by $u(A)$ the number of unclaimed elements of A, and, for $v \in V$, we set

$$\Phi = \sum_{A \text{ alive}} 2^{-u(A)}, \qquad w(v) = \sum_{A \text{ alive}, v \in A} 2^{-u(A)}.$$

If the remaining elements were assigned to the two players by fair coin tosses, the quantity Φ would be the expected number of sets completed by Player I. A claim of v by Player I increases Φ by $w(v)$, and a claim of v by Player II decreases it by $w(v)$. By hypothesis the potential starts below $\frac{1}{2}$, and after the first move of Player I it is below 1 (see Exercise 1 at the end of this section). Player II's strategy is to claim, at each move, an unclaimed element v of maximal weight $w(v)$. Her claim of an element kills the alive sets that contain the element and leaves the other terms unchanged, so it increases no weight; Player I's next claim therefore raises Φ by at most the weight that Player II has just removed. Over each pair of moves the potential does not increase, and it stays below 1. If Player I completed a set A, the term of A in Φ would be $2^0 = 1$, which is impossible, since $\Phi < 1$ and all terms are nonnegative; Breaker thus wins. For part (2) one player colors the elements one at a time, choosing at each step a color that does not increase the expected number of monochromatic sets (such a color exists: see Exercise 4). The expectation starts below 1, never increases, and at the end is an integer, hence 0. Each conditional expectation is a sum of $|\mathcal{F}|$ explicit terms, and the whole computation takes time polynomial in $|V|$ and $|\mathcal{F}|$. The proof of part (2) is more than an existence proof: it is an explicit algorithm, short enough to write in a few lines of Python.

The game adds robustness to the probabilistic method. A probabilistic argument shows that a random play is good on average; a strategy forces a good outcome one move at a time, even against an adversary who chooses half of the moves. Forcing the outcome move by move is what makes the argument constructive, and the simplicity of each move is what makes it efficient.

Exercises.

(1) Check the bookkeeping in the proof of Theorem 17.1(1): a claim of v by Player I increases Φ by exactly $w(v)$, a claim of v by Player II decreases Φ by exactly $w(v)$, and, under the hypothesis of part (1), we have $\Phi < 1$ after the first move of Player I.

(2) Show that the board $\{1,\dots,n\}^2$ has $2n+2 = \frac{1}{2}\big((n+2)^2 - n^2\big)$ lines, and find the least n for which Theorem 17.1(3) applies with $d=2$. Do the same for $d=3$, where the number of lines is $3n^2+6n+4$.

(3) Consider the Maker–Breaker game on the 3×3 tic-tac-toe board, with $\mathcal{F}$ the set of its 8 lines. Show that $\sum_{A\in\mathcal{F}} 2^{-|A|} = 1$, so Theorem 17.1(1) does not apply, and show that Maker has a winning strategy. (Hint: Maker takes the center and then creates two threats at once.) Explain why Maker's win does not contradict the fact that ordinary tic-tac-toe is a draw.

(4) Let $\mathcal{F}$ be a family of subsets of a finite set V with $\sum_{A\in\mathcal{F}} 2^{1-|A|} < 1$, as in Theorem 17.1(2). Explain why, in the proof of Theorem 17.1(2), one of the two colors for the next element never increases the conditional expected number of monochromatic members of $\mathcal{F}$.

(5) (∗) Let $n \geq 1$ be an integer, let V be the set of vertices of a complete binary tree with n levels, and let $\mathcal{F}$ be the family of the 2^{n-1} paths from the root to a leaf. Show that $\sum_{A\in\mathcal{F}} 2^{-|A|} = \frac{1}{2}$ and that Maker has a winning strategy. Thus the strict inequality in Theorem 17.1(1) cannot be weakened.

18. Applications to fixed point theory

The problem. The *Brouwer fixed point theorem* says that, for every positive integer n, every continuous map $f\colon [0,1]^n \to [0,1]^n$ has a fixed point. The original proofs used algebraic topology. The problem is to find an elementary, combinatorial proof that exposes the finite core of the theorem and, ideally, yields a method for locating approximate fixed points.

18.1. **History of the problem and of the solution using combinatorial games.** The theorem was proved in general by Hadamard in 1910 and Brouwer in 1912 [Bro12], with the methods of the new algebraic topology. A combinatorial route was opened by Sperner's lemma of 1928 on labeled triangulations [Spe28], from which Knaster, Kuratowski and Mazurkiewicz derived the fixed point theorem in 1929. Scarf turned this line of thought into an algorithm for computing approximate fixed points in 1967 [Sca67]. Meanwhile the game of Hex was invented by Piet Hein in 1942 and again, apparently independently, by John Nash at Princeton in 1948 or 1949. Nash observed that the game cannot end in a draw and showed, with a strategy-stealing argument, that the first player has a winning strategy. The argument shows that a winning strategy for the second player could be adapted by the first, since an extra stone never hurts its owner, so that the second player has none (see Exercise 4 at the end of this section).

David Gale found the game-theoretic solution in 1979 [Gal79]. He proved that the Brouwer fixed point theorem is an easy consequence of the fact that Hex cannot end in a draw, and he extended both statements to n dimensions.

Recognition. Gale's paper, in the *American Mathematical Monthly*, received a Lester R. Ford Award of the Mathematical Association of America in 1980, the year in which Gale shared the John von Neumann Theory Prize. Nash received the Nobel Memorial Prize in Economic Sciences in 1994 and the Abel Prize in 2015.

18.2. **The game.** Hex is played on a rhombus-shaped board of hexagonal cells, and anyone who has played it knows the feeling that somebody *must* win: a complete blockade of one player seems to be, at the same time, a connection for the other. To make this precise, we fix an integer $k \geq 1$ and describe the board by the centers of its cells, the points (i,j) with

integers $1 \le i, j \le k$. The cell (i, j) is adjacent to those of the cells $(i \pm 1, j)$, $(i, j \pm 1)$, $(i+1, j+1)$ and $(i-1, j-1)$ that lie on the board. Player I owns the bottom and top rows (where $j = 1$ and $j = k$), and Player II the left and right columns (where $i = 1$ and $i = k$). The players alternately place stones on empty cells, Player I first. A *chain* is a sequence of cells, each adjacent to the next. Player I wins if his stones contain a chain from the bottom row to the top row, and Player II wins if hers contain one from the left column to the right column.

18.3. **The Main Result of this section.** The main result is the Hex theorem and, as a consequence, the fixed point theorem in dimension two.

Theorem 18.1 (Gale, 1979).

(1) The Hex theorem. *If $k \ge 1$ is an integer and every cell of the $k \times k$ Hex board is occupied by a stone of Player I or of Player II, then exactly one of the players has a chain of his or her stones joining his or her two sides of the board.*
(2) *The Hex theorem implies the Brouwer fixed point theorem for $[0,1]^2$.*

Part (1) has an n-dimensional counterpart, which Gale also proved [Gal79]: in n-dimensional Hex, for integers $n \ge 2$ and $k \ge 1$, played by n players on the grid $\{1, \dots, k\}^n$, a full board has at least one winner, and this statement implies the Brouwer fixed point theorem for $[0,1]^n$.

18.4. **Solving the problem with the game.**

No draws. Gale's idea is to follow the boundary between the stones of the two players, which we picture in two colors, until it can go no further. We add an extra row of Player I's stones above and below the board, and an extra column of Player II's stones on either side. Starting at a corner of the enlarged board, between the two colors, we walk along the edges of the hexagons that separate a stone of Player I from one of Player II. At each vertex of the walk three cells meet, two of one color and one of the other; the walk therefore continues in exactly one way, and it never returns to a vertex, since a vertex already visited has no unused separating edge. The board is finite, so the walk ends, and it can end only at another corner of the enlarged board; the stones of one color along one side of the walk then form a winning chain. The two players cannot both win: a bottom-to-top chain and a left-to-right chain would have to cross, and on the hexagonal board crossing chains share a cell. The walk on a full board is shown in Figure 23.

Brouwer's theorem. The idea is that a winning chain must cross from a side where the map pushes points one way to a side where it pushes them the other way, and continuity forbids such a reversal between neighboring cells. Suppose that a continuous map $f = (f_1, f_2)\colon [0,1]^2 \to [0,1]^2$ has no fixed point. For points $z = (z_1, z_2)$ and $z' = (z_1', z_2')$ of $[0,1]^2$, we write $|z - z'|_\infty = \max(|z_1 - z_1'|, |z_2 - z_2'|)$. By compactness there is $\varepsilon > 0$ with $\max(|f_1(z) - z_1|, |f_2(z) - z_2|) \ge \varepsilon$ for all $z \in [0,1]^2$, and by uniform continuity there is δ with $0 < \delta < \varepsilon$ such that $|f_1(z) - f_1(z')| < \varepsilon$ and $|f_2(z) - f_2(z')| < \varepsilon$ whenever $|z - z'|_\infty \le \delta$. We choose an integer $k \ge 2$ with $1/(k-1) \le \delta$, place at each cell (i, j) the point $z = ((i-1)/(k-1), (j-1)/(k-1))$, and give the cell to Player II if $|f_1(z) - z_1| \ge \varepsilon$, and to Player I otherwise, in which case $|f_2(z) - z_2| \ge \varepsilon$. According to the Hex theorem, one of the players has a winning chain. Suppose first that Player II has a chain from left to right. On the left side we have $z_1 = 0 \le f_1(z)$, whence $f_1(z) - z_1 \ge \varepsilon$; on the right side, the inequality $z_1 = 1 \ge f_1(z)$ forces $z_1 - f_1(z) \ge \varepsilon$. The sign of $f_1(z) - z_1$ must change along the chain, so two adjacent cells of the chain, carrying points z and z', satisfy $f_1(z) - z_1 \ge \varepsilon$ and $z_1' - f_1(z') \ge \varepsilon$. Adding these inequalities, we obtain $(f_1(z) - f_1(z')) + (z_1' - z_1) \ge 2\varepsilon$,

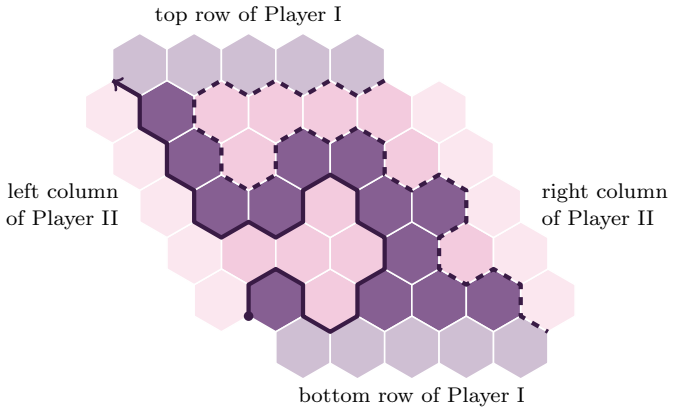


FIGURE 23. The no-draw argument on a full 5×5 board (see Theorem 18.1). The stones of Player I are plum and those of Player II are pink; the paler cells are the extra rows and columns. The walk starts at the lower left corner, between the two colors, and follows the edges that separate plum from pink. It cannot stop or turn back, so it leaves the board at another corner, and the plum stones along its right side form a chain from bottom to top. The dashed line is the walk that starts at the lower right corner.

which is impossible because both terms are less than ε. A chain of Player I leads to the same contradiction, with f_2 in place of f_1. Hence every continuous map of $[0,1]^2$ into itself has a fixed point.

Seen through the game, the topological content of the theorem is the fact that Hex cannot end in a draw. The argument also has a finite form. Taken in the contrapositive, it shows that for every continuous map $f\colon [0,1]^2 \to [0,1]^2$ and every $\varepsilon > 0$, some point z of a sufficiently fine grid satisfies $|f(z) - z|_\infty < \varepsilon$; by compactness these approximate fixed points have a limit point as $\varepsilon \to 0$, and by continuity it is a fixed point. Combinatorial proofs of fixed point theorems, from Sperner's lemma to Scarf's algorithm, enter computation in exactly this finite form.

Exercises.

(1) List the adjacent pairs of cells of the 2×2 Hex board, and check directly that in each of the 16 ways of filling the board exactly one player has a winning chain, as Theorem 18.1(1) asserts.

(2) Show that on the 3×3 board Player I has a winning strategy. (Hint: open with the center cell $(2,2)$; its neighbors $(1,1)$ and $(2,1)$ lie in the bottom row and its neighbors $(2,3)$ and $(3,3)$ in the top row, and Player I answers a move of Player II in one of these pairs with the other cell of the pair.)

(3) In the proof of Theorem 18.1(2), check that the points z and z' placed at adjacent cells satisfy $|z - z'|_\infty \le \delta$, and carry out the argument for a winning chain of Player I, using f_2.

(4) $(*)$ Show that Player I has a winning strategy in Hex on the $k \times k$ board for every $k \ge 1$. (Hint: the reflection $(i,j) \mapsto (j,i)$ preserves adjacency and exchanges the sides of the two players; combine Theorem 18.1(1), the determinacy of finite games, and the fact that an extra stone never hurts its owner, to show that Player II cannot have a winning strategy.)

Part
V

Ehrenfeucht–Fraïssé games

The games of Part IV were finite games played on boards, and the game of this part is finite as well. Its board, however, is a pair of mathematical structures, two linear orders, say, or two graphs, and either of them may be infinite. The purpose of the game is to compare the two structures through the eyes of first-order logic. First-order logic can say that a linear order has exactly seventeen elements, or that it has no largest element; as we show in Corollary 21.5 below, it cannot say that a finite linear order has an even number of elements. A proof of such a negative statement must account for every sentence at once, and there are infinitely many sentences, some of them very long.

The game deals with the sentences by turning their quantifiers into moves. Suppose that a sentence $\exists x\, \varphi(x)$ holds in one structure and fails in the other. Player I then plays a witness for x in the structure where the sentence holds; whatever element Player II chooses in the other structure, the formula φ fails there, and the contest moves on to the simpler formula φ, with one more pair of elements matched. Each round thus answers one quantifier, and a sentence whose quantifiers are nested n deep is played out in n rounds. The Ehrenfeucht–Fraïssé theorem says that this correspondence is exact: Player II can survive n rounds precisely when the two structures satisfy the same sentences with at most n nested quantifiers (see Theorem 20.3). A statement about infinitely many sentences thereby becomes a statement about a single strategy, much as, in Part I, the largeness of a set in the sense of Baire category became a fact about what Player II can enforce.

Like the Banach–Mazur game, the game in the form in which we play it comes from Poland, but from Warsaw rather than Lwów. Between the wars the two cities shared Polish mathematics: Lwów, around Banach and Steinhaus and the journal *Studia Mathematica*, was the home of functional analysis, and Warsaw, around Sierpiński and Mazurkiewicz and the journal *Fundamenta Mathematicae*, the home of set theory, topology and logic. Ehrenfeucht found the game in Warsaw, by recasting as a game a characterization of elementary equivalence that Fraïssé had given in his thesis, and he published it in *Fundamenta Mathematicae* a year before Mycielski and Steinhaus proposed the axiom of determinacy (see Section 26).

Roadmap

In Section 19 we define the game and watch it separate a triangle from a path in two rounds. In Section 20 we prove the Ehrenfeucht–Fraïssé theorem (see Theorem 20.3); the one delicate point is the Hintikka formula of a tuple, a single formula that says everything about the tuple that can be said with a given number of nested quantifiers. In Section 21 we put the theorem to work. Player II wins on any two dense linear orders without endpoints, and on any two long finite linear orders by halving gaps; the second strategy yields the result on parity announced above and its counterpart for the connectivity of finite graphs. In Section 22 we first take the players away, which leaves Fraïssé's back-and-forth systems, and then let the game run for infinitely many rounds, which yields Cantor's theorem on countable dense orders and, with ultrafilters, a theorem of Keisler and Shelah: elementarily equivalent structures have isomorphic ultrapowers. In Section 23 we survey the games that do for other logics what this one does for first-order logic, from infinitary and finite-variable logics to the game comonads of current research. The history, which in the earlier parts accompanied each problem, is told at the end, in Section 24, because a single story runs through all the sections. From Part I the part needs only the language of games of Section 2; the paragraph on ultrapowers also uses Łoś's theorem and saturated models, and in Section 23 we state results and give references rather than proofs.

19. Elementary equivalence and the game

The problem. Two structures $\mathcal{M}$ and $\mathcal{N}$ in the same signature are *elementarily equivalent*, written $\mathcal{M} \equiv \mathcal{N}$, if they satisfy the same first-order sentences. Isomorphic structures are elementarily equivalent, but the converse fails: the orders $(\mathbb{Q}, <)$ and $(\mathbb{R}, <)$ are not isomorphic, since only the first is countable, and yet, as we show in Proposition 21.1 below, no first-order sentence tells them apart. The problem is to decide whether two given structures are elementarily equivalent without examining the sentences one by one, and to prove, with the same tool, that a given property of structures cannot be expressed by a first-order sentence.

The answer is the game sketched at the opening of this part, and in this section we make it precise. Two notions come first: the structures on which the game is played, and the partial isomorphisms that Player II must preserve. We work with relational signatures, which keep the bookkeeping of the game to a minimum; function and constant symbols can be traded for relations, as we explain in Remark 20.9 below.

Definition 19.1. A *relational signature* is a set Σ of relation symbols, each with an *arity*, which is an integer $r \geq 1$; the signature is *finite* if the set Σ is finite. A Σ-*structure* $\mathcal{M}$ consists of a non-empty set M, the *domain* of $\mathcal{M}$, together with a relation $R^{\mathcal{M}} \subseteq M^r$ for each symbol $R \in \Sigma$ of arity r.

Throughout this part, $\mathcal{M}$ and $\mathcal{N}$ denote structures with domains M and N, and we use the letters $a, a', a_1, \ldots$ for elements of M and $b, b', b_1, \ldots$ for elements of N. Our examples are linear orders and graphs: a linear order is a structure in the signature $\{<\}$ whose binary relation is a strict linear order, and a graph is a structure $\mathcal{G}$ in the signature $\{E\}$ whose relation $E^{\mathcal{G}}$ is symmetric and irreflexive. All tuples are finite; for a tuple $\bar{a} = (a_1, \ldots, a_k) \in M^k$ and an element $a \in M$ we write $\bar{a}a$ for the tuple $(a_1, \ldots, a_k, a)$, and similarly $\bar{a}\bar{a}^*$ for the concatenation of two tuples.

Definition 19.2. Let $\mathcal{M}$ and $\mathcal{N}$ be structures in a relational signature Σ, let $k \geq 0$, and let $\bar{a} = (a_1, \ldots, a_k) \in M^k$ and $\bar{b} = (b_1, \ldots, b_k) \in N^k$. The map $\bar{a} \mapsto \bar{b}$ is a *partial isomorphism* if

(1) for all $i, l \in \{1, \ldots, k\}$, we have $a_i = a_l$ if and only if $b_i = b_l$, and

(2) for every symbol $R \in \Sigma$ of arity r and all indices $i_1, \dots, i_r \in \{1, \dots, k\}$,

$$(a_{i_1}, \dots, a_{i_r}) \in R^{\mathcal{M}} \iff (b_{i_1}, \dots, b_{i_r}) \in R^{\mathcal{N}}.$$

Condition (1) says that $a_i \mapsto b_i$ is a well-defined bijection between the sets $\{a_1, \dots, a_k\}$ and $\{b_1, \dots, b_k\}$, and condition (2) says that this bijection is an isomorphism between the substructures with these domains. For $k = 0$ both conditions are vacuous, and the empty map is a partial isomorphism. Partial isomorphisms survive reindexing: for $h \geq 0$ and a map $\pi\colon \{1, \dots, h\} \to \{1, \dots, k\}$, write $\bar{a}_\pi = (a_{\pi(1)}, \dots, a_{\pi(h)})$ and $\bar{b}_\pi = (b_{\pi(1)}, \dots, b_{\pi(h)})$; if $\bar{a} \mapsto \bar{b}$ is a partial isomorphism, so is $\bar{a}_\pi \mapsto \bar{b}_\pi$, since each condition for the new tuples is one of the conditions for the old ones.

Definition 19.3. Let $\mathcal{M}$ and $\mathcal{N}$ be structures in a relational signature, let $k, n \geq 0$, and let $\bar{a} = (a_1, \dots, a_k) \in M^k$ and $\bar{b} = (b_1, \dots, b_k) \in N^k$. The *Ehrenfeucht–Fraïssé game* $\mathrm{EF}_n(\mathcal{M}, \bar{a}, \mathcal{N}, \bar{b})$ has n rounds. In round t, where $1 \leq t \leq n$, Player I chooses an element of M or an element of N, and Player II answers with an element of the other structure. We write $a_{k+t} \in M$ and $b_{k+t} \in N$ for the two elements chosen in round t, whichever player chose them. Player II wins the play if the map $(a_1, \dots, a_{k+n}) \mapsto (b_1, \dots, b_{k+n})$ is a partial isomorphism, and Player I wins otherwise. For $k = 0$ we write $\mathrm{EF}_n(\mathcal{M}, \mathcal{N})$.

Player I is often called *Spoiler*, since he tries to expose a difference, and Player II *Duplicator*, since she tries to copy his moves. Player I may switch structures from one round to the next, and this freedom is essential. The tuples $\bar{a}$ and $\bar{b}$ record elements matched before the game starts; they cost nothing, and they make induction on the number of rounds possible, because after one round of $\mathrm{EF}_n(\mathcal{M}, \bar{a}, \mathcal{N}, \bar{b})$ what remains is a game of the same kind with $n-1$ rounds. For $n = 0$ there are no rounds, and Player II wins $\mathrm{EF}_0(\mathcal{M}, \bar{a}, \mathcal{N}, \bar{b})$ if and only if $\bar{a} \mapsto \bar{b}$ is a partial isomorphism.

Example 19.4. Let $\mathcal{M}$ be the triangle, the graph with three pairwise adjacent vertices, and let $\mathcal{N}$ be the path $b_1 - b_2 - b_3$, in which b_1 and b_3 are not adjacent. Player II wins $\mathrm{EF}_1(\mathcal{M}, \mathcal{N})$, since a single vertex matched with a single vertex is always a partial isomorphism (the relation E is irreflexive in both graphs). Player I wins $\mathrm{EF}_2(\mathcal{M}, \mathcal{N})$. He chooses b_1 and then b_3 in $\mathcal{N}$. If Player II answers with two different vertices of the triangle, they are adjacent, while b_1 and b_3 are not; if she answers twice with the same vertex, condition (1) of Definition 19.2 fails. The winning strategy of Player I mirrors the sentence $\exists x\, \exists y\, (\neg x = y \wedge \neg E(x, y))$, which holds in $\mathcal{N}$ and fails in $\mathcal{M}$: each quantifier becomes a move, played in the structure where the sentence is true. In the next section we turn this correspondence into a theorem (see Theorem 20.3).

Remark 19.5. The game is an instance of the games (T, W) of Definition 2.2. A move of Player I is a pair $(1, a)$ with $a \in M$ or a pair $(2, b)$ with $b \in N$, the first coordinate naming the structure, and Player II answers $(1, a)$ with some $(2, b)$, $b \in N$, and $(2, b)$ with some $(1, a)$, $a \in M$; the coordinates keep the structures apart even when M and N meet. After the $2n$-th move both players play a dummy move $*$ forever, and W consists of the plays whose first $2n$ moves satisfy the winning condition. Two consequences are used below. First, the tree of positions depends only on $\mathcal{M}$, $\mathcal{N}$ and n; the tuples $\bar{a}$ and $\bar{b}$ enter only through the payoff set. Second, a play is decided by its first $2n$ moves, so the game is determined (see Exercise 4 at the end of Section 2), and exactly one player has a winning strategy (see Remarks 2.3(3)).

Exercises.

(1) Let $\mathcal{M}$ and $\mathcal{N}$ be structures in a relational signature, and let $f\colon M \to N$ be an isomorphism of $\mathcal{M}$ onto $\mathcal{N}$. Show that Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{M}, \mathcal{N})$ for every $n \geq 0$, by answering each move with its image under f or under f^{-1}.

(2) Let $\Sigma = \varnothing$, so that a Σ-structure is just a non-empty set, and let $\mathcal{M}$ and $\mathcal{N}$ be Σ-structures with m and m' elements, where m and m' are positive integers. Show that, for every $n \geq 0$, Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{M}, \mathcal{N})$ if and only if $m = m'$ or $m, m' \geq n$.
(3) In the setting of Example 19.4, suppose that Player I must choose all his elements in the triangle $\mathcal{M}$. Show that Player II then wins the game with two rounds. Thus, in the example, the choice of the structure in which Player I plays is essential.
(4) Show that Player I wins $\mathrm{EF}_2((\omega, <), (\mathbb{Z}, <))$, but that Player II wins if Player I must choose both of his elements in the same structure. Thus the freedom to switch structures from one round to the next is essential. (Hint: Player I starts with the least element of ω.)
(5) Let C_4 be the cycle with four vertices and P_4 the path with four vertices. Show that Player II wins $\mathrm{EF}_2(C_4, P_4)$ and that Player I wins $\mathrm{EF}_3(C_4, P_4)$. (Hint: an end of the path has only one neighbor.)

20. The Ehrenfeucht–Fraïssé theorem

In Example 19.4 the winning strategy of Player I followed a sentence quantifier by quantifier. The Ehrenfeucht–Fraïssé theorem, which we prove in this section, says that this is no accident: Player I wins the game with n rounds exactly when some sentence of quantifier rank at most n holds in one of the structures and fails in the other (see Theorem 20.3). Since the theorem counts quantifiers, we first fix the conventions of syntax that matter when formulas are counted; for everything else we use the standard syntax and semantics of first-order logic.

Definition 20.1. Let Σ be a relational signature. The variables are $v_1, v_2, v_3, \ldots$, and the letters x, y, z, u, w, with or without indices, stand for variables.

(1) The *atomic formulas* are the constant $\top$ (true), the equations $x = y$, and the formulas $R(x_1, \ldots, x_r)$ with $R \in \Sigma$ of arity r. The *formulas* are built from the atomic formulas by negation $\neg$, conjunction $\wedge$ and existential quantification $\exists x$; the formulas $\varphi \vee \psi$, $\varphi \to \psi$ and $\forall x\, \varphi$ abbreviate $\neg(\neg\varphi \wedge \neg\psi)$, $\neg(\varphi \wedge \neg\psi)$ and $\neg\exists x\, \neg\varphi$.
(2) The *quantifier rank* $\mathrm{qr}(\varphi)$ is defined by recursion: $\mathrm{qr}(\alpha) = 0$ for atomic α, $\mathrm{qr}(\neg\varphi) = \mathrm{qr}(\varphi)$, $\mathrm{qr}(\varphi \wedge \psi) = \max\{\mathrm{qr}(\varphi), \mathrm{qr}(\psi)\}$, and $\mathrm{qr}(\exists x\, \varphi) = \mathrm{qr}(\varphi) + 1$.

The quantifier rank measures the depth of nesting of quantifiers, not their number. For example, the sentence $\exists x\, \exists y\, (x < y)$ has rank 2, while the sentence $\exists x\, (x < x) \wedge \exists y\, (y = y)$, which also has two quantifiers, has rank 1. For the abbreviations,

$$\mathrm{qr}(\forall x\, \varphi) = \mathrm{qr}(\varphi) + 1 \qquad \text{and} \qquad \mathrm{qr}(\varphi \vee \psi) = \max\{\mathrm{qr}(\varphi), \mathrm{qr}(\psi)\}.$$

If $\bar{x} = (x_1, \ldots, x_k)$ is a list of distinct variables containing the free variables of φ, we write $\varphi(\bar{x})$, and for $\bar{a} \in M^k$ we write $\mathcal{M} \models \varphi(\bar{a})$ if φ holds in $\mathcal{M}$ when each x_i is interpreted as a_i. In this notation Tarski's clause for $\exists$ reads as follows: if $\varphi = \exists z\, \psi$, and $\bar{x}'$ and $\bar{a}'$ are obtained from $\bar{x}$ and $\bar{a}$ by deleting the entry of z when z occurs in $\bar{x}$, then the free variables of ψ lie in $\bar{x}'z$, and $\mathcal{M} \models \varphi(\bar{a})$ if and only if $\mathcal{M} \models \psi(\bar{a}', a)$ for some $a \in M$. Two formulas $\varphi(\bar{x})$ and $\psi(\bar{x})$ are *logically equivalent* if they hold of the same tuples in every Σ-structure.

Definition 20.2. Let $\mathcal{M}$ and $\mathcal{N}$ be structures in a relational signature, let $k, n \geq 0$, and let $\bar{a} \in M^k$ and $\bar{b} \in N^k$. We write $(\mathcal{M}, \bar{a}) \equiv_n (\mathcal{N}, \bar{b})$ if every formula $\varphi(\bar{x})$ of quantifier rank at most n, with $\bar{x}$ a list of k distinct variables, satisfies

$$\mathcal{M} \models \varphi(\bar{a}) \iff \mathcal{N} \models \varphi(\bar{b}).$$

For $k = 0$ we write $\mathcal{M} \equiv_n \mathcal{N}$: the two structures satisfy the same sentences of quantifier rank at most n.

Since every sentence has some quantifier rank, $\mathcal{M} \equiv \mathcal{N}$ if and only if $\mathcal{M} \equiv_n \mathcal{N}$ for every $n \geq 0$.

Theorem 20.3 (Fraïssé, 1954; Ehrenfeucht, 1961). *Let $\mathcal{M}$ and $\mathcal{N}$ be structures in a finite relational signature.*

(1) *For all $k, n \geq 0$, $\bar{a} \in M^k$ and $\bar{b} \in N^k$, we have $(\mathcal{M}, \bar{a}) \equiv_n (\mathcal{N}, \bar{b})$ if and only if Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{M}, \bar{a}, \mathcal{N}, \bar{b})$.*
(2) *For every $n \geq 0$, we have $\mathcal{M} \equiv_n \mathcal{N}$ if and only if Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{M}, \mathcal{N})$.*
(3) *$\mathcal{M} \equiv \mathcal{N}$ if and only if Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{M}, \mathcal{N})$ for every $n \geq 0$.*

In Example 19.4, the theorem says that the triangle and the path of three vertices satisfy the same sentences of quantifier rank 1, and that some sentence of rank 2 separates them. The rest of this section is the proof. The implication from a strategy to $\equiv_n$ is an induction on formulas, in which each quantifier is matched by a round. The converse needs, for each tuple, a single formula of rank n that says everything formulas of rank at most n can say about it (Lemma 20.7); this is where the finiteness of the signature is used. We begin with two lemmas about the game.

Lemma 20.4. *Let $\mathcal{M}$ and $\mathcal{N}$ be structures in a relational signature, let $k, n \geq 0$, and let $\bar{a} \in M^k$ and $\bar{b} \in N^k$. Suppose that Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{M}, \bar{a}, \mathcal{N}, \bar{b})$.*

(1) *The map $\bar{a} \mapsto \bar{b}$ is a partial isomorphism.*
(2) *For every $h \geq 0$ and every map $\pi\colon \{1, \dots, h\} \to \{1, \dots, k\}$, Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{M}, \bar{a}_\pi, \mathcal{N}, \bar{b}_\pi)$.*

Proof. Let τ be a winning strategy for Player II. A play consistent with τ produces tuples $\bar{a}^* \in M^n$ and $\bar{b}^* \in N^n$ such that $\bar{a}\bar{a}^* \mapsto \bar{b}\bar{b}^*$ is a partial isomorphism. (1) Such plays exist, since the domains are non-empty, and $\bar{a} \mapsto \bar{b}$ is obtained from $\bar{a}\bar{a}^* \mapsto \bar{b}\bar{b}^*$ by reindexing along the inclusion $\{1, \dots, k\} \subseteq \{1, \dots, k+n\}$. (2) By Remark 19.5, τ is also a strategy in $\mathrm{EF}_n(\mathcal{M}, \bar{a}_\pi, \mathcal{N}, \bar{b}_\pi)$, with the same consistent plays. In such a play, $\bar{a}_\pi\bar{a}^*$ and $\bar{b}_\pi\bar{b}^*$ are obtained from $\bar{a}\bar{a}^*$ and $\bar{b}\bar{b}^*$ by reindexing along the map sending $i \leq h$ to $\pi(i)$ and $h+t$ to $k+t$. Hence $\bar{a}_\pi\bar{a}^* \mapsto \bar{b}_\pi\bar{b}^*$ is a partial isomorphism, and τ is winning in the new game. ☺

The second lemma plays the game one round at a time. Its two conditions are the *forth* and *back* steps of the back-and-forth method.

Lemma 20.5. *Let $\mathcal{M}$ and $\mathcal{N}$ be structures in a relational signature, let $n \geq 1$ and $k \geq 0$, and let $\bar{a} \in M^k$ and $\bar{b} \in N^k$. Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{M}, \bar{a}, \mathcal{N}, \bar{b})$ if and only if*

(forth) *for every $a \in M$ there is $b \in N$ such that Player II has a winning strategy in $\mathrm{EF}_{n-1}(\mathcal{M}, \bar{a}a, \mathcal{N}, \bar{b}b)$, and*
(back) *for every $b \in N$ there is $a \in M$ such that Player II has a winning strategy in $\mathrm{EF}_{n-1}(\mathcal{M}, \bar{a}a, \mathcal{N}, \bar{b}b)$.*

Proof. Suppose that the first round has produced $a \in M$ and $b \in N$, in either order. The continuations of the play are exactly the plays of $\mathrm{EF}_{n-1}(\mathcal{M}, \bar{a}a, \mathcal{N}, \bar{b}b)$, and a continuation is won by Player II in one game if and only if it is won by her in the other, because both winning conditions say that the same map $\bar{a}a\bar{a}^* \mapsto \bar{b}b\bar{b}^*$ is a partial isomorphism.

Let τ be a winning strategy in $\mathrm{EF}_n(\mathcal{M}, \bar{a}, \mathcal{N}, \bar{b})$, and let $a \in M$. Let $b \in N$ be the answer of τ to the opening move a. In $\mathrm{EF}_{n-1}(\mathcal{M}, \bar{a}a, \mathcal{N}, \bar{b}b)$, let Player II play at each position p the move that τ assigns to the round (a, b) followed by p. Every play consistent with this rule,

preceded by the round (a, b), is consistent with τ and hence won by Player II, so the rule is a winning strategy. This proves (forth); opening moves in N give (back).

Conversely, assume (forth) and (back). For each $a \in M$ choose $b_a \in N$ and a winning strategy τ_a in $\mathrm{EF}_{n-1}(\mathcal{M}, \bar{a}a, \mathcal{N}, \bar{b}b_a)$, and for each $b \in N$ choose $a_b \in M$ and a winning strategy τ'_b in $\mathrm{EF}_{n-1}(\mathcal{M}, \bar{a}a_b, \mathcal{N}, \bar{b}b)$. Player II answers an opening move a with b_a and then follows τ_a, and she answers an opening move b with a_b and then follows τ'_b. (At positions that cannot arise in this way her moves are arbitrary.) Every play consistent with this strategy is a first round followed by a play consistent with some τ_a or τ'_b, and it is therefore won by Player II. ☺

Hintikka formulas. The formula of rank $n+1$ describing a tuple lists the rank-n descriptions of all its one-element extensions twice: under $\exists$, saying that each extension is realized (the forth step), and under $\forall$, saying that no other extension occurs (the back step). For this list to be a formula it must be finite, and this is where the finiteness of the signature enters.

Fix a finite relational signature Σ and an enumeration of all Σ-formulas. For a finite set Φ of formulas, $\bigwedge \Phi$ denotes the conjunction of its members in the order of the enumeration ($\top$ if $\Phi = \varnothing$), and $\bigvee \Phi$ denotes $\neg \bigwedge\{\neg\varphi : \varphi \in \Phi\}$. These are definite formulas, of quantifier rank the largest rank of a member of Φ (0 if $\Phi = \varnothing$), and they hold of a tuple exactly when every member, respectively some member, of Φ does. For $k \geq 0$, let Λ_k be the set of atomic formulas with variables among $v_1, \ldots, v_k$, together with their negations.

Definition 20.6. Let Σ be a finite relational signature. For $n, k \geq 0$, the sets $H_{n,k}$ of formulas are defined by recursion on n:

$$H_{0,k} = \Big\{ \bigwedge \Delta : \Delta \subseteq \Lambda_k \Big\},$$
$$H_{n+1,k} = \Big\{ \bigwedge \big\{ \exists v_{k+1}\, \chi : \chi \in S \big\} \wedge \forall v_{k+1} \bigvee S \ : \ S \subseteq H_{n,k+1} \Big\}.$$

For a Σ-structure $\mathcal{M}$ and $\bar{a} \in M^k$, the *Hintikka formula of rank n* of $\bar{a}$ in $\mathcal{M}$ is defined by recursion on n:

$$\varphi^0_{\bar{a}} = \bigwedge \big\{ \alpha \in \Lambda_k : \mathcal{M} \models \alpha(\bar{a}) \big\},$$
$$\varphi^{n+1}_{\bar{a}} = \bigwedge \big\{ \exists v_{k+1}\, \varphi^n_{\bar{a}a} : a \in M \big\} \wedge \forall v_{k+1} \bigvee \big\{ \varphi^n_{\bar{a}a} : a \in M \big\}.$$

The definition and part (1) of the next lemma go together, by induction on n: part (1) guarantees at each stage that the conjunctions and disjunctions range over finite sets. The structure $\mathcal{M}$ is omitted from the notation $\varphi^n_{\bar{a}}$.

Lemma 20.7. *Let $\mathcal{M}$ and $\mathcal{N}$ be structures in a finite relational signature Σ, let $n, k \geq 0$, and let $\bar{a} \in M^k$ and $\bar{b} \in N^k$.*

(1) *$H_{n,k}$ is a finite set of formulas of quantifier rank at most n with free variables among $v_1, \ldots, v_k$, and $\varphi^n_{\bar{a}} \in H_{n,k}$.*
(2) *$\mathcal{M} \models \varphi^n_{\bar{a}}(\bar{a})$.*
(3) *$\mathcal{N} \models \varphi^n_{\bar{a}}(\bar{b})$ if and only if Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{M}, \bar{a}, \mathcal{N}, \bar{b})$.*

Proof. All three parts are proved by induction on n, simultaneously for all k, all structures and all tuples.

(1) The set Λ_k consists of $\top$, the k^2 equations $v_i = v_l$, the k^r formulas $R(v_{i_1}, \ldots, v_{i_r})$ for each $R \in \Sigma$ of arity r, and their negations; it is finite because Σ is finite. Hence $H_{0,k}$ has at most $2^{|\Lambda_k|}$ members, all of rank 0, and it contains $\varphi^0_{\bar{a}}$. Assume the statement for n and every k. Every $S \subseteq H_{n,k+1}$ is finite, and $\bigwedge\{\exists v_{k+1}\chi : \chi \in S\} \wedge \forall v_{k+1} \bigvee S$ has rank at most $n+1$ and free variables among $v_1, \ldots, v_k$. So $H_{n+1,k}$ has at most $2^{|H_{n,k+1}|}$ members. By the induction

hypothesis, $\{\varphi^n_{\bar a a} : a \in M\} \subseteq H_{n,k+1}$, and $\varphi^{n+1}_{\bar a}$ is the member of $H_{n+1,k}$ corresponding to this subset.

(2) Every conjunct of $\varphi^0_{\bar a}$ holds of $\bar a$ by definition. For the inductive step, $\mathcal{M} \models \varphi^n_{\bar a a}(\bar a a)$ for each $a \in M$ by the induction hypothesis. Thus a witnesses $\exists v_{k+1}\varphi^n_{\bar a a}$ at $\bar a$, and $\bar a a$ satisfies the disjunct $\varphi^n_{\bar a a}$ of $\bigvee\{\varphi^n_{\bar a a'} : a' \in M\}$ for every a; so both conjuncts of $\varphi^{n+1}_{\bar a}$ hold at $\bar a$.

(3) For each atomic formula α with variables among $v_1, \dots, v_k$, exactly one of α and $\neg\alpha$ is a conjunct of $\varphi^0_{\bar a}$, so $\mathcal{N} \models \varphi^0_{\bar a}(\bar b)$ holds exactly when $\bar a$ and $\bar b$ satisfy the same atomic formulas in $v_1, \dots, v_k$. These atomic formulas express the conditions of Definition 19.2, so this happens exactly when $\bar a \mapsto \bar b$ is a partial isomorphism, that is, when Player II wins $\mathrm{EF}_0(\mathcal{M}, \bar a, \mathcal{N}, \bar b)$. For the inductive step, by the definition of $\varphi^{n+1}_{\bar a}$, $\mathcal{N} \models \varphi^{n+1}_{\bar a}(\bar b)$ holds if and only if

(i) for every $a \in M$ there is $b \in N$ with $\mathcal{N} \models \varphi^n_{\bar a a}(\bar b b)$, and
(ii) for every $b \in N$ there is $a \in M$ with $\mathcal{N} \models \varphi^n_{\bar a a}(\bar b b)$.

By the induction hypothesis, $\mathcal{N} \models \varphi^n_{\bar a a}(\bar b b)$ holds exactly when Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{M}, \bar a a, \mathcal{N}, \bar b b)$. So (i) and (ii) are the conditions (forth) and (back) of Lemma 20.5 for $\mathrm{EF}_{n+1}(\mathcal{M}, \bar a, \mathcal{N}, \bar b)$, and that lemma completes the induction. ☺

Proof of Theorem 20.3. (1) Suppose first that $(\mathcal{M}, \bar a) \equiv_n (\mathcal{N}, \bar b)$. By Lemma 20.7(1),(2), the formula $\varphi^n_{\bar a}(v_1, \dots, v_k)$ has rank at most n and holds of $\bar a$ in $\mathcal{M}$; hence it holds of $\bar b$ in $\mathcal{N}$, and Lemma 20.7(3) gives a winning strategy for Player II in $\mathrm{EF}_n(\mathcal{M}, \bar a, \mathcal{N}, \bar b)$.

For the converse we prove, by induction on φ, the following claim, which does not use the finiteness of the signature: if $\bar x$ is a list of distinct variables containing the free variables of φ, $j \geq \mathrm{qr}(\varphi)$, and Player II has a winning strategy in $\mathrm{EF}_j(\mathcal{M}, \bar a, \mathcal{N}, \bar b)$, where $\bar a$ and $\bar b$ have the length of $\bar x$, then

$$\mathcal{M} \models \varphi(\bar a) \iff \mathcal{N} \models \varphi(\bar b).$$

For atomic φ this holds because $\bar a \mapsto \bar b$ is a partial isomorphism (Lemma 20.4(1)) and atomic formulas express the conditions of Definition 19.2. The cases $\neg\varphi$ and $\varphi \wedge \psi$ follow from the induction hypothesis with the same $\bar x$, j and tuples.

Let $\varphi = \exists z\, \psi$, and let $\bar x'$, $\bar a'$ and $\bar b'$ be obtained from $\bar x$, $\bar a$ and $\bar b$ by deleting the entry of z, if z occurs in $\bar x$. Then $j \geq \mathrm{qr}(\psi) + 1 \geq 1$, and by Lemma 20.4(2) Player II has a winning strategy in $\mathrm{EF}_j(\mathcal{M}, \bar a', \mathcal{N}, \bar b')$. Suppose $\mathcal{M} \models \varphi(\bar a)$, and choose $a \in M$ with $\mathcal{M} \models \psi(\bar a', a)$. By (forth) in Lemma 20.5 there is $b \in N$ such that Player II has a winning strategy in $\mathrm{EF}_{j-1}(\mathcal{M}, \bar a' a, \mathcal{N}, \bar b' b)$. The induction hypothesis for ψ, with the list $\bar x' z$ and $j - 1 \geq \mathrm{qr}(\psi)$, gives $\mathcal{N} \models \psi(\bar b', b)$, hence $\mathcal{N} \models \varphi(\bar b)$. The implication from $\mathcal{N}$ to $\mathcal{M}$ is the same, with (back) in place of (forth). This proves the claim, and applying it to formulas of rank at most n proves (1).

Part (2) is the case $k = 0$ of (1), and part (3) follows from (2), since $\mathcal{M} \equiv \mathcal{N}$ if and only if $\mathcal{M} \equiv_n \mathcal{N}$ for all n. ☺

The proof also describes the *rank-n type* of a tuple, the set of formulas of rank at most n that it satisfies, and it bounds the number of such formulas up to logical equivalence.

Corollary 20.8. *Let Σ be a finite relational signature, and let $n, k \geq 0$.*

(1) *For Σ-structures $\mathcal{M}$, $\mathcal{N}$ and tuples $\bar a \in M^k$, $\bar b \in N^k$, we have $(\mathcal{M}, \bar a) \equiv_n (\mathcal{N}, \bar b)$ if and only if $\mathcal{N} \models \varphi^n_{\bar a}(\bar b)$.*
(2) *Every formula $\varphi(v_1, \dots, v_k)$ of quantifier rank at most n is logically equivalent to $\bigvee S$ for some $S \subseteq H_{n,k}$. Hence, up to logical equivalence, there are at most $2^{|H_{n,k}|}$ formulas of rank at most n with free variables among $v_1, \dots, v_k$.*

Proof. Part (1) combines Theorem 20.3(1) with Lemma 20.7(3). For (2), let S be the set of Hintikka formulas $\varphi^n_{\bar a}$, where $\mathcal{M}$ ranges over all Σ-structures and $\bar a$ over the tuples in M^k with $\mathcal{M} \models \varphi(\bar a)$; then $S \subseteq H_{n,k}$. If $\mathcal{M} \models \varphi(\bar a)$, then $\varphi^n_{\bar a} \in S$ holds of $\bar a$ (Lemma 20.7(2)), so $\bar a$ satisfies $\bigvee S$. Conversely, if $\bar b \in N^k$ satisfies $\bigvee S$, there are $\mathcal{M}$ and $\bar a$ with $\mathcal{M} \models \varphi(\bar a)$ and $\mathcal{N} \models \varphi^n_{\bar a}(\bar b)$; by (1), $(\mathcal{M}, \bar a) \equiv_n (\mathcal{N}, \bar b)$, and so $\mathcal{N} \models \varphi(\bar b)$. ☺

Remark 20.9. Finiteness of the signature is used only in the implication from $\equiv_n$ to a winning strategy, through the finiteness of Λ_k in Lemma 20.7(1); the converse holds in every relational signature. The implication from $\equiv_n$ to a winning strategy fails for infinite signatures: two structures may be elementarily equivalent while Player I wins in a single round (Exercise 8 at the end of Section 21). Function and constant symbols can be absorbed into relational signatures: a function symbol f of arity r is replaced by a relation symbol R_f of arity $r+1$, interpreted as the graph $\{(\bar a, a) : f(\bar a) = a\}$, and a constant symbol by a unary relation symbol. Formulas translate back and forth, so two structures are elementarily equivalent exactly when their relational versions are, and Theorem 20.3(3) characterizes elementary equivalence in every finite signature. The quantifier rank, however, is not preserved, since nested terms must be unnested by quantifiers: $f(f(x)) = y$ becomes $\exists z\,(R_f(x,z) \wedge R_f(z,y))$.

21. Linear orders, parity and connectivity

The Ehrenfeucht–Fraïssé theorem turns the task of showing that two structures satisfy the same sentences of quantifier rank at most n into the task of describing a winning strategy for Player II in the game with n rounds (see Theorem 20.3). In this section we describe two such strategies, both on linear orders. On dense linear orders without endpoints she survives any number of rounds; on finite linear orders she survives n rounds as soon as both orders have at least $2^n - 1$ elements. The second strategy has two consequences for finite structures: no first-order sentence says that a finite linear order has an even number of elements, and none says that a finite graph is connected.

Dense linear orders. For tuples $\bar a$ and $\bar b$ of the same length in two linear orders, $\bar a \mapsto \bar b$ is a partial isomorphism exactly when $a_i < a_l \iff b_i < b_l$ and $a_i = a_l \iff b_i = b_l$ for all i, l, that is, when the two tuples have the same order type. Player II's strategy is the one of the back-and-forth proof of Cantor's theorem: answer every move with an element in the same position relative to the elements already chosen.

Proposition 21.1. *Let $\mathcal{M}$ and $\mathcal{N}$ be dense linear orders without endpoints, let $k \geq 0$, and let $\bar a \in M^k$ and $\bar b \in N^k$ be such that $\bar a \mapsto \bar b$ is a partial isomorphism. Then Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{M}, \bar a, \mathcal{N}, \bar b)$ for every $n \geq 0$. In particular $\mathcal{M} \equiv \mathcal{N}$; for example, $(\mathbb{Q}, <) \equiv (\mathbb{R}, <)$.*

Proof. By induction on n, for all tuples at once. For $n = 0$ the hypothesis is the winning condition. Let $n \geq 1$. By Lemma 20.5 and the induction hypothesis, it suffices to show that for every $a \in M$ there is $b \in N$ with $\bar a a \mapsto \bar b b$ a partial isomorphism, and symmetrically; by symmetry we treat only the first. If $a = a_i$ for some i, take $b = b_i$. Otherwise let $P^- = \{i : a_i < a\}$ and $P^+ = \{i : a_i > a\}$, which partition $\{1, \dots, k\}$. For $i \in P^-$ and $l \in P^+$ we have $a_i < a_l$, hence $b_i < b_l$. So there is $b \in N$ with $b_i < b < b_l$ for all $i \in P^-$ and $l \in P^+$: by density if both sets are non-empty, by the absence of endpoints if exactly one is, and trivially if both are empty, which happens only when $k = 0$. Then $a_i < a \iff b_i < b$ and $a_i > a \iff b_i > b$ for every i, and neither $a_i = a$ nor $b_i = b$, so $\bar a a \mapsto \bar b b$ is a partial isomorphism.

Applied to the empty tuples, this shows that Player II wins $\mathrm{EF}_n(\mathcal{M},\mathcal{N})$ for every n, and Theorem 20.3(3) gives $\mathcal{M}\equiv\mathcal{N}$. ☺

This settles the example given in the problem at the beginning of Section 19.

Finite linear orders. For an integer $m\geq 1$, we write L_m for the linear order $(\{1,\dots,m\},<)$. In $\mathrm{EF}_n(L_m,L_{m'})$ Player II must keep track of distances. She adds an imaginary element 0 below each order, and imaginary elements $m+1$ and $m'+1$ above L_m and $L_{m'}$, and she watches the gaps between consecutive chosen elements. The idea is that two elements at distance at least 2^j look infinitely far apart to a formula with j quantifiers left, since each quantifier can at best halve a distance.

Definition 21.2. Let $j\geq 0$. For integers d and d', we write $d\approx_j d'$ if $d=d'$ or $d,d'\geq 2^j$. Let $m,m'\geq 1$, $k\geq 0$, $\bar a\in\{1,\dots,m\}^k$ and $\bar b\in\{1,\dots,m'\}^k$. Put $a_-=0$, $a_+=m+1$, $b_-=0$, $b_+=m'+1$, and $K=\{-,1,\dots,k,+\}$. Indices $p,q\in K$ are *consecutive* for $\bar a$ if $a_p<a_q$ and no $s\in K$ satisfies $a_p<a_s<a_q$. The tuples $\bar a$ and $\bar b$ are *j-similar* if

(1) for all $p,q\in K$, $a_p<a_q \iff b_p<b_q$ and $a_p=a_q \iff b_p=b_q$, and
(2) for all $p,q\in K$ consecutive for $\bar a$, $a_q-a_p\approx_j b_q-b_p$.

By (1), p and q are consecutive for $\bar a$ exactly when they are consecutive for $\bar b$, so the definition is symmetric. For $j\geq 1$, $d\approx_j d'$ implies $d\approx_{j-1} d'$. The heart of the strategy is arithmetic: when Player I cuts a gap into two pieces, Player II can cut the corresponding gap so that corresponding pieces are similar one level down.

Lemma 21.3. *Let $j\geq 1$, and let d, d', e be positive integers with $d\approx_j d'$ and $e<d$. Then there is an integer e' with $0<e'<d'$ such that*

$$e\approx_{j-1} e' \qquad \text{and} \qquad d-e\approx_{j-1} d'-e'.$$

Proof. Copy a short piece, measured from the corresponding end, and cut at distance 2^{j-1} from the left end when both pieces are long. If $d=d'$, take $e'=e$. Otherwise $d,d'\geq 2^j$, and there are three cases.

If $e<2^{j-1}$, take $e'=e$. Then $0<e'<2^{j-1}<d'$, and $d-e>2^{j-1}$ and $d'-e'>2^{j-1}$.

If $e\geq 2^{j-1}$ and $d-e<2^{j-1}$, take $e'=d'-(d-e)$. Then $d'-e'=d-e>0$, so $e'<d'$, and $e'>d'-2^{j-1}\geq 2^{j-1}$; thus $e,e'\geq 2^{j-1}$.

If $e\geq 2^{j-1}$ and $d-e\geq 2^{j-1}$, take $e'=2^{j-1}$. Then $0<e'<2^j\leq d'$, $e,e'\geq 2^{j-1}$, and $d-e\geq 2^{j-1}$, $d'-e'\geq 2^j-2^{j-1}=2^{j-1}$. ☺

Proposition 21.4. *Let $m,m'\geq 1$.*

(1) *If $j,k\geq 0$ and $\bar a\in\{1,\dots,m\}^k$ and $\bar b\in\{1,\dots,m'\}^k$ are j-similar, then Player II has a winning strategy in $\mathrm{EF}_j(L_m,\bar a,L_{m'},\bar b)$.*
(2) *If $n\geq 0$ and $m,m'\geq 2^n-1$, then $L_m\equiv_n L_{m'}$.*

Proof. (1) By induction on j, for all tuples at once. For $j=0$, condition (1) of Definition 21.2, restricted to the indices $1,\dots,k$, says that $\bar a\mapsto\bar b$ is a partial isomorphism.

Let $j\geq 1$. By Lemma 20.5 and the induction hypothesis, it suffices to answer every move of Player I so that the extended tuples are $(j-1)$-similar. By symmetry, assume that Player I plays $a\in\{1,\dots,m\}$.

If $a=a_i$ for some $i\in\{1,\dots,k\}$, Player II answers $b=b_i$. The new index then behaves exactly like i: it has the same comparisons, and the consecutive pairs containing it have the same gaps as those containing i. So the extended tuples are $(j-1)$-similar.

Otherwise a differs from every a_p, $p\in K$. Let $p,q\in K$ be indices such that a_p is the largest value below a and a_q the smallest value above a; then p and q are consecutive for $\bar a$. Put $d=a_q-a_p$, $d'=b_q-b_p$ and $e=a-a_p$, so that $0<e<d$ and $d\approx_j d'$. Let e' be given

by Lemma 21.3, and let Player II answer $b = b_p + e'$; since $0 \le b_p < b < b_q \le m' + 1$, this is an element of $\{1, \dots, m'\}$.

Every $s \in K$ satisfies $a_s \le a_p$ or $a_s \ge a_q$, hence, by condition (1), $b_s \le b_p$ or $b_s \ge b_q$. Therefore

$$a_s < a \iff a_s \le a_p \iff b_s \le b_p \iff b_s < b,$$

likewise $a_s > a \iff b_s > b$, and neither $a_s = a$ nor $b_s = b$; this is condition (1) for the extended tuples. The consecutive pairs of the extended tuples are the old ones, except those with values a_p and a_q, together with the new pairs joining the new index to the indices with values a_p and a_q. The old pairs satisfy condition (2) for $j - 1$ because they satisfy it for j, and the new pairs have gaps $e, d - e$ and $e', d' - e'$, which satisfy it by the choice of e'.

(2) The empty tuples have the single consecutive pair $(-, +)$, with gaps $m+1, m'+1 \ge 2^n$, so they are n-similar. By (1) Player II wins $\mathrm{EF}_n(L_m, L_{m'})$, and Theorem 20.3(2) gives $L_m \equiv_n L_{m'}$. ☺

In the play of $\mathrm{EF}_2(L_6, L_5)$ shown in Figure 24, Player II follows this strategy. The bound $2^n - 1$ cannot be lowered: if $m \ne m'$ and $\min\{m, m'\} < 2^n - 1$, then Player I wins $\mathrm{EF}_n(L_m, L_{m'})$ (Exercises 2 and 3).

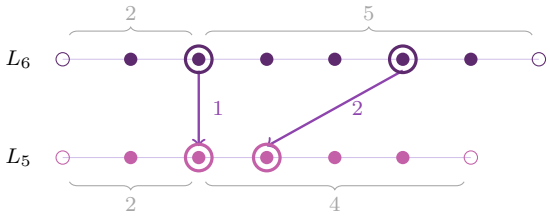


FIGURE 24. A play of $\mathrm{EF}_2(L_6, L_5)$ (see Proposition 21.4); the imaginary elements are drawn as open circles. Before the play, the single gap has length 7 in L_6 and 6 in L_5, both at least $2^2 = 4$. In round 1 Player I chooses 2 in L_6; this is the third case of Lemma 21.3 ($e = 2$, $d - e = 5$, both at least 2), so Player II cuts at $e' = 2$ and answers 2. The gaps become $2, 5$ in L_6 and $2, 4$ in L_5: equal, or both at least $2^1 = 2$, as 1-similarity requires (braces). In round 2 Player I chooses 5 in the long gap of L_6; again the third case applies ($e = 3$, $d - e = 2$, both at least 1), so Player II cuts at $e' = 1$ and answers 3. The chosen elements are in the same order in both structures, and Player II wins. (In the last round any element of the corresponding gap would do, since $d \approx_0 d'$ for all positive d, d'.)

Parity and connectivity. Every first-order sentence has some quantifier rank n, and by Proposition 21.4(2) it takes the same truth value in all finite linear orders with at least $2^n - 1$ elements. A property that changes from each length to the next, as parity does, therefore cannot be expressed by a sentence.

Corollary 21.5. *No first-order sentence θ in the signature $\{<\}$ holds in exactly the finite linear orders with an even number of elements.*

Proof. Suppose θ were such a sentence, of quantifier rank n. Since $2^n, 2^n + 1 \ge 2^n - 1$, Proposition 21.4(2) gives $L_{2^n} \equiv_n L_{2^n+1}$, so the two orders agree on θ, although exactly one of them has an even number of elements. ☺

For connectivity we define, inside a linear order, a graph that is connected exactly when the order has odd size, and we transfer sentences about the graph to sentences about the order. Let $\delta(x, y)$ be a formula in the signature $\{<\}$ with free variables among the distinct variables x and y. For a $\{<\}$-structure $\mathcal{M}$, let $\mathcal{M}^\delta$ be the $\{E\}$-structure with domain M and

$$E^{\mathcal{M}^\delta} = \{(a, a') \in M^2 : \mathcal{M} \models \delta(a, a')\}.$$

Lemma 21.6. *Let $\delta(x, y)$ be a formula in the signature $\{<\}$. For every sentence θ in the signature $\{E\}$ there is a sentence θ^* in the signature $\{<\}$ such that, for every $\{<\}$-structure $\mathcal{M}$, $\mathcal{M} \models \theta^*$ if and only if $\mathcal{M}^\delta \models \theta$.*

Proof. First rename the bound variables of δ so that they do not occur in θ. For variables u, w of θ, let $\delta(u, w)$ be the result of substituting u for x and w for y simultaneously; by the substitution lemma, $\delta(u, w)$ holds in $\mathcal{M}$ under the assignment $u \mapsto a$, $w \mapsto a'$ if and only if $\mathcal{M} \models \delta(a, a')$. For each subformula φ of θ, let φ^* be obtained by replacing every atomic subformula $E(u, w)$ with $\delta(u, w)$; the free variables of φ^* lie among those of φ. We show by induction on φ that, for every list $\bar{x}$ of distinct variables containing the free variables of φ and every tuple $\bar{a}$ in M of the same length,

$$\mathcal{M} \models \varphi^*(\bar{a}) \iff \mathcal{M}^\delta \models \varphi(\bar{a}).$$

For $\varphi = E(u, w)$ this is the substitution lemma together with the definition of $E^{\mathcal{M}^\delta}$; $\top$ and equations are unchanged, and their truth does not depend on the relations. The steps for $\neg$ and $\wedge$ are immediate. For $\varphi = \exists z\, \psi$ we have $\varphi^* = \exists z\, \psi^*$, and since $\mathcal{M}$ and $\mathcal{M}^\delta$ have the same domain, Tarski's clause reduces the claim to the induction hypothesis for ψ at the tuples $\bar{a}'a$, $a \in M$. The case $\varphi = \theta$, with the empty list, is the lemma. ☺

The translation may raise the quantifier rank, but only the rank of θ^* matters below. For $m \geq 3$, let G_m be the graph with vertex set $\{1, \ldots, m\}$ in which i is joined to $i + 2$ for $1 \leq i \leq m - 2$, m is joined to 2, and $m - 1$ is joined to 1 (see Figure 25).

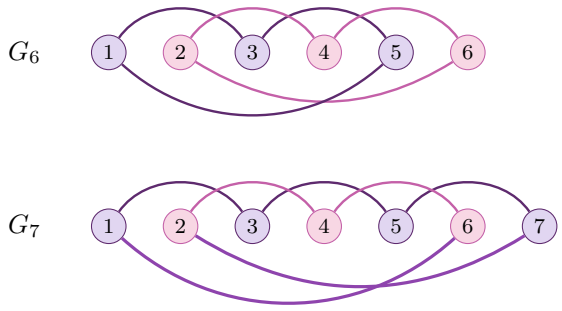


FIGURE 25. The graphs G_6 and G_7 of Lemma 21.7, with the odd vertices in lavender and the even vertices in pink. The arcs above the vertices join i to $i + 2$, and the arcs below are the extra edges, which join 6 to 2 and 5 to 1 in G_6, and 7 to 2 and 6 to 1 in G_7. In G_6 the extra edges close the odd and the even vertices into two separate triangles; in G_7 they join an odd vertex to an even one (orchid), and the graph is the single cycle $1, 3, 5, 7, 2, 4, 6$.

Lemma 21.7. (1) *For $m \geq 3$, the graph G_m is connected if and only if m is odd.*
(2) *There is a formula $\delta(x, y)$ in the signature $\{<\}$ such that $L_m^\delta = G_m$ for every $m \geq 3$.*

Proof. (1) The edges joining i to $i+2$ form a path through the odd vertices $1, 3, 5, \ldots$ and a path through the even vertices $2, 4, 6, \ldots$, both non-empty since $m \geq 3$. If m is odd, the edge joining the odd vertex m to the even vertex 2 connects the two paths. If m is even, every edge joins vertices of the same parity (the extra edges join m to 2 and $m-1$ to 1), so no path leads from 1 to 2.

(2) Successor, the first two elements and the last two elements of a finite linear order are first-order definable:

$$\begin{aligned} \mathrm{succ}(x,y) &= x < y \wedge \neg\exists z\,(x < z \wedge z < y), & \mathrm{plus}_2(x,y) &= \exists z\,\big(\mathrm{succ}(x,z) \wedge \mathrm{succ}(z,y)\big), \\ \mathrm{first}(x) &= \neg\exists z\,(z < x), & \mathrm{second}(x) &= \exists z\,\big(\mathrm{first}(z) \wedge \mathrm{succ}(z,x)\big), \\ \mathrm{last}(x) &= \neg\exists z\,(x < z), & \mathrm{penult}(x) &= \exists z\,\big(\mathrm{last}(z) \wedge \mathrm{succ}(x,z)\big), \end{aligned}$$

with nested bound variables renamed apart from each other and from x and y. In L_m, $\mathrm{succ}(i,i')$ and $\mathrm{plus}_2(i,i')$ hold exactly when $i' = i+1$ and $i' = i+2$, respectively, and first, second, last, penult define 1, 2, m, $m-1$. So

$$\varepsilon(x,y) = \mathrm{plus}_2(x,y) \vee \big(\mathrm{last}(x) \wedge \mathrm{second}(y)\big) \vee \big(\mathrm{penult}(x) \wedge \mathrm{first}(y)\big)$$

holds of (i,i') exactly when $i' = i+2$, or $(i,i') = (m,2)$, or $(i,i') = (m-1,1)$, and $\delta(x,y) = \varepsilon(x,y) \vee \varepsilon(y,x)$ defines the edges of G_m. ☺

Corollary 21.8. *No first-order sentence θ in the signature $\{E\}$ holds in exactly the connected finite graphs.*

Proof. Suppose θ were such a sentence. Let δ be as in Lemma 21.7(2) and θ^* as in Lemma 21.6. For every $m \geq 3$,

$$L_m \models \theta^* \iff G_m \models \theta \iff G_m \text{ is connected} \iff m \text{ is odd}.$$

Let $n = \mathrm{qr}(\theta^*)$ and $m = 2^n + 2$. Then $m \geq 3$ and $m, m+1 \geq 2^n - 1$, so $L_m \equiv_n L_{m+1}$ by Proposition 21.4(2), and L_m and L_{m+1} agree on θ^*, although exactly one of m and $m+1$ is odd. ☺

Neither corollary mentions a game; in both, the game reduced a statement about all sentences to a single strategy, the halving of gaps in Lemma 21.3.

Exercises.

(1) Show that $L_m \equiv_1 L_{m'}$ for all $m, m' \geq 1$, by describing a winning strategy for Player II in $\mathrm{EF}_1(L_m, L_{m'})$. Show that Player I wins $\mathrm{EF}_2(L_1, L_2)$, and find a sentence of quantifier rank 2 true in L_2 and false in L_1.

(2) Show that Player I wins $\mathrm{EF}_2(L_2, L_3)$ and that Player II wins $\mathrm{EF}_2(L_3, L_4)$, describing the strategies directly. Thus the orders L_3 and L_4 satisfy the same sentences of quantifier rank 2, while L_2 and L_3 do not, and for $n = 2$ the bound $2^n - 1$ of Proposition 21.4(2) cannot be lowered.

(3) (∗) Let $n \geq 1$ be an integer, and let $m \neq m'$ be positive integers with $\min\{m, m'\} < 2^n - 1$. Show that Player I wins $\mathrm{EF}_n(L_m, L_{m'})$, and conclude that $L_m \equiv_n L_{m'}$ holds if and only if $m = m'$ or $m, m' \geq 2^n - 1$. (Hint: show by induction on j that Player I wins in j rounds from any pair of tuples that violates condition (1) of Definition 21.2, or that has a consecutive pair of indices whose gaps have lengths $d \neq d'$ with $\min\{d, d'\} < 2^j$; if $d < d'$, he plays in the longer gap, at distance $\min\{2^{j-1}, \lfloor d'/2 \rfloor\}$ from its left end.)

(4) Show that $(\mathbb{Z}, <) \equiv_2 (\mathbb{Q}, <)$, but that Player I wins $\mathrm{EF}_3((\mathbb{Z}, <), (\mathbb{Q}, <))$. Find a sentence of quantifier rank 3 that distinguishes the two orders.

(5) Draw the graphs G_5 and G_8 of Lemma 21.7. Show that for odd $m \geq 3$ the graph G_m is a single cycle through all its vertices, and that for even $m \geq 6$ it consists of two disjoint cycles, one through the odd and one through the even vertices. Compute the quantifier rank of the formula δ in the proof of Lemma 21.7(2).

(6) Show that there is no first-order sentence in the signature $\{E\}$ that holds in a finite graph if and only if the graph is bipartite. (Hint: argue as in the proof of Corollary 21.8, with the cycle obtained from L_m, for $m \geq 3$, by joining each $i < m$ to $i+1$ and m to 1, which is bipartite exactly when m is even.)

(7) (∗) Let $(\mathbb{Z} + \mathbb{Z}, <)$ denote the order obtained by placing one copy of $(\mathbb{Z}, <)$ after another. Show that $(\mathbb{Z}, <) \equiv (\mathbb{Z} + \mathbb{Z}, <)$. (Hint: adapt the invariant used for finite linear orders, regarding two elements in different copies as being infinitely far apart.)

(8) (∗) Let $\Sigma = \{P_0, P_1, P_2, \dots\}$ be an infinite signature of unary relation symbols. Let $\mathcal{M}$ be the Σ-structure with domain $\mathbb{N}$ in which P_i is interpreted as $\{a \in \mathbb{N} : a > i\}$, and let $\mathcal{N}$ be obtained from $\mathcal{M}$ by adding a new element ∞ that lies in every P_i. Show that $\mathcal{M} \equiv \mathcal{N}$, but that Player I wins $\mathrm{EF}_1(\mathcal{M}, \mathcal{N})$. (Hint: for every integer $r \geq 0$, the reducts of $\mathcal{M}$ and $\mathcal{N}$ to the signature $\{P_0, \dots, P_r\}$ are isomorphic.) Thus parts (2) and (3) of Theorem 20.3 fail for infinite signatures, as stated in Remark 20.9.

22. Back-and-forth systems and ultrapowers

So far the game has lasted a fixed finite number of rounds. In this section we look at it from two further sides. Without the players, the game becomes Fraïssé's original formulation, a family of partial isomorphisms that can be extended back and forth; played for infinitely many rounds, it builds isomorphisms, first between countable dense linear orders and then, with the help of ultrafilters, between ultrapowers.

Back-and-forth systems. We begin with the static form. Let $\mathcal{M}$ and $\mathcal{N}$ be structures in the same finite relational signature. A *finite partial isomorphism* from $\mathcal{M}$ to $\mathcal{N}$ is a map p from a finite subset of M to N such that $\bar{a} \mapsto p(\bar{a})$ is a partial isomorphism for an enumeration $\bar{a}$ of the domain of p. A *back-and-forth system* of length n between $\mathcal{M}$ and $\mathcal{N}$ is a sequence $I_0, \dots, I_n$ of sets of finite partial isomorphisms, with $I_n \neq \varnothing$, such that for $1 \leq j \leq n$ and $p \in I_j$, every $a \in M$ has some $b \in N$ with $p \cup \{(a, b)\} \in I_{j-1}$, and every $b \in N$ has some $a \in M$ with the same property. Strategies of Player II and back-and-forth systems carry the same information, one dynamically and the other statically. If Player II wins $\mathrm{EF}_n(\mathcal{M}, \mathcal{N})$, the sets I_j of those p for which she wins $\mathrm{EF}_j(\mathcal{M}, \bar{a}, \mathcal{N}, p(\bar{a}))$, with $\bar{a}$ enumerating the domain of p, form such a system: the enumeration does not matter by Lemma 20.4(2), and the conditions are those of Lemma 20.5. Conversely, given a system, induction on j shows that Player II wins $\mathrm{EF}_j(\mathcal{M}, \bar{a}, \mathcal{N}, p(\bar{a}))$ for every $p \in I_j$: for $j = 0$ because p is a partial isomorphism, and for $j \geq 1$ by Lemma 20.5, since $\bar{a}a \mapsto p(\bar{a})b$ is a reindexing of an enumeration of $p \cup \{(a, b)\}$ and Lemma 20.4(2) applies. For some $p \in I_n$, Lemma 20.4(2) with $h = 0$ then gives a winning strategy in $\mathrm{EF}_n(\mathcal{M}, \mathcal{N})$. So Theorem 20.3(2) is Fraïssé's theorem: $\mathcal{M} \equiv_n \mathcal{N}$ if and only if there is a back-and-forth system of length n between them. For dense linear orders without endpoints, the set of all finite partial isomorphisms serves as every I_j, as the proof of Proposition 21.1 shows. If both orders are countable, Player II can keep the matched pairs a partial isomorphism for ω rounds; if Player I enumerates both domains, alternating between them, the union of the matched pairs is an isomorphism, and Cantor's theorem appears as the infinite form of the game.

Ultrapowers. Ultrafilters give a second, algebraic answer to the key problem: two structures are elementarily equivalent if and only if they have isomorphic ultrapowers. Keisler proved

this in 1961 under the generalized continuum hypothesis (GCH) [Kei61], and Shelah proved it in ZFC in 1971 [She71]. One direction is Łoś's theorem: every structure embeds elementarily, by the diagonal map, in each of its ultrapowers. For the other, elementarily equivalent finite structures are isomorphic; for infinite ones, here is the proof under GCH as it is usually given today, with Keisler's good ultrafilters. Let λ be an infinite cardinal at least the cardinalities of the structures and of the signature, and let $\mathcal{D}$ be a countably incomplete λ^+-good ultrafilter on λ; such ultrafilters exist under GCH by Keisler's theorem of 1964 [Kei64], and in ZFC by Kunen's theorem of 1972 [Kun72], which builds them from an independent family of 2^λ subsets of λ. For elementarily equivalent infinite $\mathcal{M}$ and $\mathcal{N}$, the ultrapowers $\mathcal{M}^\lambda/\mathcal{D}$ and $\mathcal{N}^\lambda/\mathcal{D}$ are λ^+-saturated and have cardinality 2^λ, which is λ^+ under GCH. Now play the game for λ^+ rounds between the two ultrapowers: Player II keeps the map between the chosen elements elementary: she answers each move by realizing, in the other structure, the image under this map of the type of the new element over the fewer than λ^+ elements already chosen, which saturation allows, and at limit stages the union of elementary maps is elementary. If Player I enumerates both structures, alternating between them, the completed play is an isomorphism. Shelah removed GCH by building the ultrafilter and the isomorphism together, in a single transfinite construction [She71]; for structures of cardinality at most λ his ultrafilter lies on $\mu = 2^\lambda$ rather than on λ, and the construction has length 2^μ. At each stage the filter is enlarged just enough to carry out one more back-and-forth step, or to decide one more subset of μ, and a family of 2^μ functions on μ, independent modulo the filter, guarantees that the construction can continue. The method generalizes Kunen's proof that good ultrafilters exist in ZFC [Kun72].

23. Extensions to other logics

The proof of Theorem 20.3 matches one feature of the logic, the existential quantifier, with one feature of the game, a round. Change the logic, and the game changes with it: for most logics studied today there is a game of this kind, and it characterizes equivalence in that logic as EF_n characterizes $\equiv_n$. In this section we describe the principal examples, without proofs; the details are in the surveys of Hodges and Väänänen [HV19] and of Abramsky and Reggio [AR24], and in the books of Ebbinghaus and Flum, Libkin, Immerman, Grohe, and Väänänen [EF99, Lib04, Imm99, Gro17, Vää11].

Infinitary logic. Karp showed that two structures are equivalent in the infinitary logic $L_{\infty\omega}$, which allows conjunctions and disjunctions of arbitrary sets of formulas, exactly when there is a non-empty set of finite partial isomorphisms with the back and forth properties of Lemma 20.5, that is, exactly when Player II can survive the game for ω rounds [Kar65]. Games of transfinite length, and *dynamic* games in which Player I fixes the length as play proceeds, measure equivalence in stronger infinitary logics and are a central tool for uncountable structures [Vää11].

Pebble games and counting. If there are only k pairs of pebbles, Player I placing one pebble of a pair on an element of either structure and Player II its partner on the other, and pebbles may be lifted and reused, the resulting *pebble game* with n rounds characterizes equivalence for formulas of quantifier rank at most n with at most k variables, each of which may be quantified many times; the game of unbounded length characterizes the logic L^k of all such formulas on finite structures [Bar77, Imm82, Poi82]. In the *bijective pebble game* of Hella [Hel96], Player II must answer each move with a bijection between the two domains,

from which Player I then chooses a pair; it characterizes the logic C^k, which adds the counting quantifiers $\exists^{\geq i} x$ to L^k. Cai, Fürer and Immerman proved that C^{k+1}-equivalence coincides with indistinguishability by the k-dimensional Weisfeiler–Leman algorithm, a standard heuristic for graph isomorphism, and used a counting version of the pebble game to construct, for each k, non-isomorphic graphs that C^k does not distinguish [CFI92]. The same hierarchy now bounds the power of the graph neural networks of machine learning: message-passing networks distinguish no more graphs than the 1-dimensional algorithm [MRF+19, XHLJ19], and their logical expressiveness is measured by fragments of C^2 [BKM+20, Gro21]. In the *existential* pebble game, Player I moves only in $\mathcal{M}$ and Player II must maintain a partial homomorphism; Kolaitis and Vardi introduced it to study the expressive power of Datalog, and later showed that its winning strategies correspond to the consistency algorithms for constraint satisfaction problems [KV00]. Pebble games for logics with generalized quantifiers were developed by Kolaitis and Väänänen [KV95] and by Hella [Hel96].

Monadic second-order logic and locality. For monadic second-order logic, Player I may also choose a subset of one structure, and Player II answers with a subset of the other. A variant in which the players first choose colorings and then play the first-order game is due to Fagin, who used it to prove that connectivity of finite graphs is not definable in monadic existential second-order logic [Fag75]; Ajtai and Fagin refined the method to separate directed from undirected reachability [AF90]. Many non-definability proofs avoid explicit strategies by means of *locality*: by the theorems of Hanf [Han65] and Gaifman [Gai82], a first-order formula of quantifier rank n sees only neighborhoods of bounded radius, and on finite structures these theorems can be proved with games [FSV95, Lib04]. In modal logic the game starts from two pointed structures, and each move must follow the accessibility relation from the element last chosen in the same structure. With finitely many propositional letters, the n-round game characterizes equivalence for modal formulas of modal depth at most n, and Player II survives forever exactly when the two pointed structures are bisimilar. Bisimulations appear, as p-relations, in van Benthem's thesis, which proves that modal logic is the bisimulation-invariant fragment of first-order logic [vB76].

Current directions. Abramsky, Dawar and Wang showed that the k-pebble game is encoded by a comonad on the category of relational structures [ADW17], and Abramsky and Shah found comonads for the Ehrenfeucht–Fraïssé and modal games as well [AS18, AS21]. In this *game comonad* approach, equivalence in the extension of each logic by counting quantifiers becomes isomorphism in the Kleisli category of the comonad, and equivalence in the logic itself becomes a form of bisimulation between coalgebras. The coalgebras also capture combinatorial parameters: a structure admits a coalgebra for the k-pebble comonad exactly when its treewidth is less than k, and for the Ehrenfeucht–Fraïssé comonad of depth k exactly when its tree-depth is at most k. Theorems are thus proved for all the games at once; examples are the Lovász-type theorems on homomorphism counts of Dawar, Jakl and Reggio [DJR21] and the axiomatic theory of arboreal categories of Abramsky and Reggio [AR23] (see the survey [AR24]). A second line of research counts quantifiers instead of nesting them: Immerman's separability game [Imm81], rediscovered as the multi-structural game, is played on two sets of structures and gives lower bounds on the number of quantifiers needed to express a property [FLRV21, CFI+24]. The game itself is computationally hard: deciding which player wins $\mathrm{EF}_n(\mathcal{M}, \mathcal{N})$, for finite structures and n given as input, is PSPACE-complete, as Pezzoli proved for signatures containing a ternary relation symbol [Pez99]. For a fixed n and a fixed finite signature, by contrast, it suffices to check which of the finitely many sentences of $H_{n,0}$ hold in $\mathcal{M}$ and in $\mathcal{N}$ (Corollary 20.8), and this can be done in logarithmic space. Finally, Ehrenfeucht–Fraïssé games have been adapted to the continuous logic of metric structures,

where Player II must keep the values of formulas close rather than equal. In Hart's version, two structures are elementarily equivalent exactly when Player II wins for every finite set of atomic formulas and every tolerance $\varepsilon > 0$ [Har23]. In the version of Hirvonen and Puljujärvi, as in Theorem 20.3, the n-round game characterizes equivalence up to quantifier rank n [HP24].

24. Notes and historical remarks

Elementary equivalence was introduced by Tarski in Warsaw around 1930 (see [HV19]) and studied systematically by Tarski and Vaught [TV57]. The program of Tarski's school, to classify the complete first-order theories of familiar structures, needed methods for proving that two structures satisfy the same sentences, and the method that Fraïssé would find has its roots in an argument of Cantor.

Cantor proved in 1895 that any two countable dense linear orders without endpoints are isomorphic [Can95]. His proof enumerates both orders but builds the isomorphism in one direction only: each element of the first order is sent, in turn, to the first element of the second enumeration that fits, and a separate argument shows that every element of the second order is reached; the symmetric back-and-forth argument appears first in Huntington [Hun04] and Hausdorff [Hau14] (see [Sil94]).

Fraïssé, who was then teaching in Algiers, characterized elementary equivalence by systems of partial isomorphisms in his thesis, written under René de Possel, defended in Paris in 1953 and published in Algiers in 1954 [Fra54]. Ehrenfeucht, a student of Mostowski at the University of Warsaw, recast the characterization as a game, announcing it in 1957 [Ehr57] and publishing it in 1961 [Ehr61]; among his applications was a game proof of the results of Mostowski and Tarski on the elementary equivalence of well-orders. The formulas $\varphi^n_{\bar a}$ of Definition 20.6 go back to Hintikka's distributive normal forms of 1953 [Hin53]. Karp's theorem of 1965 [Kar65] extended Fraïssé's characterization to infinitary logic. The names Spoiler and Duplicator are due to Joel Spencer, in the early 1990s (see [HV19]).

In finite model theory, which grew from the 1970s onward in connection with databases and computational complexity, the games became one of the principal methods for proving that properties are not definable; an early example is Fagin's proof of 1975 that connectivity is not expressible in monadic existential second-order logic [Fag75]. The ultrapower characterization of elementary equivalence has a story of its own. Keisler proved it in 1961 under GCH [Kei61], and the hypothesis stood for ten years, until Shelah removed it in a paper submitted in February 1971 [She71]; Kunen's construction of good ultrafilters in ZFC [Kun72] was then still to appear, and the two arguments were found at nearly the same time (see [She71, pp. 224–225]).

Part
VI

The Scottish Book, von Neumann, Ulam, Texas, Toronto, and the triumph of games, logic, topology and combinatorics

The epilogue returns from Warsaw to Lwów, and from the mathematics of these notes to the people behind it: where they worked, what happened to them, and how their problems traveled. Many of them were very young. When the Scottish Book was started, in the summer of 1935, Stanisław Ulam was twenty-six, Stanisław Mazur was thirty, Mark Kac was twenty, and Stefan Banach, who led the discussions at the café table, was forty-three. The journey of the Scottish Book and of the people who carried it is shown in Figure 26, and the timeline in Figure 27 is worth keeping at hand throughout.

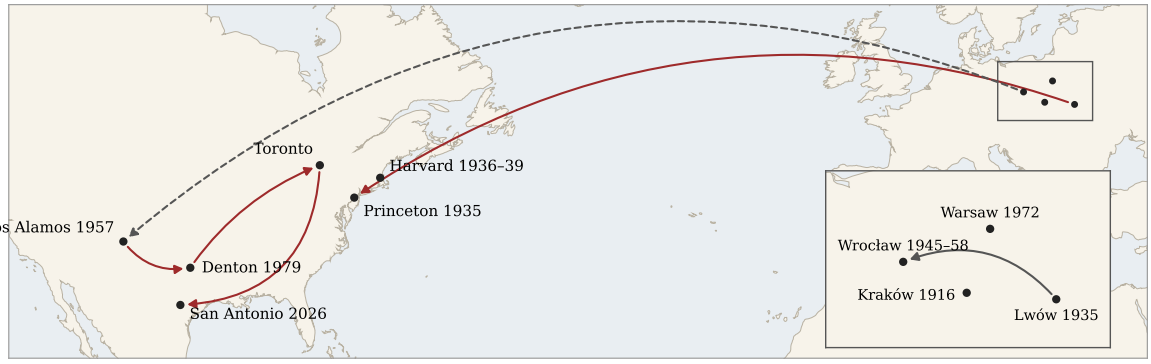


Figure 26. The journey of the Scottish Book. Red: Ulam's crossing in 1935, and the road from Los Alamos to Denton, Toronto and San Antonio. Dashed: the typewritten copy that Hugo Steinhaus sent from Wrocław to Ulam in 1956. The inset shows Poland: Kraków, where Steinhaus met Banach in 1916; Lwów, home of the Scottish Café; Wrocław, home of the New Scottish Book after the war; and Warsaw, where Per Enflo received a live goose from Mazur in 1972 (see Section 31).

1916	Steinhaus meets Banach and Nikodým in the Planty park, Kraków.
1920	Banach's doctorate in Lwów, without a university degree.
1930	Ulam on measures defined on all subsets of a set (measurable cardinals).
1932	Banach's *Théorie des opérations linéaires*.
1935	17 July: Problem 1, by Banach. Mazur's Problem 43; 4 August: Banach's answer. End of the year: Ulam goes to Princeton.
1936	6 November: Mazur's Problem 153, the goose problem.
1939	20 August: Ulam leaves Poland for the last time. September: war.
1941	31 May: last problem, number 193, by Steinhaus. July: the Lwów professors are murdered.
1945	31 August: Banach dies in Lwów. Steinhaus moves to Wrocław.
1946	2 July: first entry in the New Scottish Book, by Steinhaus.
1946	Ulam's idea of the Monte Carlo method.
1957	Ulam's English translation, mimeographed at Los Alamos; Oxtoby publishes the first proof of Banach's solution of Problem 43.
1962	Mycielski and Steinhaus propose the Axiom of Determinacy.
1972	Mazur gives Enflo a live goose in Warsaw.
1975	Martin proves Borel determinacy.
1979	The Scottish Book Conference in Denton, organized by Mauldin.
1981	*The Scottish Book*, edited by Mauldin (second edition 2015).
1994–98	Distortion problem (Odell and Schlumprecht); Gowers's dichotomy; Fields Medal for Gowers.
2026	The joint UT San Antonio–Toronto seminar.

FIGURE 27. A timeline of the Scottish Book.

25. The marble tables of the Scottish Café

Between the two world wars, the city of Lwów (then in southeastern Poland, now Lviv, in Ukraine) was home to one of the liveliest mathematical communities of the century. Led by Stefan Banach and Hugo Steinhaus, with Stanisław Mazur, Juliusz Schauder, Stanisław Ulam, Mark Kac and many others, the Lwów School did much to create modern functional analysis and, through Mazur's Problem 43, gave topology its first infinite game.

The school began with a chance meeting. One evening in 1916, walking in the Planty, the park around the old town of Kraków, Steinhaus overheard the words "Lebesgue integral" and stopped to talk to the two young men who had said them. They were Stefan Banach and Otto Nikodým. Banach, then twenty-four, had studied engineering in Lwów without finishing his degree, and was teaching himself mathematics. Steinhaus mentioned a problem on which he had been working; a few days later Banach came back with the solution, and their first joint paper followed. In 1920, with Steinhaus's support, Banach was allowed to submit a doctoral thesis in Lwów without a university degree; the thesis, published in 1922, contains what we now call Banach spaces. Steinhaus, whose own contributions to mathematics were many, later said that Banach was his greatest scientific discovery. In 1935, fifteen years after the thesis, Banach opened the Scottish Book with Problem 1, which asks when a metric space can be remetrized so as to become complete and compact while every sequence that converged in the old metric still converges in the new one; within weeks he had also answered Mazur's Problem 43.

The school's real seminar room was the *Kawiarnia Szkocka*, the Scottish Café, on Plac Akademicki (see Figure 28). The mathematicians spent hours at its tables over coffee and, as the stories go, wrote their formulas directly on the marble tabletops, which were wiped

clean when the café closed. To save at least the problems, Banach's wife, Łucja Banachowa, is said to have provided a large notebook, which was kept by the headwaiter and brought to the table on request. Thus was born the *Scottish Book* (*Księga Szkocka*); its first problem, by Banach, is dated 17 July 1935. The sessions could be long: Ulam recalled one with Mazur and Banach that lasted seventeen hours, interrupted only for meals [Ula76]. He also recorded Banach's view of what a mathematician does: "Good mathematicians see analogies between theorems or theories, the very best ones see analogies between analogies." Problems came with prizes: a cup of coffee, a small beer, five beers, a bottle of wine, 100 grams of caviar, a dinner at the Hotel George, a fondue in Geneva, a bottle of whisky "of measure > 0" (John von Neumann, 1937), a kilo of bacon (Stanisław Saks, 1940), and a live goose (see Section 31).



FIGURE 28. Plac Akademicki in Lwów, the square of the Scottish Café, on a hand-colored postcard of 1909 (Kraków: Wydawnictwo Salonu Malarzy Polskich). National Library of Poland (Polona), public domain.

26. PROBLEM 43

Mazur's Problem 43, the subject of these notes, was written in the summer of 1935, in the first weeks of the book. In the Introduction we saw what it asks and how Banach answered it on 4 August (see Figure 1), and in Section 7 we recalled Oxtoby's paper of 1957, whose proof, the first to be published, turned the answer into a theorem. Right after Problem 43 in the book, Ulam proposed a variant in which the players choose binary digits, one digit at a time. This variant later inspired Steinhaus and Jan Mycielski to propose, in 1962, a new axiom for set theory, the *Axiom of Determinacy* [MS62]; the game of Section 14 that bears Ulam's name is the variant in which the players choose non-empty finite binary strings. A

question about the size of sets of real numbers, asked over coffee, had become a question in the foundations of mathematics.

27. Von Neumann's invitation and the bridge to America

John von Neumann met Ulam in Warsaw in 1935 and invited him to spend a few months at the Institute for Advanced Study in Princeton. Ulam sailed for America in December 1935.

Ulam was the most prolific author of the Scottish Book: forty problems are his alone, and about twenty more are joint with others, among them Banach, Mazur and Józef Schreier. His problems have had a long life. Problem 19 asks whether a solid of uniform density that floats in water in equilibrium in every position must be a ball. In 2022 Dmitry Ryabogin showed that the answer is no: there is a strictly convex body of revolution in $\mathbb{R}^3$, of density $1/2$ and not a ball, that floats in equilibrium in every position [Rya22]. Ulam's earlier work, of 1930, on measures defined on all subsets of a set led to the measurable cardinals (see Section 34 below).

At a tea in von Neumann's house, Ulam met G. D. Birkhoff, who asked whether he would like to join the Society of Fellows at Harvard. From 1936 to 1939 Ulam divided his time between Harvard and Lwów, returning each summer to Poland and to the tables of the Scottish Café. On 20 August 1939 he left Poland for the last time, sailing from Gdynia with his seventeen-year-old brother Adam. Twelve days later Germany invaded Poland; on 17 September the Soviet Union followed. The brothers' father, Józef, who had seen them off at the quay, was killed in the Holocaust, together with their sister Stefania and much of the family.

28. War, and the survival of the book

The war destroyed the Lwów School. From September 1939 the city was under Soviet occupation, and the café meetings went on: the problem for which Stanisław Saks offered a kilo of bacon dates from February 1940. German troops entered Lwów on 30 June 1941. On the night of 3–4 July, in a planned action against the Polish intelligentsia, they arrested professors of the city's universities together with members of their families, and shot them on the Wuleckie Hills: twenty-five professors and twenty-five of their relatives died. Among them were the mathematicians Antoni Łomnicki and Włodzimierz Stożek, together with two of Stożek's sons. Kazimierz Bartel, geometer and former prime minister of Poland, was arrested on 2 July and killed later that month.

The killing did not stop there. Saks was shot by the Gestapo in Warsaw in 1942. Herman Auerbach died the same year, deported from the Lwów ghetto to the Bełżec death camp. Juliusz Schauder, of the Schauder fixed point theorem, was murdered in 1943. Mark Kac had left for the United States in 1938; his parents and his brother were killed in 1942. Steinhaus survived by hiding in the countryside under a false name.

Banach survived because of Rudolf Weigl, a biologist who had developed a vaccine against typhus. The vaccine was grown in lice, and the lice had to be fed on human blood: "louse feeders" let cages of lice feed on their legs. The German army needed the vaccine, and the papers of Weigl's institute protected those who worked there; Weigl used that protection to employ mathematicians, scientists, members of the underground and Jews. Banach fed lice there, and so did several other members of the school. Weigl also smuggled vaccine into the Warsaw and Lwów ghettos. In 2003 Yad Vashem recognized him as Righteous Among the

Nations. Banach lived to see the end of the war; he died of lung cancer in Lwów on 31 August 1945.

The book survived too. Its last entry, Problem 193, had been written by Steinhaus on 31 May 1941, a month before the Germans arrived. Ulam recalled that Mazur had once proposed, in case the city was bombed, to put the manuscripts and the Scottish Book in a case and to bury the case near a goal post of a football field outside the city; whether he did so is not known. What is known is that after the war Banach's son, Stefan Banach Jr., brought the book to Wrocław, where Steinhaus had moved.

29. Ulam after Lwów

The mathematics of the café went on in Ulam's life (see Figure 29) in unexpected ways. In 1943, after asking von Neumann for war work, he received an invitation, signed by Hans Bethe, to an unnamed project near Santa Fe; in February 1944 he arrived at Los Alamos and joined the Manhattan Project. In 1946, while recovering from encephalitis and brain surgery, Ulam played a great deal of solitaire. He wondered what the chance of winning a game was, and saw that it was easier to estimate that chance by playing many random games than by counting. With von Neumann he turned this idea into the *Monte Carlo method*, named by their colleague Nicholas Metropolis after Monte Carlo, where an uncle of Ulam's liked to gamble [MU49]; it is now among the most widely used methods in science. With Edward Teller he conceived the Teller–Ulam design of the hydrogen bomb. In the early 1950s, with Enrico Fermi and John Pasta, he simulated on the MANIAC computer a chain of oscillators that, unexpectedly, did not reach thermal equilibrium; their report appeared in 1955 [FPU55]. The programs were written by Mary Tsingou, whose role was acknowledged only decades later, when the problem became known as the Fermi–Pasta–Ulam–Tsingou problem [Dau08].

30. The 1957 Los Alamos mimeograph

Steinhaus had a typewritten copy of the book made and in 1956 sent it to Ulam, who was by then at the Los Alamos Scientific Laboratory in New Mexico. Ulam translated it into English and, in his words, "had some three hundred mimeographed copies of the translation made. I had to pay for this myself—Los Alamos is a government laboratory, and one cannot use taxpayers' money for such frivolous purposes." The copies circulated among mathematicians around the world, and for a generation they were how many mathematicians came to know the problems.

31. Problem 153: a live goose

The best-known prize of the Scottish Book was paid thirty-six years after it was offered. Mazur's Problem 153 of 6 November 1936 asked a question about approximating continuous functions of two variables and offered a live goose. The problem turned out to be equivalent to the approximation problem formulated by Grothendieck in 1955, the question whether every Banach space has the *approximation property*. A Banach space has this property if every compact operator from a Banach space into it is a norm limit of operators of finite rank. A negative answer would also settle Banach's basis problem, the question whether every separable Banach space has a *Schauder basis*, a sequence of vectors such that every vector is the sum of a unique norm-convergent series of scalar multiples of them. Indeed, a space with a Schauder basis has the approximation property. In 1972 the young Swedish mathematician Per Enflo constructed a separable Banach space without the approximation

property, and therefore without a Schauder basis [Enf73]. Mazur, then in Warsaw, kept his word: that same year, in a ceremony at the Stefan Banach Center, he presented the twenty-eight-year-old Enflo with a live goose. According to several accounts, the ceremony was broadcast on Polish television.

Enflo is also a concert pianist. In 1956, at the age of eleven, he won the Swedish competition for young pianists, and he won it again in 1961; among his teachers was Géza Anda. He has gone on performing ever since.

32. The road to Denton: Dan Mauldin and North Texas

In the 1970s the story moved to Texas. R. Daniel Mauldin, a measure theorist and descriptive set theorist at North Texas State University (now the University of North Texas) in Denton, took on the task of finding out what had become of the 193 problems. In May 1979 he organized the *Scottish Book Conference* in Denton.

The list of speakers reads like a family reunion. Ulam, Mark Kac, Antoni Zygmund, Paul Erdős and Andrzej Granas all lectured, Ulam on the anecdotal history of the book, Kac on his personal history of it, and Zygmund on Steinhaus and the development of Polish mathematics. R. D. Anderson and D. A. Martin spoke about work that went back to the Scottish Book. Martin had proved Borel determinacy in 1975: every infinite game of perfect information in which the players choose natural numbers, and whose payoff set is a Borel subset of the space ω^ω of infinite sequences of natural numbers, is determined (see Section 16). John Oxtoby, who had published the first proof of Banach's solution of Problem 43, toasted a "condensation of Poles" at the conference; R. H. Bing, a pillar of Texas topology, also offered a toast.

From the conference came the book edited by Mauldin, *The Scottish Book: Mathematics from the Scottish Café* (Birkhäuser, 1981) [Mau81]. It contains lectures from the conference, Ulam's "anecdotal history" of the book among them, and the problems, with commentaries on what became of them. A second edition appeared in 2015 [Mau15].

33. The New Scottish Book

The tradition did not die in Lwów. In Wrocław, where much of the Lwów university was re-established after the war, Steinhaus started the *New Scottish Book* (*Nowa Księga Szkocka*). Its first entry, by Steinhaus, is dated 2 July 1946; problems were entered in it regularly until 1979, and occasionally until 1987. The second edition of Mauldin's book includes a short history of the New Scottish Book and a selection of its problems.

34. Set-theoretic topology and Ramsey spaces at Toronto

The questions of the Scottish Book on measure, category and infinite games also run through set-theoretic topology, a field in which the University of Toronto has long been a leading center.

In 1930 Ulam proved that a countably additive probability measure that is defined on all subsets of a set and vanishes on points can exist only if the set is very large: at least as large as a weakly inaccessible cardinal, and, if the measure takes only the values 0 and 1, at least as large as a strongly inaccessible one [Ula30]. In particular, such a measure on all subsets of the real line can exist only if the continuum is at least as large as a weakly inaccessible cardinal; in fact, the least cardinal that carries such a measure is then weakly inaccessible and no larger than the continuum. Along the way Ulam introduced what are now called Ulam matrices. A set carries a two-valued measure of this kind exactly when it is at least

as large as a measurable cardinal, and measurable cardinals, with the larger cardinals that followed, became a standard tool for consistency proofs in topology. At Toronto, Franklin D. Tall and his collaborators applied such set-theoretic methods to classical problems of point-set topology, most notably the normal Moore space problem. Infinite games of the Banach–Mazur type, and their relatives such as the Menger game and the point-open game of Part III, are now everyday tools in the field.

Stevo Todorčević, at Toronto, developed the theory of Ramsey spaces, which unifies the Galvin–Prikry theorem, Ellentuck's theorem, Gowers's FIN_k theorem and much else in a single axiomatic framework [Tod10]. Here FIN_k, for a positive integer k, is the set of finitely supported functions from $\mathbb{N}$ to $\{0, 1, \ldots, k\}$ that take the value k. The circle closes in the present: Tall is one of the authors, with Eduardo Dueñez and José Iovino of UT San Antonio and Tonatiuh Matos-Wiederhold and Luciano Salvetti of Toronto, of recent work on the complexity of deep computations through the topology of function spaces [DIMW$^+$26]; that work is the subject of the seminar for which these notes were written.

35. The triumph of games: combinatorics, logic and the real-valued setting

Texas also became a center of the Banach space theory that Banach's 1932 monograph, *Théorie des opérations linéaires*, and the Scottish Book had launched. Edward Odell at the University of Texas at Austin and Thomas Schlumprecht at Texas A&M University were at the heart of it.

In 1991 Schlumprecht constructed the first arbitrarily distortable Banach space [Sch91], a space that admits, for every $\lambda > 1$, an equivalent norm whose supremum on the original unit sphere of every infinite-dimensional subspace exceeds λ times its infimum there. In 1994 Odell and Schlumprecht used his ideas to solve the distortion problem, the question whether ℓ_2 admits, for some $\lambda > 1$, an equivalent norm of this kind: the Hilbert space ℓ_2 is itself arbitrarily distortable [OS94]. Schlumprecht's space was also the starting point of W. T. Gowers and B. Maurey, who in 1991 (published in 1993) answered in the negative the old unconditional basic sequence problem, the question whether every infinite-dimensional Banach space contains an unconditional basic sequence, a sequence that is a Schauder basis of the closed subspace it spans and whose expansions converge whatever the order of their terms (see Section 13). They constructed a "hereditarily indecomposable" space, one in which no infinite-dimensional closed subspace is the direct sum of two infinite-dimensional closed subspaces; in particular, no infinite-dimensional subspace has an unconditional basis [GM93].

The last step needed a different kind of mathematics. Banach's homogeneous space problem, from his 1932 monograph, asks whether every *homogeneous* Banach space, an infinite-dimensional space that is isomorphic to all of its infinite-dimensional closed subspaces, is isomorphic to ℓ_2. To answer it, Gowers turned to infinite combinatorics and logic.

The bridge was infinite-dimensional Ramsey theory, the subject of Section 12: the Galvin–Prikry theorem, Silver's theorem and Ellentuck's theorem, all proved by logicians, say that definable colorings of the infinite subsets of ω have infinite homogeneous sets. Gowers saw that this machinery could be imported into Banach space theory if it were translated into a continuous, real-valued setting, with block sequences of vectors in place of infinite sets of integers, and with a winning strategy in a two-player game in place of a homogeneous set. His Ramsey theorem for this game yields his dichotomy theorem: every infinite-dimensional Banach space contains an infinite-dimensional closed subspace that either has an unconditional basis or is hereditarily indecomposable. The dichotomy, combined with the Gowers–Maurey

theorem that a hereditarily indecomposable space is isomorphic to none of its proper subspaces and with a theorem of Komorowski and Tomczak-Jaegermann, implies that every infinite-dimensional homogeneous Banach space is isomorphic to ℓ_2 [Gow96] (see Section 13).

When Gowers received the Fields Medal in 1998, the cycle was complete. Infinite games, born at a café table in Lwów in 1935, had passed through the foundations of mathematics (determinacy) and through the hands of logicians and combinatorialists (infinite Ramsey theory); by way of Texas, Paris and Cambridge, they had come back to analysis to settle a problem posed by Banach himself. The notebook that survived a war, and the people who survived it, or did not, and the colleagues who carried their problems across an ocean, are part of that triumph.

36. Epilogue: A Scottish Book of our own

The Scottish Book was not a journal. It was a notebook on a café table, open to anyone who sat down, where a problem could be written before it was understood and a prize could be a beer or a goose. That spirit is easy to revive. We propose a Scottish Book of our own for the participants of this seminar, in San Antonio and in Toronto: a shared notebook of the problems that come up in our discussions, each signed and dated, with a prize chosen by its author, and with room for the answers, partial results and counterexamples that follow. Students are especially welcome: some of the first authors of the original book were in their twenties, and some of its best problems are short questions that anyone can state.

Further reading

Ulam's autobiography [Ula76] and Kac's [Kac85] are the best first-hand accounts of the Lwów School and of what came after. Kałuża's biography of Banach [Kał96], written with a reporter's eye, tells the story of his life and of his city. Mauldin's book [Mau81, Mau15] contains the problems themselves, with commentaries on what became of them.

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FIGURE 29. Stanisław Ulam in 1962. Photograph by Alfred Eisenstaedt for *LIFE*, from a series of portraits of mathematicians that Eisenstaedt made for the magazine in 1962–63, which also includes Kurt Gödel and Claude Shannon.

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